

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12396851
B2
August 26, 2025
Hariton; Ilia et al.

Prosthetic valve with arms and flanges

Abstract

A valve frame includes a tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around a longitudinal axis of the valve frame, and a plurality of arms, extending radially outward from an upstream portion of the tubular portion to define an arm span, and configured to engage tissue in the atrium. An outer frame is coupled to the valve frame, and includes a ring circumscribing the tubular portion, and a plurality of flanges that extend radially outward from the ring to define a flange span, and are configured to engage ventricular tissue. Each arm defines a rigid portion and a flexible portion that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion. The majority of the flexible portion is disposed further radially outward than the flange span. Other embodiments are also described.

Inventors: Hariton; Ilia (Zichron Yaackov, IL), Harari; Boaz (Ganey Tikva, IL), Iamberger; Meni (Kfar Saba, IL)

Applicant: CARDIOVALVE LTD. (Or Yehuda, IL)

Family ID: 1000008778040

Assignee: CARDIOVALVE LTD. (Or Yehuda, IL)

Appl. No.: 18/368250

Filed: September 14, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240074856 A1	Mar. 07, 2024

Related U.S. Application Data

continuation parent-doc US 17141853 20210105 US 11801135 child-doc US 18368250
continuation parent-doc US 15668659 20170803 US 10918481 20210216 child-doc US 17141853
continuation-in-part parent-doc US 15541783 US 9974651 20180522 WO PCT/IL2016/050125 20160203
child-doc US 15668659
us-provisional-application US 62112343 20150205

Publication Classification

Int. Cl.: A61F2/24 (20060101); A61F2/915 (20130101); A61F2/95 (20130101)

U.S. Cl.:

CPC A61F2/2436 (20130101); A61F2/2409 (20130101); A61F2/2418 (20130101); A61F2/243 (20130101); A61F2002/91575 (20130101); A61F2/9517 (20200501); A61F2220/0016 (20130101); A61F2220/0025 (20130101)

Field of Classification Search

CPC: A61F (2/243); A61F (2/2436); A61F (2/2409); A61F (2/2418); A61F (2/2403)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3604488	12/1970	Wishart et al.	N/A	N/A
3656185	12/1971	Carpentier	N/A	N/A
3840018	12/1973	Heifetz	N/A	N/A
3874388	12/1974	King et al.	N/A	N/A
3898701	12/1974	La Russa	N/A	N/A
4042979	12/1976	Angell	N/A	N/A
4118805	12/1977	Reimels	N/A	N/A
4214349	12/1979	Munch	N/A	N/A
4222126	12/1979	Boretos et al.	N/A	N/A
4261342	12/1980	Aranguren	N/A	N/A
4275469	12/1980	Gabbay	N/A	N/A
4340091	12/1981	Skelton et al.	N/A	N/A
4423525	12/1983	Vallana et al.	N/A	N/A
4434828	12/1983	Trincia	N/A	N/A
4473928	12/1983	Johnson	N/A	N/A
4602911	12/1985	Ahmadi et al.	N/A	N/A
4625727	12/1985	Leiboff	N/A	N/A
4712549	12/1986	Peters et al.	N/A	N/A
4778468	12/1987	Hunt et al.	N/A	N/A
4853986	12/1988	Allen	N/A	N/A
4892541	12/1989	Alonso	N/A	N/A
4917698	12/1989	Carpenter et al.	N/A	N/A
4961738	12/1989	Mackin	N/A	N/A
4972494	12/1989	White et al.	N/A	N/A
4994077	12/1990	Dobben	N/A	N/A
5061277	12/1990	Carpentier et al.	N/A	N/A
5078739	12/1991	Martin	N/A	N/A
5089006	12/1991	Stiles	N/A	N/A
5104407	12/1991	Lam et al.	N/A	N/A
5108420	12/1991	Marks	N/A	N/A
5201757	12/1992	Heyn et al.	N/A	N/A
5201880	12/1992	Wright	N/A	N/A
5258008	12/1992	Wilk	N/A	N/A
5275622	12/1993	Lazarus et al.	N/A	N/A
5300034	12/1993	Behnke	N/A	N/A
5306296	12/1993	Wright et al.	N/A	N/A

5314473	12/1993	Godin	N/A	N/A
5325845	12/1993	Adair	N/A	N/A
5332402	12/1993	Teitelbaum	N/A	N/A
5397351	12/1994	Pavcnik et al.	N/A	N/A
5405378	12/1994	Strecker	N/A	N/A
5443500	12/1994	Sigwart	N/A	N/A
5450860	12/1994	O'Connor	N/A	N/A
5473812	12/1994	Morris et al.	N/A	N/A
5477856	12/1994	Lundquist	N/A	N/A
5593424	12/1996	Northrup III	N/A	N/A
5601572	12/1996	Middleman et al.	N/A	N/A
5607444	12/1996	Lam	N/A	N/A
5607470	12/1996	Milo	N/A	N/A
5626609	12/1996	Zvenyatsky et al.	N/A	N/A
5647857	12/1996	Anderson et al.	N/A	N/A
5669919	12/1996	Sanders et al.	N/A	N/A
5674279	12/1996	Wright et al.	N/A	N/A
5683402	12/1996	Cosgrove et al.	N/A	N/A
5702397	12/1996	Goble et al.	N/A	N/A
5702398	12/1996	Tarabishy	N/A	N/A
5709695	12/1997	Northrup, III	N/A	N/A
5713948	12/1997	Uflacker	N/A	N/A
5716370	12/1997	Williamson et al.	N/A	N/A
5716397	12/1997	Myers	N/A	N/A
5716417	12/1997	Girard et al.	N/A	N/A
5728116	12/1997	Rosenman	N/A	N/A
5730150	12/1997	Peppel et al.	N/A	N/A
5741297	12/1997	Simon	N/A	N/A
5749371	12/1997	Zadini et al.	N/A	N/A
5749920	12/1997	Quiachon et al.	N/A	N/A
5765682	12/1997	Bley et al.	N/A	N/A
5766151	12/1997	Valley et al.	N/A	N/A
5776140	12/1997	Cottone	N/A	N/A
5810882	12/1997	Bolduc	N/A	N/A
5824066	12/1997	Gross	N/A	N/A
5830221	12/1997	Stein et al.	N/A	N/A
5843120	12/1997	Israel et al.	N/A	N/A
5855614	12/1998	Stevens et al.	N/A	N/A
5868777	12/1998	Lam	N/A	N/A
5873906	12/1998	Lau et al.	N/A	N/A
5876373	12/1998	Giba et al.	N/A	N/A
5906619	12/1998	Olson et al.	N/A	N/A
5935098	12/1998	Blaisdell et al.	N/A	N/A
5954766	12/1998	Zadno-Azizi et al.	N/A	N/A
5957949	12/1998	Leonhardt et al.	N/A	N/A
5957953	12/1998	DiPoto et al.	N/A	N/A
5961440	12/1998	Schweich et al.	N/A	N/A
5961539	12/1998	Northrup, III et al.	N/A	N/A
5961549	12/1998	Nguyen et al.	N/A	N/A
5980565	12/1998	Jayaraman	N/A	N/A
5984959	12/1998	Robertson	N/A	N/A
6010530	12/1999	Goicoechea	N/A	N/A
6019787	12/1999	Richard et al.	N/A	N/A
6042554	12/1999	Rosenman	N/A	N/A
6042607	12/1999	Williamson, IV	N/A	N/A

6045497	12/1999	Schweich et al.	N/A	N/A
6050936	12/1999	Schweich et al.	N/A	N/A
6059715	12/1999	Schweich et al.	N/A	N/A
6059827	12/1999	Fenton	N/A	N/A
6074401	12/1999	Gardiner et al.	N/A	N/A
6074417	12/1999	Peredo	N/A	N/A
6102945	12/1999	Campbell	N/A	N/A
6106550	12/1999	Magovern	N/A	N/A
6110200	12/1999	Hinnenkamp	N/A	N/A
6113612	12/1999	Swanson et al.	N/A	N/A
6120534	12/1999	Ruiz	N/A	N/A
6126686	12/1999	Badylak et al.	N/A	N/A
6143024	12/1999	Campbell et al.	N/A	N/A
6152937	12/1999	Peterson et al.	N/A	N/A
6159240	12/1999	Sparer	N/A	N/A
6165119	12/1999	Schweich et al.	N/A	N/A
6165183	12/1999	Kuehn et al.	N/A	N/A
6165210	12/1999	Lau et al.	N/A	N/A
6174332	12/2000	Loch	N/A	N/A
6183411	12/2000	Mortier et al.	N/A	N/A
6187020	12/2000	Zegdi et al.	N/A	N/A
6187040	12/2000	Wright	N/A	N/A
6193686	12/2000	Estrada et al.	N/A	N/A
6193745	12/2000	Fogarty et al.	N/A	N/A
6217610	12/2000	Carpentier et al.	N/A	N/A
6221102	12/2000	Baker et al.	N/A	N/A
6231602	12/2000	Carpentier et al.	N/A	N/A
6251092	12/2000	Qin et al.	N/A	N/A
6254609	12/2000	Vrba et al.	N/A	N/A
6264700	12/2000	Kilcoyne et al.	N/A	N/A
6271278	12/2000	Park et al.	N/A	N/A
6287339	12/2000	Vasquez et al.	N/A	N/A
6296656	12/2000	Bodluc et al.	N/A	N/A
6312465	12/2000	Griffin et al.	N/A	N/A
6315784	12/2000	Djurovic	N/A	N/A
6319281	12/2000	Patel	N/A	N/A
6332893	12/2000	Mortier et al.	N/A	N/A
6334873	12/2001	Lane et al.	N/A	N/A
6346074	12/2001	Roth	N/A	N/A
6350278	12/2001	Lenker et al.	N/A	N/A
6352561	12/2001	Leopold et al.	N/A	N/A
6391036	12/2001	Berg et al.	N/A	N/A
6398758	12/2001	Jacobsen et al.	N/A	N/A
6402780	12/2001	Williamson, IV	N/A	N/A
6406420	12/2001	McCarthy et al.	N/A	N/A
6406493	12/2001	Tu et al.	N/A	N/A
6409755	12/2001	Vrba	N/A	N/A
6419696	12/2001	Ortiz et al.	N/A	N/A
6428550	12/2001	Vargas et al.	N/A	N/A
6440164	12/2001	Dimatteo et al.	N/A	N/A
6451054	12/2001	Stevens	N/A	N/A
6454799	12/2001	Schreck	N/A	N/A
6458153	12/2001	Bailey et al.	N/A	N/A
6461366	12/2001	Seguin	N/A	N/A
6470892	12/2001	Forsell	N/A	N/A

6478807	12/2001	Foreman et al.	N/A	N/A
6482228	12/2001	Norred	N/A	N/A
6491711	12/2001	Durcan	N/A	N/A
6503274	12/2002	Howanec et al.	N/A	N/A
6511491	12/2002	Grudem et al.	N/A	N/A
6524338	12/2002	Gundry	N/A	N/A
6530952	12/2002	Vesely	N/A	N/A
6533772	12/2002	Sherts et al.	N/A	N/A
6537314	12/2002	Langberg et al.	N/A	N/A
6540782	12/2002	Snyders	N/A	N/A
6547801	12/2002	Dargent et al.	N/A	N/A
6551350	12/2002	Thornton et al.	N/A	N/A
6554845	12/2002	Fleenor et al.	N/A	N/A
6558396	12/2002	Inoue	N/A	N/A
6558418	12/2002	Carpentier et al.	N/A	N/A
6564805	12/2002	Garrison et al.	N/A	N/A
6565603	12/2002	Cox	N/A	N/A
6569196	12/2002	Vesely	N/A	N/A
6569198	12/2002	Wilson et al.	N/A	N/A
6579297	12/2002	Bicek et al.	N/A	N/A
6582464	12/2002	Gabbay	N/A	N/A
6589160	12/2002	Schweich et al.	N/A	N/A
6602263	12/2002	Swanson et al.	N/A	N/A
6602288	12/2002	Cosgrove et al.	N/A	N/A
6602289	12/2002	Colvin et al.	N/A	N/A
6613078	12/2002	Barone	N/A	N/A
6613079	12/2002	Wolinsky et al.	N/A	N/A
6616675	12/2002	Evard et al.	N/A	N/A
6619291	12/2002	Hlavka et al.	N/A	N/A
6626899	12/2002	Houser et al.	N/A	N/A
6626917	12/2002	Craig	N/A	N/A
6626930	12/2002	Allen et al.	N/A	N/A
6629534	12/2002	St. Goar et al.	N/A	N/A
6629921	12/2002	Schweich et al.	N/A	N/A
6651671	12/2002	Donlon et al.	N/A	N/A
6652556	12/2002	VanTassel et al.	N/A	N/A
6669724	12/2002	Park et al.	N/A	N/A
6682558	12/2003	Tu et al.	N/A	N/A
6689125	12/2003	Keith et al.	N/A	N/A
6689164	12/2003	Seguin	N/A	N/A
6695866	12/2003	Kuehn et al.	N/A	N/A
6699256	12/2003	Logan et al.	N/A	N/A
6702826	12/2003	Liddicoat et al.	N/A	N/A
6702846	12/2003	Mikus et al.	N/A	N/A
6706065	12/2003	Langberg et al.	N/A	N/A
6709456	12/2003	Langberg et al.	N/A	N/A
6711444	12/2003	Koblish	N/A	N/A
6712836	12/2003	Berg et al.	N/A	N/A
6716244	12/2003	Klaco	N/A	N/A
6718985	12/2003	Hlavka et al.	N/A	N/A
6719781	12/2003	Kim	N/A	N/A
6719786	12/2003	Ryan et al.	N/A	N/A
6719788	12/2003	Cox	N/A	N/A
6723038	12/2003	Schroeder et al.	N/A	N/A
6726716	12/2003	Marquez	N/A	N/A

6726717	12/2003	Alfieri et al.	N/A	N/A
6730118	12/2003	Spenser et al.	N/A	N/A
6730121	12/2003	Ortiz et al.	N/A	N/A
6733525	12/2003	Yang et al.	N/A	N/A
6749630	12/2003	McCarthy et al.	N/A	N/A
6752813	12/2003	Goldfarb et al.	N/A	N/A
6755857	12/2003	Peterson et al.	N/A	N/A
6764310	12/2003	Ichihashi et al.	N/A	N/A
6764510	12/2003	Vidlund et al.	N/A	N/A
6764514	12/2003	Li et al.	N/A	N/A
6764518	12/2003	Godin	N/A	N/A
6767362	12/2003	Schreck	N/A	N/A
6770083	12/2003	Seguin	N/A	N/A
6786924	12/2003	Ryan et al.	N/A	N/A
6786925	12/2003	Schoon et al.	N/A	N/A
6790231	12/2003	Liddicoat et al.	N/A	N/A
6797001	12/2003	Mathis et al.	N/A	N/A
6797002	12/2003	Spence et al.	N/A	N/A
6802319	12/2003	Stevens et al.	N/A	N/A
6805710	12/2003	Bolling et al.	N/A	N/A
6805711	12/2003	Quijano et al.	N/A	N/A
6821297	12/2003	Snyders	N/A	N/A
6830585	12/2003	Artof et al.	N/A	N/A
6830638	12/2003	Boylan et al.	N/A	N/A
6849084	12/2004	Rabkin et al.	N/A	N/A
6855126	12/2004	Flinchbaugh	N/A	N/A
6858039	12/2004	McCarthy	N/A	N/A
6884250	12/2004	Monassevitch et al.	N/A	N/A
6884257	12/2004	Cox	N/A	N/A
6893459	12/2004	Macoviak	N/A	N/A
6893460	12/2004	Spenser et al.	N/A	N/A
6908482	12/2004	McCarthy et al.	N/A	N/A
6918917	12/2004	Nguyen et al.	N/A	N/A
6926715	12/2004	Hauck et al.	N/A	N/A
6926730	12/2004	Nguyen et al.	N/A	N/A
6939370	12/2004	Hartley et al.	N/A	N/A
6951571	12/2004	Srivastava	N/A	N/A
6960217	12/2004	Bolduc	N/A	N/A
6964684	12/2004	Ortiz et al.	N/A	N/A
6964686	12/2004	Gordon	N/A	N/A
6974476	12/2004	McGuckin et al.	N/A	N/A
6976995	12/2004	Mathis et al.	N/A	N/A
6986775	12/2005	Morales et al.	N/A	N/A
6989028	12/2005	Lashinski et al.	N/A	N/A
6997918	12/2005	Soltesz et al.	N/A	N/A
6997951	12/2005	Solem et al.	N/A	N/A
7004176	12/2005	Lau	N/A	N/A
7011669	12/2005	Kimblad	N/A	N/A
7011681	12/2005	Vesely	N/A	N/A
7011682	12/2005	Lashinski et al.	N/A	N/A
7018406	12/2005	Seguin et al.	N/A	N/A
7037334	12/2005	Hlavka et al.	N/A	N/A
7041132	12/2005	Quijano et al.	N/A	N/A
7074236	12/2005	Rabkin et al.	N/A	N/A
7077850	12/2005	Kortenbach	N/A	N/A

7077861	12/2005	Spence	N/A	N/A
7077862	12/2005	Vidlund et al.	N/A	N/A
7087064	12/2005	Hyde	N/A	N/A
7101336	12/2005	Miller	N/A	N/A
7101395	12/2005	Tremulis et al.	N/A	N/A
7101396	12/2005	Artof et al.	N/A	N/A
7112207	12/2005	Allen et al.	N/A	N/A
7118595	12/2005	Ryan et al.	N/A	N/A
7125421	12/2005	Tremulis et al.	N/A	N/A
7137184	12/2005	Schreck	N/A	N/A
7150737	12/2005	Purdy et al.	N/A	N/A
7159593	12/2006	McCarthy et al.	N/A	N/A
7166127	12/2006	Spence et al.	N/A	N/A
7169187	12/2006	Datta et al.	N/A	N/A
7172625	12/2006	Shu et al.	N/A	N/A
7175656	12/2006	Khairkhahan	N/A	N/A
7175660	12/2006	Cartledge et al.	N/A	N/A
7186262	12/2006	Saadat	N/A	N/A
7186264	12/2006	Liddicoat et al.	N/A	N/A
7189199	12/2006	McCarthy et al.	N/A	N/A
7192443	12/2006	Solem et al.	N/A	N/A
7198646	12/2006	Figulla et al.	N/A	N/A
7201772	12/2006	Schwammenthal	N/A	N/A
7220277	12/2006	Arru et al.	N/A	N/A
7226467	12/2006	Lucatero et al.	N/A	N/A
7226477	12/2006	Cox	N/A	N/A
7226647	12/2006	Kasperchik et al.	N/A	N/A
7229452	12/2006	Kayan	N/A	N/A
7238191	12/2006	Bachmann	N/A	N/A
7252682	12/2006	Seguin	N/A	N/A
7261686	12/2006	Couvillon, Jr.	N/A	N/A
7288097	12/2006	Seguin	N/A	N/A
7288111	12/2006	Holloway et al.	N/A	N/A
7294148	12/2006	McCarthy	N/A	N/A
7297150	12/2006	Cartledge et al.	N/A	N/A
7311728	12/2006	Solem et al.	N/A	N/A
7311729	12/2006	Mathis et al.	N/A	N/A
7314485	12/2007	Mathis	N/A	N/A
7316710	12/2007	Cheng et al.	N/A	N/A
7316716	12/2007	Egan	N/A	N/A
7329279	12/2007	Haug et al.	N/A	N/A
7329280	12/2007	Bolling et al.	N/A	N/A
7335213	12/2007	Hyde et al.	N/A	N/A
7351256	12/2007	Hojeibane et al.	N/A	N/A
7361190	12/2007	Shouljian et al.	N/A	N/A
7364588	12/2007	Mathis et al.	N/A	N/A
7374571	12/2007	Pease et al.	N/A	N/A
7374573	12/2007	Gabbay	N/A	N/A
7377938	12/2007	Sarac et al.	N/A	N/A
7377941	12/2007	Rhee et al.	N/A	N/A
7381218	12/2007	Schreck	N/A	N/A
7381219	12/2007	Salahieh et al.	N/A	N/A
7390329	12/2007	Westra et al.	N/A	N/A
7404824	12/2007	Webler et al.	N/A	N/A
7422603	12/2007	Lane	N/A	N/A

7429269	12/2007	Schwammenthal	N/A	N/A
7431692	12/2007	Zollinger et al.	N/A	N/A
7442204	12/2007	Schwammenthal	N/A	N/A
7442207	12/2007	Rafiee	N/A	N/A
7445630	12/2007	Lashinski et al.	N/A	N/A
7452376	12/2007	Lim et al.	N/A	N/A
7455677	12/2007	Vargas et al.	N/A	N/A
7455688	12/2007	Furst et al.	N/A	N/A
7455690	12/2007	Cartledge et al.	N/A	N/A
7462162	12/2007	Phan et al.	N/A	N/A
7481838	12/2008	Carpentier et al.	N/A	N/A
7485142	12/2008	Milo	N/A	N/A
7500989	12/2008	Solem et al.	N/A	N/A
7507252	12/2008	Lashinski et al.	N/A	N/A
7510575	12/2008	Spenser et al.	N/A	N/A
7510577	12/2008	Moaddeb et al.	N/A	N/A
7513909	12/2008	Lane et al.	N/A	N/A
7524331	12/2008	Birdsall	N/A	N/A
7527646	12/2008	Rahdert et al.	N/A	N/A
7527647	12/2008	Spence	N/A	N/A
7530995	12/2008	Quijano et al.	N/A	N/A
7549983	12/2008	Roue et al.	N/A	N/A
7556632	12/2008	Zadno	N/A	N/A
7556646	12/2008	Yang et al.	N/A	N/A
7559936	12/2008	Levine	N/A	N/A
7562660	12/2008	Saadat	N/A	N/A
7563267	12/2008	Goldfarb et al.	N/A	N/A
7563273	12/2008	Goldfarb et al.	N/A	N/A
7569062	12/2008	Kuehn et al.	N/A	N/A
7582111	12/2008	Krolik et al.	N/A	N/A
7585321	12/2008	Cribier	N/A	N/A
7588582	12/2008	Starksen et al.	N/A	N/A
7591826	12/2008	Alferness et al.	N/A	N/A
7597711	12/2008	Drews et al.	N/A	N/A
7604646	12/2008	Goldfarb et al.	N/A	N/A
7608091	12/2008	Goldfarb et al.	N/A	N/A
7608103	12/2008	McCarthy	N/A	N/A
7611534	12/2008	Kapadia et al.	N/A	N/A
7618449	12/2008	Tremulis et al.	N/A	N/A
7621948	12/2008	Hermann et al.	N/A	N/A
7625403	12/2008	Krivoruchko	N/A	N/A
7632302	12/2008	Vreeman et al.	N/A	N/A
7632303	12/2008	Stalker et al.	N/A	N/A
7635329	12/2008	Goldfarb et al.	N/A	N/A
7635386	12/2008	Gammie	N/A	N/A
7648528	12/2009	Styrc	N/A	N/A
7655015	12/2009	Goldfarb et al.	N/A	N/A
7666204	12/2009	Thornton et al.	N/A	N/A
7682319	12/2009	Martin	N/A	N/A
7682369	12/2009	Seguin	N/A	N/A
7682380	12/2009	Thornton et al.	N/A	N/A
7686822	12/2009	Shayani	N/A	N/A
7699892	12/2009	Rafiee et al.	N/A	N/A
7704269	12/2009	St. Goar et al.	N/A	N/A
7704277	12/2009	Zakay et al.	N/A	N/A

7708775	12/2009	Rowe et al.	N/A	N/A
7717951	12/2009	Flagle et al.	N/A	N/A
7717952	12/2009	Case et al.	N/A	N/A
7717955	12/2009	Lane et al.	N/A	N/A
7722666	12/2009	Lafontaine	N/A	N/A
7731741	12/2009	Eidenschink	N/A	N/A
7731742	12/2009	Schlick et al.	N/A	N/A
7736388	12/2009	Goldfarb et al.	N/A	N/A
7748389	12/2009	Salahieh et al.	N/A	N/A
7753922	12/2009	Starksen	N/A	N/A
7753924	12/2009	Starksen et al.	N/A	N/A
7753949	12/2009	Lamphere et al.	N/A	N/A
7758595	12/2009	Allen et al.	N/A	N/A
7758632	12/2009	Hojeibane et al.	N/A	N/A
7758640	12/2009	Vesely	N/A	N/A
7771467	12/2009	Svensson	N/A	N/A
7771469	12/2009	Liddicoat	N/A	N/A
7776080	12/2009	Bei et al.	N/A	N/A
7776083	12/2009	Vesely	N/A	N/A
7780726	12/2009	Seguin	N/A	N/A
7785341	12/2009	Forster et al.	N/A	N/A
7799069	12/2009	Bailey et al.	N/A	N/A
7803181	12/2009	Furst et al.	N/A	N/A
7811296	12/2009	Goldfarb et al.	N/A	N/A
7811316	12/2009	Kalmann et al.	N/A	N/A
7824442	12/2009	Salahieh et al.	N/A	N/A
7837645	12/2009	Bessler et al.	N/A	N/A
7837727	12/2009	Goetz et al.	N/A	N/A
7842081	12/2009	Yadin	N/A	N/A
7850725	12/2009	Vardi et al.	N/A	N/A
7871368	12/2010	Zollinger et al.	N/A	N/A
7871432	12/2010	Bergin	N/A	N/A
7871433	12/2010	Lattouf	N/A	N/A
7871436	12/2010	Ryan et al.	N/A	N/A
7887583	12/2010	Macoviak	N/A	N/A
7892281	12/2010	Seguin et al.	N/A	N/A
7896915	12/2010	Guyenot et al.	N/A	N/A
7914544	12/2010	Nguyen et al.	N/A	N/A
7914569	12/2010	Nguyen et al.	N/A	N/A
7927370	12/2010	Webler et al.	N/A	N/A
7927371	12/2010	Navia et al.	N/A	N/A
7942927	12/2010	Kaye et al.	N/A	N/A
7947072	12/2010	Yang et al.	N/A	N/A
7947075	12/2010	Goetz et al.	N/A	N/A
7951195	12/2010	Antonsson et al.	N/A	N/A
7955375	12/2010	Agnew	N/A	N/A
7955377	12/2010	Melsheimer	N/A	N/A
7955384	12/2010	Rafiee et al.	N/A	N/A
7959666	12/2010	Salahieh et al.	N/A	N/A
7959672	12/2010	Salahieh et al.	N/A	N/A
7967833	12/2010	Sterman et al.	N/A	N/A
7967857	12/2010	Lane	N/A	N/A
7981151	12/2010	Rowe	N/A	N/A
7981153	12/2010	Fogarty et al.	N/A	N/A
7988725	12/2010	Gross et al.	N/A	N/A

7992567	12/2010	Hirotosuka et al.	N/A	N/A
7993368	12/2010	Gambale et al.	N/A	N/A
7993392	12/2010	Righini et al.	N/A	N/A
7993393	12/2010	Carpentier et al.	N/A	N/A
7993397	12/2010	Lashinski	N/A	N/A
8002825	12/2010	Letac et al.	N/A	N/A
8002826	12/2010	Seguin	N/A	N/A
8012201	12/2010	Lashinski et al.	N/A	N/A
8016877	12/2010	Seguin et al.	N/A	N/A
8016882	12/2010	Macoviak	N/A	N/A
8021420	12/2010	Dolan	N/A	N/A
8021421	12/2010	Fogarty et al.	N/A	N/A
8025695	12/2010	Fogarty et al.	N/A	N/A
8029518	12/2010	Goldfarb et al.	N/A	N/A
8029557	12/2010	Sobrinho-Serrano et al.	N/A	N/A
8029564	12/2010	Johnson et al.	N/A	N/A
8034103	12/2010	Burriesci	N/A	N/A
8034104	12/2010	Carpentier et al.	N/A	N/A
8038720	12/2010	Wallace et al.	N/A	N/A
8043360	12/2010	McNamara et al.	N/A	N/A
8048138	12/2010	Sullivan et al.	N/A	N/A
8048140	12/2010	Purdy	N/A	N/A
8048153	12/2010	Salahieh et al.	N/A	N/A
8052592	12/2010	Goldfarb et al.	N/A	N/A
8052741	12/2010	Bruszewski et al.	N/A	N/A
8052749	12/2010	Salahieh et al.	N/A	N/A
8057493	12/2010	Goldfarb et al.	N/A	N/A
8057532	12/2010	Hoffman	N/A	N/A
8057540	12/2010	Letac et al.	N/A	N/A
8062355	12/2010	Figulla et al.	N/A	N/A
8062359	12/2010	Marquez et al.	N/A	N/A
8070708	12/2010	Rottenberg et al.	N/A	N/A
8070800	12/2010	Lock et al.	N/A	N/A
8070802	12/2010	Lamphere et al.	N/A	N/A
8070804	12/2010	Hyde	N/A	N/A
8070805	12/2010	Vidlund	N/A	N/A
8075611	12/2010	Milwee et al.	N/A	N/A
8075616	12/2010	Solem	N/A	N/A
8080054	12/2010	Rowe	N/A	N/A
8083793	12/2010	Lane et al.	N/A	N/A
D652927	12/2011	Braido et al.	N/A	N/A
D653341	12/2011	Braido et al.	N/A	N/A
8092518	12/2011	Schreck	N/A	N/A
8092520	12/2011	Quadri	N/A	N/A
8092521	12/2011	Figulla et al.	N/A	N/A
8100964	12/2011	Spence	N/A	N/A
8105377	12/2011	Liddicoat	N/A	N/A
8109996	12/2011	Stacchino et al.	N/A	N/A
8118866	12/2011	Herrmann et al.	N/A	N/A
8123800	12/2011	McCarthy	N/A	N/A
8123801	12/2011	Milo	N/A	N/A
8133270	12/2011	Kheradvar et al.	N/A	N/A
8136218	12/2011	Millwee et al.	N/A	N/A
8137398	12/2011	Tuval et al.	N/A	N/A
8142492	12/2011	Forster et al.	N/A	N/A

8142493	12/2011	Spence et al.	N/A	N/A
8142494	12/2011	Rahdert et al.	N/A	N/A
8142495	12/2011	Hasenkam et al.	N/A	N/A
8142496	12/2011	Berrekouw	N/A	N/A
8142497	12/2011	Friedman	N/A	N/A
8147504	12/2011	Ino et al.	N/A	N/A
8147542	12/2011	Maisano et al.	N/A	N/A
8152844	12/2011	Rao	N/A	N/A
8157852	12/2011	Bloom et al.	N/A	N/A
8157853	12/2011	Laske et al.	N/A	N/A
8157860	12/2011	McNamara et al.	N/A	N/A
8163008	12/2011	Wilson et al.	N/A	N/A
8163013	12/2011	Machold et al.	N/A	N/A
8163014	12/2011	Lane et al.	N/A	N/A
D660433	12/2011	Braido et al.	N/A	N/A
D660967	12/2011	Braido et al.	N/A	N/A
8167894	12/2011	Miles et al.	N/A	N/A
8167932	12/2011	Bourang et al.	N/A	N/A
8167935	12/2011	McGuckin, Jr. et al.	N/A	N/A
8172896	12/2011	McNamara et al.	N/A	N/A
8172898	12/2011	Alferness et al.	N/A	N/A
8177836	12/2011	Lee et al.	N/A	N/A
8182528	12/2011	Salahieh et al.	N/A	N/A
8187299	12/2011	Goldfarb et al.	N/A	N/A
8187324	12/2011	Webler et al.	N/A	N/A
8202315	12/2011	Hlavka et al.	N/A	N/A
8206439	12/2011	Gomez-Duran	N/A	N/A
8211169	12/2011	Lane et al.	N/A	N/A
8216256	12/2011	Raschdorf, Jr. et al.	N/A	N/A
8216301	12/2011	Bonhoeffer et al.	N/A	N/A
8221492	12/2011	Case et al.	N/A	N/A
8221493	12/2011	Boyle et al.	N/A	N/A
8226710	12/2011	Nguyen et al.	N/A	N/A
8226711	12/2011	Mortier et al.	N/A	N/A
8231670	12/2011	Salahieh et al.	N/A	N/A
8231671	12/2011	Kim	N/A	N/A
8235933	12/2011	Keren et al.	N/A	N/A
8236045	12/2011	Benichou et al.	N/A	N/A
8236049	12/2011	Rowe et al.	N/A	N/A
8241351	12/2011	Cabiri	N/A	N/A
8252042	12/2011	McNamara et al.	N/A	N/A
8252050	12/2011	Maisano et al.	N/A	N/A
8252051	12/2011	Chau et al.	N/A	N/A
8252052	12/2011	Salahieh et al.	N/A	N/A
8257390	12/2011	Carley et al.	N/A	N/A
8262725	12/2011	Subramanian	N/A	N/A
8267988	12/2011	Hamer et al.	N/A	N/A
8277501	12/2011	Chalekian et al.	N/A	N/A
8277502	12/2011	Miller et al.	N/A	N/A
8287584	12/2011	Salahieh et al.	N/A	N/A
8287591	12/2011	Keidar et al.	N/A	N/A
8298280	12/2011	Yadin et al.	N/A	N/A
8303608	12/2011	Goldfarb et al.	N/A	N/A
8303653	12/2011	Bonhoeffer et al.	N/A	N/A
8308798	12/2011	Pintor et al.	N/A	N/A

8313525	12/2011	Tuval et al.	N/A	N/A
8317853	12/2011	Agnew	N/A	N/A
8317855	12/2011	Gregorich et al.	N/A	N/A
8323334	12/2011	Deem et al.	N/A	N/A
8323335	12/2011	Rowe et al.	N/A	N/A
8328868	12/2011	Paul et al.	N/A	N/A
8333777	12/2011	Schaller et al.	N/A	N/A
8337541	12/2011	Quadri et al.	N/A	N/A
8343173	12/2012	Starksen et al.	N/A	N/A
8343174	12/2012	Goldfarb et al.	N/A	N/A
8343213	12/2012	Salahieh et al.	N/A	N/A
8348999	12/2012	Kheradvar et al.	N/A	N/A
8349002	12/2012	Milo	N/A	N/A
8353956	12/2012	Miller et al.	N/A	N/A
8357195	12/2012	Kuehn	N/A	N/A
8361144	12/2012	Fish et al.	N/A	N/A
8366767	12/2012	Zhang	N/A	N/A
8372140	12/2012	Hoffman et al.	N/A	N/A
8377119	12/2012	Drews et al.	N/A	N/A
8382829	12/2012	Call et al.	N/A	N/A
8388680	12/2012	Starksen et al.	N/A	N/A
8393517	12/2012	Milo	N/A	N/A
8398708	12/2012	Meiri et al.	N/A	N/A
8403981	12/2012	Forster et al.	N/A	N/A
8403983	12/2012	Quadri et al.	N/A	N/A
8408214	12/2012	Spenser	N/A	N/A
8414644	12/2012	Quadri et al.	N/A	N/A
8425593	12/2012	Braido et al.	N/A	N/A
8430926	12/2012	Kirson	N/A	N/A
8430934	12/2012	Das	N/A	N/A
8444689	12/2012	Zhang	N/A	N/A
8449599	12/2012	Chau et al.	N/A	N/A
8449625	12/2012	Campbell et al.	N/A	N/A
8454686	12/2012	Alkhatib	N/A	N/A
8460365	12/2012	Haverkost et al.	N/A	N/A
8460370	12/2012	Zakay et al.	N/A	N/A
8460371	12/2012	Hlavka et al.	N/A	N/A
8460372	12/2012	Mcnamara et al.	N/A	N/A
8474460	12/2012	Barrett et al.	N/A	N/A
8475491	12/2012	Milo	N/A	N/A
8480732	12/2012	Subramanian	N/A	N/A
8500800	12/2012	Maisano et al.	N/A	N/A
8500821	12/2012	Sobrinho-Serrano et al.	N/A	N/A
8512400	12/2012	Tran et al.	N/A	N/A
8518107	12/2012	Tsukashima et al.	N/A	N/A
8523881	12/2012	Cabiri et al.	N/A	N/A
8523940	12/2012	Richardson et al.	N/A	N/A
8529431	12/2012	Baker et al.	N/A	N/A
8539662	12/2012	Stacchino et al.	N/A	N/A
8540767	12/2012	Zhang	N/A	N/A
8545544	12/2012	Spenser et al.	N/A	N/A
8545553	12/2012	Zipory et al.	N/A	N/A
8551160	12/2012	Figulla et al.	N/A	N/A
8551161	12/2012	Dolan	N/A	N/A
8562672	12/2012	Bonhoeffer et al.	N/A	N/A

8568475	12/2012	Nguyen et al.	N/A	N/A
8579964	12/2012	Lane et al.	N/A	N/A
8579965	12/2012	Bonhoeffer et al.	N/A	N/A
8585755	12/2012	Chau et al.	N/A	N/A
8585756	12/2012	Bonhoeffer et al.	N/A	N/A
8591460	12/2012	Wilson et al.	N/A	N/A
8591570	12/2012	Revuelta et al.	N/A	N/A
8591576	12/2012	Hasenkam et al.	N/A	N/A
8608797	12/2012	Gross et al.	N/A	N/A
8623075	12/2013	Murray et al.	N/A	N/A
8623080	12/2013	Fogarty et al.	N/A	N/A
8628569	12/2013	Benichou et al.	N/A	N/A
8628570	12/2013	Seguin	N/A	N/A
8628571	12/2013	Hacohen et al.	N/A	N/A
8641727	12/2013	Starksen et al.	N/A	N/A
8652202	12/2013	Alon et al.	N/A	N/A
8652203	12/2013	Quadri et al.	N/A	N/A
8652204	12/2013	Quill et al.	N/A	N/A
8657872	12/2013	Seguin	N/A	N/A
8663322	12/2013	Keranen	N/A	N/A
8673020	12/2013	Sobrinho-Serrano et al.	N/A	N/A
8679174	12/2013	Ottma et al.	N/A	N/A
8685086	12/2013	Navia et al.	N/A	N/A
8690939	12/2013	Miller et al.	N/A	N/A
8696611	12/2013	Nitzan et al.	N/A	N/A
8696742	12/2013	Pintor et al.	N/A	N/A
8715342	12/2013	Zipory et al.	N/A	N/A
8728097	12/2013	Sugimoto et al.	N/A	N/A
8728155	12/2013	Montorfano et al.	N/A	N/A
8734467	12/2013	Miller et al.	N/A	N/A
8734507	12/2013	Keranen	N/A	N/A
8740920	12/2013	Goldfarb et al.	N/A	N/A
8740962	12/2013	Finch et al.	N/A	N/A
8747460	12/2013	Tuval et al.	N/A	N/A
8771345	12/2013	Tuval et al.	N/A	N/A
8778021	12/2013	Cartledge	N/A	N/A
8784472	12/2013	Eidenschink	N/A	N/A
8784479	12/2013	Antonsson et al.	N/A	N/A
8784481	12/2013	Alkhatib et al.	N/A	N/A
8790367	12/2013	Nguyen et al.	N/A	N/A
8790394	12/2013	Miller et al.	N/A	N/A
8795298	12/2013	Hernlund et al.	N/A	N/A
8795355	12/2013	Alkhatib	N/A	N/A
8795356	12/2013	Quadri et al.	N/A	N/A
8795357	12/2013	Yohanan et al.	N/A	N/A
8801776	12/2013	House et al.	N/A	N/A
8808366	12/2013	Braido et al.	N/A	N/A
8808368	12/2013	Maisano et al.	N/A	N/A
8808371	12/2013	Cartledge	N/A	N/A
8834551	12/2013	McGuckin, Jr.	N/A	N/A
8840663	12/2013	Salahieh et al.	N/A	N/A
8840664	12/2013	Karapetian et al.	N/A	N/A
8845717	12/2013	Khairkhahan et al.	N/A	N/A
8845722	12/2013	Gabbay	N/A	N/A
8845723	12/2013	Spence et al.	N/A	N/A

8852261	12/2013	White	N/A	N/A
8852272	12/2013	Gross et al.	N/A	N/A
8858623	12/2013	Miller et al.	N/A	N/A
8864822	12/2013	Spence et al.	N/A	N/A
8870948	12/2013	Erzberger et al.	N/A	N/A
8870949	12/2013	Rowe	N/A	N/A
8870950	12/2013	Hacohen	N/A	N/A
8876800	12/2013	Behan	N/A	N/A
8888843	12/2013	Khairkhahan et al.	N/A	N/A
8894702	12/2013	Quadri et al.	N/A	N/A
8900294	12/2013	Paniagua et al.	N/A	N/A
8900295	12/2013	Migliazza et al.	N/A	N/A
8906083	12/2013	Obermiller et al.	N/A	N/A
8911455	12/2013	Quadri et al.	N/A	N/A
8911461	12/2013	Traynor et al.	N/A	N/A
8911489	12/2013	Ben-Muvhar	N/A	N/A
8911493	12/2013	Rowe et al.	N/A	N/A
8911494	12/2013	Hammer et al.	N/A	N/A
8926695	12/2014	Gross et al.	N/A	N/A
8926696	12/2014	Cabiri et al.	N/A	N/A
8926697	12/2014	Gross et al.	N/A	N/A
8932343	12/2014	Alkhatib et al.	N/A	N/A
8932348	12/2014	Solem et al.	N/A	N/A
8940042	12/2014	Miller et al.	N/A	N/A
8940044	12/2014	Hammer et al.	N/A	N/A
8945177	12/2014	Dell et al.	N/A	N/A
8945211	12/2014	Sugimoto	N/A	N/A
8951285	12/2014	Sugimoto et al.	N/A	N/A
8951286	12/2014	Sugimoto et al.	N/A	N/A
8961595	12/2014	Alkhatib	N/A	N/A
8979922	12/2014	Jayasinghe et al.	N/A	N/A
8986370	12/2014	Annest	N/A	N/A
8986373	12/2014	Chau et al.	N/A	N/A
8986375	12/2014	Garde et al.	N/A	N/A
8992599	12/2014	Thubrikar et al.	N/A	N/A
8992604	12/2014	Gross et al.	N/A	N/A
8992608	12/2014	Haug et al.	N/A	N/A
8998982	12/2014	Richter et al.	N/A	N/A
9005155	12/2014	Sugimoto	N/A	N/A
9005273	12/2014	Salahieh et al.	N/A	N/A
9011468	12/2014	Ketai et al.	N/A	N/A
9011520	12/2014	Miller et al.	N/A	N/A
9011527	12/2014	Li et al.	N/A	N/A
9011530	12/2014	Reich et al.	N/A	N/A
9017399	12/2014	Gross et al.	N/A	N/A
D730520	12/2014	Braido et al.	N/A	N/A
D730521	12/2014	Braido et al.	N/A	N/A
9023100	12/2014	Quadri et al.	N/A	N/A
9034032	12/2014	McLean et al.	N/A	N/A
9034033	12/2014	McLean et al.	N/A	N/A
9039757	12/2014	McLean et al.	N/A	N/A
D732666	12/2014	Nguyen et al.	N/A	N/A
9050188	12/2014	Schweich et al.	N/A	N/A
9060858	12/2014	Thornton et al.	N/A	N/A
9072603	12/2014	Tuval et al.	N/A	N/A

9084676	12/2014	Chau et al.	N/A	N/A
9095434	12/2014	Rowe	N/A	N/A
9119719	12/2014	Zipory et al.	N/A	N/A
9125632	12/2014	Loulmet et al.	N/A	N/A
9125738	12/2014	Figulla et al.	N/A	N/A
9125740	12/2014	Morriss et al.	N/A	N/A
9132006	12/2014	Spenser et al.	N/A	N/A
9132009	12/2014	Hacohen et al.	N/A	N/A
9138312	12/2014	Tuval et al.	N/A	N/A
9155619	12/2014	Liu et al.	N/A	N/A
9173646	12/2014	Fabro	N/A	N/A
9173659	12/2014	Bodewadt et al.	N/A	N/A
9173738	12/2014	Murray et al.	N/A	N/A
9180005	12/2014	Lashinski et al.	N/A	N/A
9180009	12/2014	Majkrzak et al.	N/A	N/A
9192472	12/2014	Gross et al.	N/A	N/A
9216076	12/2014	Mitra et al.	N/A	N/A
9220594	12/2014	Braido et al.	N/A	N/A
9226820	12/2015	Braido et al.	N/A	N/A
9226825	12/2015	Starksen et al.	N/A	N/A
9226839	12/2015	Kariniemi et al.	N/A	N/A
9232995	12/2015	Kovalsky et al.	N/A	N/A
9241790	12/2015	Lane et al.	N/A	N/A
9241791	12/2015	Braido et al.	N/A	N/A
9241792	12/2015	Benichou et al.	N/A	N/A
9241794	12/2015	Braido et al.	N/A	N/A
9248014	12/2015	Lane et al.	N/A	N/A
9265608	12/2015	Miller et al.	N/A	N/A
9277994	12/2015	Miller et al.	N/A	N/A
9277995	12/2015	Celermajer et al.	N/A	N/A
9289290	12/2015	Alkhatib et al.	N/A	N/A
9289291	12/2015	Gorman et al.	N/A	N/A
9295550	12/2015	Nguyen et al.	N/A	N/A
9295551	12/2015	Straubinger et al.	N/A	N/A
9295552	12/2015	McLean et al.	N/A	N/A
9301836	12/2015	Buchbinder et al.	N/A	N/A
9308087	12/2015	Lane et al.	N/A	N/A
9320591	12/2015	Bolduc	N/A	N/A
D755384	12/2015	Pesce et al.	N/A	N/A
9326852	12/2015	Spenser	N/A	N/A
9326876	12/2015	Acosta et al.	N/A	N/A
9345573	12/2015	Nyuli et al.	N/A	N/A
9351830	12/2015	Gross et al.	N/A	N/A
9358107	12/2015	Nguyen et al.	N/A	N/A
9387078	12/2015	Gross et al.	N/A	N/A
9393110	12/2015	Levi et al.	N/A	N/A
9393111	12/2015	Ma et al.	N/A	N/A
9421098	12/2015	Gifford et al.	N/A	N/A
9427303	12/2015	Liddy et al.	N/A	N/A
9427316	12/2015	Schweich, Jr. et al.	N/A	N/A
9439757	12/2015	Wallace et al.	N/A	N/A
9445893	12/2015	Vaturi	N/A	N/A
9463102	12/2015	Kelly	N/A	N/A
9474599	12/2015	Keränen	N/A	N/A
9474638	12/2015	Robinson et al.	N/A	N/A

9480559	12/2015	Vidlund et al.	N/A	N/A
9492273	12/2015	Wallace et al.	N/A	N/A
9498314	12/2015	Behan	N/A	N/A
9498332	12/2015	Hacohen et al.	N/A	N/A
9510947	12/2015	Straubinger et al.	N/A	N/A
9532870	12/2016	Cooper et al.	N/A	N/A
9554897	12/2016	Lane et al.	N/A	N/A
9554899	12/2016	Granada et al.	N/A	N/A
9561103	12/2016	Granada et al.	N/A	N/A
9566152	12/2016	Schweich et al.	N/A	N/A
9572665	12/2016	Lane et al.	N/A	N/A
9597182	12/2016	Straubinger et al.	N/A	N/A
9629715	12/2016	Nitzan et al.	N/A	N/A
9629716	12/2016	Seguin	N/A	N/A
9649480	12/2016	Sugimoto et al.	N/A	N/A
9662203	12/2016	Sheahan et al.	N/A	N/A
9681948	12/2016	Levi et al.	N/A	N/A
9681952	12/2016	Hacohen et al.	N/A	N/A
9707382	12/2016	Nitzan et al.	N/A	N/A
9713696	12/2016	Yacoby et al.	N/A	N/A
9717591	12/2016	Chau et al.	N/A	N/A
9743932	12/2016	Amplatz et al.	N/A	N/A
9763657	12/2016	Hacohen et al.	N/A	N/A
9763817	12/2016	Roeder	N/A	N/A
9770256	12/2016	Cohen et al.	N/A	N/A
D800908	12/2016	Hariton et al.	N/A	N/A
9788941	12/2016	Hacohen	N/A	N/A
9895226	12/2017	Harari et al.	N/A	N/A
9901443	12/2017	Morriss et al.	N/A	N/A
9974651	12/2017	Hariton et al.	N/A	N/A
9987132	12/2017	Hariton et al.	N/A	N/A
9993360	12/2017	Shalev et al.	N/A	N/A
10010414	12/2017	Cooper et al.	N/A	N/A
10016271	12/2017	Morriss et al.	N/A	N/A
10039638	12/2017	Bruchman et al.	N/A	N/A
10045845	12/2017	Hacohen et al.	N/A	N/A
10076415	12/2017	Metchik et al.	N/A	N/A
10098732	12/2017	Hariton et al.	N/A	N/A
10105222	12/2017	Metchik et al.	N/A	N/A
10111751	12/2017	Metchik et al.	N/A	N/A
10117744	12/2017	Ratz et al.	N/A	N/A
10123873	12/2017	Metchik et al.	N/A	N/A
10130475	12/2017	Metchik et al.	N/A	N/A
10136993	12/2017	Metchik et al.	N/A	N/A
10143552	12/2017	Wallace et al.	N/A	N/A
10149761	12/2017	Granada et al.	N/A	N/A
10154903	12/2017	Albitov et al.	N/A	N/A
10154906	12/2017	Granada et al.	N/A	N/A
10159570	12/2017	Metchik et al.	N/A	N/A
10182908	12/2018	Tubishevitz et al.	N/A	N/A
10206668	12/2018	Mcgoldrick et al.	N/A	N/A
10226341	12/2018	Gross et al.	N/A	N/A
10231831	12/2018	Hacohen	N/A	N/A
10231837	12/2018	Metchik et al.	N/A	N/A
10238493	12/2018	Metchik et al.	N/A	N/A

10245143	12/2018	Gross et al.	N/A	N/A
10245144	12/2018	Metchik et al.	N/A	N/A
10258471	12/2018	Lutter et al.	N/A	N/A
10292816	12/2018	Raanani et al.	N/A	N/A
10299927	12/2018	McLean et al.	N/A	N/A
10321995	12/2018	Christianson et al.	N/A	N/A
10322020	12/2018	Lam et al.	N/A	N/A
10327895	12/2018	Lozonschi et al.	N/A	N/A
10335278	12/2018	McLean et al.	N/A	N/A
10350062	12/2018	Peterson et al.	N/A	N/A
10357360	12/2018	Hariton et al.	N/A	N/A
10368988	12/2018	Jones	N/A	N/A
10376361	12/2018	Gross et al.	N/A	N/A
10390952	12/2018	Hariton et al.	N/A	N/A
10426610	12/2018	Hariton et al.	N/A	N/A
10426614	12/2018	Hariton et al.	N/A	N/A
10449047	12/2018	Hariton et al.	N/A	N/A
10456256	12/2018	Braido et al.	N/A	N/A
10463487	12/2018	Hariton et al.	N/A	N/A
10463488	12/2018	Hariton et al.	N/A	N/A
10492907	12/2018	Duffy	N/A	N/A
10492908	12/2018	Hammer et al.	N/A	N/A
10507105	12/2018	Hariton et al.	N/A	N/A
10507108	12/2018	Delgado et al.	N/A	N/A
10507109	12/2018	Metchik et al.	N/A	N/A
10512456	12/2018	Hacohen et al.	N/A	N/A
10517719	12/2018	Miller et al.	N/A	N/A
10524792	12/2019	Hernandez et al.	N/A	N/A
10524903	12/2019	Hariton et al.	N/A	N/A
10524910	12/2019	Hammer et al.	N/A	N/A
10531866	12/2019	Hariton et al.	N/A	N/A
10531872	12/2019	Hacohen et al.	N/A	N/A
10537426	12/2019	Iamberger et al.	N/A	N/A
10548726	12/2019	Hacohen et al.	N/A	N/A
10548731	12/2019	Lashinski et al.	N/A	N/A
10575948	12/2019	Iamberger et al.	N/A	N/A
10595992	12/2019	Chambers	N/A	N/A
10595997	12/2019	Metchik et al.	N/A	N/A
10610358	12/2019	Vidlund et al.	N/A	N/A
10610359	12/2019	Hacohen	N/A	N/A
10631871	12/2019	Goldfarb et al.	N/A	N/A
10631982	12/2019	Hammer et al.	N/A	N/A
10631984	12/2019	Nyuli et al.	N/A	N/A
10646342	12/2019	Marr et al.	N/A	N/A
10660751	12/2019	Hacohen	N/A	N/A
10667908	12/2019	Hariton et al.	N/A	N/A
10667912	12/2019	Dixon et al.	N/A	N/A
10682227	12/2019	Hariton et al.	N/A	N/A
10695173	12/2019	Gross et al.	N/A	N/A
10695177	12/2019	Hariton et al.	N/A	N/A
10702380	12/2019	Morriss et al.	N/A	N/A
10702385	12/2019	Hacohen	N/A	N/A
10722354	12/2019	Cohen-tzemach et al.	N/A	N/A
10722360	12/2019	Hariton et al.	N/A	N/A
10736742	12/2019	Hariton et al.	N/A	N/A

10758342	12/2019	Chau et al.	N/A	N/A
10758344	12/2019	Hariton et al.	N/A	N/A
10779939	12/2019	Hariton et al.	N/A	N/A
10799345	12/2019	Hariton et al.	N/A	N/A
10813760	12/2019	Metchnik et al.	N/A	N/A
10820998	12/2019	Marr et al.	N/A	N/A
10835377	12/2019	Hacohen et al.	N/A	N/A
10842627	12/2019	Delgado et al.	N/A	N/A
10849748	12/2019	Hariton et al.	N/A	N/A
10856972	12/2019	Hariton et al.	N/A	N/A
10856975	12/2019	Hariton et al.	N/A	N/A
10856978	12/2019	Straubinger et al.	N/A	N/A
10864078	12/2019	Hariton et al.	N/A	N/A
10874514	12/2019	Dixon et al.	N/A	N/A
10881511	12/2020	Hariton et al.	N/A	N/A
10888421	12/2020	Hariton et al.	N/A	N/A
10888422	12/2020	Hariton et al.	N/A	N/A
10888425	12/2020	Delgado et al.	N/A	N/A
10888644	12/2020	Ratz et al.	N/A	N/A
10905548	12/2020	Hariton et al.	N/A	N/A
10905549	12/2020	Hariton et al.	N/A	N/A
10905552	12/2020	Dixon et al.	N/A	N/A
10905554	12/2020	Cao	N/A	N/A
10918481	12/2020	Hariton et al.	N/A	N/A
10918483	12/2020	Metchnik et al.	N/A	N/A
10925595	12/2020	Hacohen et al.	N/A	N/A
10925732	12/2020	Delgado et al.	N/A	N/A
10945843	12/2020	Delgado et al.	N/A	N/A
10945844	12/2020	McCann et al.	N/A	N/A
10952850	12/2020	Hariton et al.	N/A	N/A
10959846	12/2020	Marr et al.	N/A	N/A
10973636	12/2020	Hariton et al.	N/A	N/A
10993809	12/2020	McCann et al.	N/A	N/A
11065114	12/2020	Raanani et al.	N/A	N/A
11065117	12/2020	Zeng	N/A	N/A
11083582	12/2020	McCann et al.	N/A	N/A
11135059	12/2020	Hammer et al.	N/A	N/A
11147672	12/2020	McCann et al.	N/A	N/A
11179240	12/2020	Delgado et al.	N/A	N/A
11246704	12/2021	Hariton et al.	N/A	N/A
11291545	12/2021	Hacohen	N/A	N/A
11291546	12/2021	Gross et al.	N/A	N/A
11291547	12/2021	Gross et al.	N/A	N/A
11291844	12/2021	Gross	N/A	N/A
11298117	12/2021	Hariton et al.	N/A	N/A
11304804	12/2021	Hariton et al.	N/A	N/A
11304805	12/2021	Hariton et al.	N/A	N/A
11304806	12/2021	Hariton et al.	N/A	N/A
11318014	12/2021	Hariton et al.	N/A	N/A
11318015	12/2021	Hariton et al.	N/A	N/A
11337802	12/2021	Hariton et al.	N/A	N/A
11337803	12/2021	Hariton et al.	N/A	N/A
11337804	12/2021	Hariton et al.	N/A	N/A
11382746	12/2021	Hariton et al.	N/A	N/A
11389297	12/2021	Franklin et al.	N/A	N/A

11426155	12/2021	Hacohen et al.	N/A	N/A
11517429	12/2021	Gross et al.	N/A	N/A
11517436	12/2021	Hacohen	N/A	N/A
11534298	12/2021	Hariton et al.	N/A	N/A
11571298	12/2022	Hariton et al.	N/A	N/A
11779458	12/2022	Hariton et al.	N/A	N/A
11793633	12/2022	Hariton et al.	N/A	N/A
11793635	12/2022	Hariton et al.	N/A	N/A
11793638	12/2022	Hariton et al.	N/A	N/A
11801135	12/2022	Hariton et al.	N/A	N/A
12090048	12/2023	Hariton et al.	N/A	N/A
2001/0002445	12/2000	Vesely	N/A	N/A
2001/0005787	12/2000	Oz et al.	N/A	N/A
2001/0021872	12/2000	Bailey et al.	N/A	N/A
2001/0021874	12/2000	Carpentier et al.	N/A	N/A
2001/0044656	12/2000	Williamson et al.	N/A	N/A
2001/0056295	12/2000	Solem	N/A	N/A
2002/0013571	12/2001	Goldfarb et al.	N/A	N/A
2002/0022862	12/2001	Grafton et al.	N/A	N/A
2002/0029080	12/2001	Mortier et al.	N/A	N/A
2002/0032481	12/2001	Gabbay	N/A	N/A
2002/0042621	12/2001	Liddicoat et al.	N/A	N/A
2002/0082525	12/2001	Oslund et al.	N/A	N/A
2002/0087048	12/2001	Brock et al.	N/A	N/A
2002/0099436	12/2001	Thornton et al.	N/A	N/A
2002/0103532	12/2001	Langberg et al.	N/A	N/A
2002/0151916	12/2001	Muramatsu et al.	N/A	N/A
2002/0151961	12/2001	Lashinski et al.	N/A	N/A
2002/0151970	12/2001	Garrison et al.	N/A	N/A
2002/0169358	12/2001	Mortier et al.	N/A	N/A
2002/0173841	12/2001	Ortiz et al.	N/A	N/A
2002/0177894	12/2001	Acosta et al.	N/A	N/A
2002/0177904	12/2001	Huxel et al.	N/A	N/A
2002/0198586	12/2001	Inoue	N/A	N/A
2003/0009236	12/2002	Godin	N/A	N/A
2003/0018358	12/2002	Saadat	N/A	N/A
2003/0036791	12/2002	Philipp et al.	N/A	N/A
2003/0050693	12/2002	Quijano et al.	N/A	N/A
2003/0050694	12/2002	Yang et al.	N/A	N/A
2003/0060846	12/2002	Egnelov et al.	N/A	N/A
2003/0060875	12/2002	Wittens	N/A	N/A
2003/0069635	12/2002	Cartledge	N/A	N/A
2003/0074052	12/2002	Besselink	N/A	N/A
2003/0074059	12/2002	Nguyen et al.	N/A	N/A
2003/0078465	12/2002	Pai et al.	N/A	N/A
2003/0078653	12/2002	Vesely et al.	N/A	N/A
2003/0083742	12/2002	Spence et al.	N/A	N/A
2003/0100943	12/2002	Bolduc	N/A	N/A
2003/0105519	12/2002	Fasol et al.	N/A	N/A
2003/0114901	12/2002	Loeb et al.	N/A	N/A
2003/0120340	12/2002	Liska et al.	N/A	N/A
2003/0130731	12/2002	Vidlund et al.	N/A	N/A
2003/0158578	12/2002	Pantages et al.	N/A	N/A
2003/0167062	12/2002	Gambale et al.	N/A	N/A
2003/0171760	12/2002	Gambale	N/A	N/A

2003/0191528	12/2002	Quijano et al.	N/A	N/A
2003/0199974	12/2002	Lee et al.	N/A	N/A
2003/0204195	12/2002	Keane et al.	N/A	N/A
2003/0229350	12/2002	Kay	N/A	N/A
2003/0229395	12/2002	Cox	N/A	N/A
2003/0233142	12/2002	Morales et al.	N/A	N/A
2004/0010272	12/2003	Manetakakis et al.	N/A	N/A
2004/0019377	12/2003	Taylor et al.	N/A	N/A
2004/0024451	12/2003	Johnson et al.	N/A	N/A
2004/0030382	12/2003	St. Goar et al.	N/A	N/A
2004/0039414	12/2003	Carley et al.	N/A	N/A
2004/0039436	12/2003	Spenser et al.	N/A	N/A
2004/0039442	12/2003	St. Goar et al.	N/A	N/A
2004/0049207	12/2003	Goldfarb et al.	N/A	N/A
2004/0059413	12/2003	Argento	N/A	N/A
2004/0092962	12/2003	Thornton et al.	N/A	N/A
2004/0093060	12/2003	Seguin et al.	N/A	N/A
2004/0122448	12/2003	Levine	N/A	N/A
2004/0122503	12/2003	Campbell et al.	N/A	N/A
2004/0122514	12/2003	Fogarty et al.	N/A	N/A
2004/0127982	12/2003	Machold et al.	N/A	N/A
2004/0127983	12/2003	Mortier et al.	N/A	N/A
2004/0133220	12/2003	Lashinski et al.	N/A	N/A
2004/0133267	12/2003	Lane	N/A	N/A
2004/0133274	12/2003	Webler et al.	N/A	N/A
2004/0133374	12/2003	Kattan	N/A	N/A
2004/0138744	12/2003	Lashinski et al.	N/A	N/A
2004/0138745	12/2003	Macoviak et al.	N/A	N/A
2004/0143315	12/2003	Bruun et al.	N/A	N/A
2004/0148019	12/2003	Vidlund et al.	N/A	N/A
2004/0148020	12/2003	Vidlund et al.	N/A	N/A
2004/0148021	12/2003	Cartledge et al.	N/A	N/A
2004/0153146	12/2003	Lashinski et al.	N/A	N/A
2004/0172046	12/2003	Hlavka et al.	N/A	N/A
2004/0176788	12/2003	Opolski	N/A	N/A
2004/0176839	12/2003	Huynh et al.	N/A	N/A
2004/0181287	12/2003	Gellman	N/A	N/A
2004/0186558	12/2003	Pavcnik et al.	N/A	N/A
2004/0186565	12/2003	Schreck	N/A	N/A
2004/0186566	12/2003	Hindrichs et al.	N/A	N/A
2004/0210244	12/2003	Vargas et al.	N/A	N/A
2004/0210304	12/2003	Seguin et al.	N/A	N/A
2004/0220593	12/2003	Greenhalgh	N/A	N/A
2004/0225354	12/2003	Allen et al.	N/A	N/A
2004/0236354	12/2003	Seguin	N/A	N/A
2004/0236419	12/2003	Milo	N/A	N/A
2004/0249433	12/2003	Freitag	N/A	N/A
2004/0249453	12/2003	Cartledge et al.	N/A	N/A
2004/0260317	12/2003	Bloom et al.	N/A	N/A
2004/0260389	12/2003	Case et al.	N/A	N/A
2004/0260393	12/2003	Rahdert et al.	N/A	N/A
2004/0260394	12/2003	Douk et al.	N/A	N/A
2004/0267358	12/2003	Reitan	N/A	N/A
2005/0004668	12/2004	Aklog et al.	N/A	N/A
2005/0010287	12/2004	Macoviak et al.	N/A	N/A

2005/0010787	12/2004	Tarbouriech	N/A	N/A
2005/0016560	12/2004	Voughlohn	N/A	N/A
2005/0021056	12/2004	St. Goar et al.	N/A	N/A
2005/0027305	12/2004	Shiu et al.	N/A	N/A
2005/0027348	12/2004	Case et al.	N/A	N/A
2005/0038494	12/2004	Eidenschink	N/A	N/A
2005/0055038	12/2004	Kelleher et al.	N/A	N/A
2005/0055086	12/2004	Stobie	N/A	N/A
2005/0055087	12/2004	Starksen	N/A	N/A
2005/0060030	12/2004	Lashinski et al.	N/A	N/A
2005/0065601	12/2004	Lee et al.	N/A	N/A
2005/0070999	12/2004	Spence	N/A	N/A
2005/0075726	12/2004	Svanidze et al.	N/A	N/A
2005/0075727	12/2004	Wheatley	N/A	N/A
2005/0075731	12/2004	Artof et al.	N/A	N/A
2005/0080430	12/2004	Wright et al.	N/A	N/A
2005/0080474	12/2004	Andreas et al.	N/A	N/A
2005/0085900	12/2004	Case et al.	N/A	N/A
2005/0085903	12/2004	Lau	N/A	N/A
2005/0090827	12/2004	Gedebou	N/A	N/A
2005/0096740	12/2004	Langberg et al.	N/A	N/A
2005/0107871	12/2004	Realyvasquez et al.	N/A	N/A
2005/0119734	12/2004	Spence et al.	N/A	N/A
2005/0125002	12/2004	Baran et al.	N/A	N/A
2005/0125011	12/2004	Spence et al.	N/A	N/A
2005/0131533	12/2004	Alfieri et al.	N/A	N/A
2005/0137681	12/2004	Shoemaker et al.	N/A	N/A
2005/0137686	12/2004	Salahieh et al.	N/A	N/A
2005/0137688	12/2004	Salahieh et al.	N/A	N/A
2005/0137689	12/2004	Salahieh et al.	N/A	N/A
2005/0137690	12/2004	Salahieh et al.	N/A	N/A
2005/0137691	12/2004	Salahieh et al.	N/A	N/A
2005/0137692	12/2004	Haug et al.	N/A	N/A
2005/0137693	12/2004	Haug et al.	N/A	N/A
2005/0137695	12/2004	Salahieh et al.	N/A	N/A
2005/0137697	12/2004	Salahieh et al.	N/A	N/A
2005/0137699	12/2004	Salahieh et al.	N/A	N/A
2005/0143809	12/2004	Salahich et al.	N/A	N/A
2005/0149160	12/2004	McFerran	N/A	N/A
2005/0154443	12/2004	Linder et al.	N/A	N/A
2005/0159728	12/2004	Armour et al.	N/A	N/A
2005/0171601	12/2004	Cosgrove et al.	N/A	N/A
2005/0177180	12/2004	Kaganov et al.	N/A	N/A
2005/0177228	12/2004	Solem et al.	N/A	N/A
2005/0182483	12/2004	Osborne et al.	N/A	N/A
2005/0182486	12/2004	Gabbay	N/A	N/A
2005/0187613	12/2004	Bolduc et al.	N/A	N/A
2005/0192596	12/2004	Jugenheimer et al.	N/A	N/A
2005/0197695	12/2004	Stacchino et al.	N/A	N/A
2005/0197696	12/2004	Gomez Duran	N/A	N/A
2005/0203549	12/2004	Realyvasquez	N/A	N/A
2005/0203606	12/2004	VanCamp	N/A	N/A
2005/0203618	12/2004	Sharkawy et al.	N/A	N/A
2005/0216039	12/2004	Lederman	N/A	N/A
2005/0216079	12/2004	MaCoviak	N/A	N/A

2005/0222665	12/2004	Aranyi	N/A	N/A
2005/0222678	12/2004	Lashinski et al.	N/A	N/A
2005/0234508	12/2004	Cummins et al.	N/A	N/A
2005/0240200	12/2004	Bergheim	N/A	N/A
2005/0251251	12/2004	Cribier	N/A	N/A
2005/0256532	12/2004	Nayak et al.	N/A	N/A
2005/0256566	12/2004	Gabbay	N/A	N/A
2005/0267478	12/2004	Corradi et al.	N/A	N/A
2005/0267573	12/2004	Macoviak et al.	N/A	N/A
2005/0273138	12/2004	To et al.	N/A	N/A
2005/0288776	12/2004	Shaouliau et al.	N/A	N/A
2005/0288778	12/2004	Shaouliau et al.	N/A	N/A
2005/0288781	12/2004	Moaddeb et al.	N/A	N/A
2006/0004439	12/2005	Spenser et al.	N/A	N/A
2006/0004442	12/2005	Spenser et al.	N/A	N/A
2006/0004443	12/2005	Liddicoat et al.	N/A	N/A
2006/0004469	12/2005	Sokel	N/A	N/A
2006/0015171	12/2005	Armstrong	N/A	N/A
2006/0020275	12/2005	Goldfarb et al.	N/A	N/A
2006/0020326	12/2005	Bolduc et al.	N/A	N/A
2006/0020327	12/2005	Lashinski et al.	N/A	N/A
2006/0020333	12/2005	Lashinski et al.	N/A	N/A
2006/0020336	12/2005	Liddicoat	N/A	N/A
2006/0025787	12/2005	Morales et al.	N/A	N/A
2006/0025855	12/2005	Lashinski et al.	N/A	N/A
2006/0025858	12/2005	Alameddine	N/A	N/A
2006/0030885	12/2005	Hyde	N/A	N/A
2006/0041189	12/2005	Vancaillie	N/A	N/A
2006/0041319	12/2005	Taylor et al.	N/A	N/A
2006/0047297	12/2005	Case	N/A	N/A
2006/0052867	12/2005	Revuelta et al.	N/A	N/A
2006/0052868	12/2005	Mortier	N/A	N/A
2006/0058871	12/2005	Zakay et al.	N/A	N/A
2006/0069429	12/2005	Spence et al.	N/A	N/A
2006/0074486	12/2005	Liddicoat et al.	N/A	N/A
2006/0085012	12/2005	Dolan	N/A	N/A
2006/0089627	12/2005	Burnett et al.	N/A	N/A
2006/0095009	12/2005	Lampropoulos et al.	N/A	N/A
2006/0106423	12/2005	Weisel et al.	N/A	N/A
2006/0111773	12/2005	Rittgers et al.	N/A	N/A
2006/0116750	12/2005	Herbert et al.	N/A	N/A
2006/0116757	12/2005	Lashinski et al.	N/A	N/A
2006/0122692	12/2005	Gilad et al.	N/A	N/A
2006/0129166	12/2005	Lavelle	N/A	N/A
2006/0135964	12/2005	Vesley	N/A	N/A
2006/0149280	12/2005	Harvine et al.	N/A	N/A
2006/0149360	12/2005	Schwammenthal et al.	N/A	N/A
2006/0149368	12/2005	Spence	N/A	N/A
2006/0155357	12/2005	Melsheimer	N/A	N/A
2006/0161250	12/2005	Shaw	N/A	N/A
2006/0161265	12/2005	Levine et al.	N/A	N/A
2006/0178700	12/2005	Quinn	N/A	N/A
2006/0178740	12/2005	Stacchino et al.	N/A	N/A
2006/0184203	12/2005	Martin et al.	N/A	N/A
2006/0184240	12/2005	Jimenez et al.	N/A	N/A

2006/0184242	12/2005	Lichtenstein	N/A	N/A
2006/0190036	12/2005	Wendel et al.	N/A	N/A
2006/0190038	12/2005	Carley et al.	N/A	N/A
2006/0195134	12/2005	Crittenden	N/A	N/A
2006/0195183	12/2005	Navia et al.	N/A	N/A
2006/0195184	12/2005	Lane et al.	N/A	N/A
2006/0201519	12/2005	Frazier et al.	N/A	N/A
2006/0212107	12/2005	Case et al.	N/A	N/A
2006/0212111	12/2005	Case et al.	N/A	N/A
2006/0216404	12/2005	Seyler et al.	N/A	N/A
2006/0229708	12/2005	Powell et al.	N/A	N/A
2006/0241622	12/2005	Zergiebel	N/A	N/A
2006/0241656	12/2005	Starksen et al.	N/A	N/A
2006/0241745	12/2005	Solem	N/A	N/A
2006/0241748	12/2005	Lee et al.	N/A	N/A
2006/0247680	12/2005	Amplatz et al.	N/A	N/A
2006/0247763	12/2005	Slater	N/A	N/A
2006/0253191	12/2005	Salahieh et al.	N/A	N/A
2006/0259135	12/2005	Navia et al.	N/A	N/A
2006/0259136	12/2005	Nguyen et al.	N/A	N/A
2006/0259137	12/2005	Artof et al.	N/A	N/A
2006/0271166	12/2005	Thill et al.	N/A	N/A
2006/0271171	12/2005	McQuinn et al.	N/A	N/A
2006/0271175	12/2005	Woolfson	N/A	N/A
2006/0282150	12/2005	Olson et al.	N/A	N/A
2006/0282161	12/2005	Huyn et al.	N/A	N/A
2006/0287661	12/2005	Bolduc et al.	N/A	N/A
2006/0287716	12/2005	Banbury et al.	N/A	N/A
2006/0287719	12/2005	Rowe et al.	N/A	N/A
2007/0001627	12/2006	Lin et al.	N/A	N/A
2007/0008018	12/2006	Nagashima et al.	N/A	N/A
2007/0016286	12/2006	Herrmann et al.	N/A	N/A
2007/0016287	12/2006	Cartledge et al.	N/A	N/A
2007/0016288	12/2006	Gurskis et al.	N/A	N/A
2007/0021781	12/2006	Jervis et al.	N/A	N/A
2007/0027528	12/2006	Agnew	N/A	N/A
2007/0027533	12/2006	Douk	N/A	N/A
2007/0027536	12/2006	Mihaljevic et al.	N/A	N/A
2007/0027549	12/2006	Godin	N/A	N/A
2007/0038221	12/2006	Fine et al.	N/A	N/A
2007/0038293	12/2006	St. Goar et al.	N/A	N/A
2007/0038295	12/2006	Case et al.	N/A	N/A
2007/0043435	12/2006	Seguin et al.	N/A	N/A
2007/0049942	12/2006	Hindrichs et al.	N/A	N/A
2007/0049970	12/2006	Belef et al.	N/A	N/A
2007/0051377	12/2006	Douk et al.	N/A	N/A
2007/0055206	12/2006	To et al.	N/A	N/A
2007/0055340	12/2006	Pryor	N/A	N/A
2007/0056346	12/2006	Spenser et al.	N/A	N/A
2007/0061010	12/2006	Hauser et al.	N/A	N/A
2007/0066863	12/2006	Rafiee et al.	N/A	N/A
2007/0078297	12/2006	Rafiee et al.	N/A	N/A
2007/0078510	12/2006	Ryan	N/A	N/A
2007/0080188	12/2006	Spence et al.	N/A	N/A
2007/0083168	12/2006	Whiting et al.	N/A	N/A

2007/0106328	12/2006	Wardle et al.	N/A	N/A
2007/0112359	12/2006	Kimura et al.	N/A	N/A
2007/0112422	12/2006	Dehdashtian	N/A	N/A
2007/0112425	12/2006	Schaller et al.	N/A	N/A
2007/0118151	12/2006	Davidson	N/A	N/A
2007/0118154	12/2006	Crabtree	N/A	N/A
2007/0118213	12/2006	Loulmet	N/A	N/A
2007/0118215	12/2006	Moaddeb	N/A	N/A
2007/0142907	12/2006	Moaddeb et al.	N/A	N/A
2007/0162103	12/2006	Case et al.	N/A	N/A
2007/0162107	12/2006	Haug et al.	N/A	N/A
2007/0162111	12/2006	Fukamachi et al.	N/A	N/A
2007/0173932	12/2006	Cali et al.	N/A	N/A
2007/0197858	12/2006	Goldfarb et al.	N/A	N/A
2007/0198077	12/2006	Cully et al.	N/A	N/A
2007/0198082	12/2006	Kapadia et al.	N/A	N/A
2007/0198097	12/2006	Zegdi	N/A	N/A
2007/0208550	12/2006	Cao et al.	N/A	N/A
2007/0213582	12/2006	Zollinger et al.	N/A	N/A
2007/0213810	12/2006	Newhauser et al.	N/A	N/A
2007/0213813	12/2006	Von Segesser et al.	N/A	N/A
2007/0219558	12/2006	Deutsch	N/A	N/A
2007/0219630	12/2006	Chu	N/A	N/A
2007/0225759	12/2006	Thommen et al.	N/A	N/A
2007/0225760	12/2006	Moszner et al.	N/A	N/A
2007/0233186	12/2006	Meng	N/A	N/A
2007/0233237	12/2006	Krivoruchko	N/A	N/A
2007/0233239	12/2006	Navia et al.	N/A	N/A
2007/0239208	12/2006	Crawford	N/A	N/A
2007/0239265	12/2006	Birdsall et al.	N/A	N/A
2007/0239272	12/2006	Navia et al.	N/A	N/A
2007/0239273	12/2006	Allen	N/A	N/A
2007/0244546	12/2006	Francis	N/A	N/A
2007/0244555	12/2006	Rafiee et al.	N/A	N/A
2007/0244556	12/2006	Rafiee et al.	N/A	N/A
2007/0244557	12/2006	Rafiee et al.	N/A	N/A
2007/0250160	12/2006	Rafiee	N/A	N/A
2007/0255397	12/2006	Ryan et al.	N/A	N/A
2007/0255400	12/2006	Parravicini et al.	N/A	N/A
2007/0270755	12/2006	Von Oepen et al.	N/A	N/A
2007/0270943	12/2006	Solem et al.	N/A	N/A
2007/0276437	12/2006	Call et al.	N/A	N/A
2007/0282375	12/2006	Hindrichs et al.	N/A	N/A
2007/0282429	12/2006	Hauser et al.	N/A	N/A
2007/0295172	12/2006	Swartz	N/A	N/A
2007/0299424	12/2006	Cumming et al.	N/A	N/A
2008/0004688	12/2007	Spenser et al.	N/A	N/A
2008/0004697	12/2007	Lichtenstein et al.	N/A	N/A
2008/0027483	12/2007	Cartledge et al.	N/A	N/A
2008/0027555	12/2007	Hawkins	N/A	N/A
2008/0035160	12/2007	Woodson et al.	N/A	N/A
2008/0039935	12/2007	Buch	N/A	N/A
2008/0051703	12/2007	Thornton et al.	N/A	N/A
2008/0058595	12/2007	Snoke et al.	N/A	N/A
2008/0065011	12/2007	Marchand et al.	N/A	N/A

2008/0065204	12/2007	Mackoviak et al.	N/A	N/A
2008/0071361	12/2007	Tuval et al.	N/A	N/A
2008/0071363	12/2007	Tuval et al.	N/A	N/A
2008/0071366	12/2007	Tuval et al.	N/A	N/A
2008/0071369	12/2007	Tuval et al.	N/A	N/A
2008/0077235	12/2007	Kirson	N/A	N/A
2008/0082083	12/2007	Forde et al.	N/A	N/A
2008/0082159	12/2007	Tseng et al.	N/A	N/A
2008/0082166	12/2007	Styrc et al.	N/A	N/A
2008/0086138	12/2007	Stone et al.	N/A	N/A
2008/0086164	12/2007	Rowe et al.	N/A	N/A
2008/0086203	12/2007	Roberts	N/A	N/A
2008/0086204	12/2007	Rankin	N/A	N/A
2008/0091257	12/2007	Andreas et al.	N/A	N/A
2008/0091261	12/2007	Long et al.	N/A	N/A
2008/0097523	12/2007	Bolduc et al.	N/A	N/A
2008/0097595	12/2007	Gabbay	N/A	N/A
2008/0103581	12/2007	Goto	N/A	N/A
2008/0132989	12/2007	Snow et al.	N/A	N/A
2008/0140003	12/2007	Bei et al.	N/A	N/A
2008/0140116	12/2007	Bonutti	N/A	N/A
2008/0140189	12/2007	Nguyen et al.	N/A	N/A
2008/0147182	12/2007	Righini et al.	N/A	N/A
2008/0161910	12/2007	Revuelta et al.	N/A	N/A
2008/0167705	12/2007	Agnew	N/A	N/A
2008/0167714	12/2007	St. Goar et al.	N/A	N/A
2008/0188929	12/2007	Schreck	N/A	N/A
2008/0195126	12/2007	Solem	N/A	N/A
2008/0195200	12/2007	Vidlund et al.	N/A	N/A
2008/0200980	12/2007	Robin et al.	N/A	N/A
2008/0208265	12/2007	Frazier et al.	N/A	N/A
2008/0208328	12/2007	Antocci et al.	N/A	N/A
2008/0208330	12/2007	Keranen	N/A	N/A
2008/0208332	12/2007	Lamphere et al.	N/A	N/A
2008/0221672	12/2007	Lamphere et al.	N/A	N/A
2008/0234813	12/2007	Heuser	N/A	N/A
2008/0234814	12/2007	Salahieh et al.	N/A	N/A
2008/0243245	12/2007	Thambar et al.	N/A	N/A
2008/0255580	12/2007	Hoffman et al.	N/A	N/A
2008/0262609	12/2007	Gross et al.	N/A	N/A
2008/0269879	12/2007	Sathe et al.	N/A	N/A
2008/0275300	12/2007	Rothe et al.	N/A	N/A
2008/0275469	12/2007	Fanton et al.	N/A	N/A
2008/0275551	12/2007	Alfieri	N/A	N/A
2008/0281411	12/2007	Berrekouw	N/A	N/A
2008/0288044	12/2007	Osborne	N/A	N/A
2008/0288062	12/2007	Andrieu et al.	N/A	N/A
2008/0294234	12/2007	Hartley et al.	N/A	N/A
2008/0294248	12/2007	Yang et al.	N/A	N/A
2008/0300629	12/2007	Surti	N/A	N/A
2009/0005863	12/2008	Goetz et al.	N/A	N/A
2009/0036966	12/2008	O'Connor et al.	N/A	N/A
2009/0043153	12/2008	Zollinger et al.	N/A	N/A
2009/0043381	12/2008	Macoviak et al.	N/A	N/A
2009/0054969	12/2008	Salahieh et al.	N/A	N/A

2009/0062866	12/2008	Jackson	N/A	N/A
2009/0076586	12/2008	Hauser et al.	N/A	N/A
2009/0076600	12/2008	Quinn	N/A	N/A
2009/0082844	12/2008	Zacharias et al.	N/A	N/A
2009/0088836	12/2008	Bishop et al.	N/A	N/A
2009/0088837	12/2008	Gillinov et al.	N/A	N/A
2009/0099554	12/2008	Forster et al.	N/A	N/A
2009/0099650	12/2008	Bolduc et al.	N/A	N/A
2009/0099653	12/2008	Suri et al.	N/A	N/A
2009/0105794	12/2008	Ziarno et al.	N/A	N/A
2009/0105816	12/2008	Olsen et al.	N/A	N/A
2009/0112159	12/2008	Slattery et al.	N/A	N/A
2009/0125098	12/2008	Chuter	N/A	N/A
2009/0125102	12/2008	Cartledge	N/A	N/A
2009/0149872	12/2008	Gross et al.	N/A	N/A
2009/0157175	12/2008	Benichou	N/A	N/A
2009/0163934	12/2008	Raschdorf, Jr. et al.	N/A	N/A
2009/0171363	12/2008	Chocron	N/A	N/A
2009/0171439	12/2008	Nissl	N/A	N/A
2009/0177266	12/2008	Powell et al.	N/A	N/A
2009/0177274	12/2008	Scorsin et al.	N/A	N/A
2009/0177277	12/2008	Milo	N/A	N/A
2009/0177278	12/2008	Spence	N/A	N/A
2009/0192601	12/2008	Rafiee et al.	N/A	N/A
2009/0210052	12/2008	Forster et al.	N/A	N/A
2009/0222081	12/2008	Linder et al.	N/A	N/A
2009/0240320	12/2008	Tuval et al.	N/A	N/A
2009/0241656	12/2008	Jacquemin	N/A	N/A
2009/0248143	12/2008	Laham	N/A	N/A
2009/0248148	12/2008	Shaolian et al.	N/A	N/A
2009/0254103	12/2008	Deustch	N/A	N/A
2009/0259306	12/2008	Rowe	N/A	N/A
2009/0259307	12/2008	Gross et al.	N/A	N/A
2009/0264859	12/2008	Mas	N/A	N/A
2009/0264994	12/2008	Saadat	N/A	N/A
2009/0264995	12/2008	Subramanian	N/A	N/A
2009/0276040	12/2008	Rowe et al.	N/A	N/A
2009/0281619	12/2008	Le et al.	N/A	N/A
2009/0287299	12/2008	Tabor et al.	N/A	N/A
2009/0287304	12/2008	Dahlgren et al.	N/A	N/A
2009/0299409	12/2008	Coe et al.	N/A	N/A
2009/0299449	12/2008	Styrc	N/A	N/A
2009/0306768	12/2008	Quardi	N/A	N/A
2009/0319037	12/2008	Rowe et al.	N/A	N/A
2009/0326648	12/2008	Machold et al.	N/A	N/A
2010/0001038	12/2009	Levin et al.	N/A	N/A
2010/0010538	12/2009	Juravic et al.	N/A	N/A
2010/0022823	12/2009	Goldfarb et al.	N/A	N/A
2010/0023117	12/2009	Yoganathan et al.	N/A	N/A
2010/0023118	12/2009	Medlock et al.	N/A	N/A
2010/0023120	12/2009	Holecek et al.	N/A	N/A
2010/0030014	12/2009	Ferrazzi	N/A	N/A
2010/0036479	12/2009	Hill et al.	N/A	N/A
2010/0036484	12/2009	Hariton et al.	N/A	N/A
2010/0042147	12/2009	Janovsky et al.	N/A	N/A

2010/0049306	12/2009	House et al.	N/A	N/A
2010/0049313	12/2009	Alon et al.	N/A	N/A
2010/0063542	12/2009	Van der Burg et al.	N/A	N/A
2010/0063550	12/2009	Felix et al.	N/A	N/A
2010/0063586	12/2009	Hasenkam et al.	N/A	N/A
2010/0069852	12/2009	Kelley	N/A	N/A
2010/0076499	12/2009	McNamara et al.	N/A	N/A
2010/0076548	12/2009	Konno	N/A	N/A
2010/0082094	12/2009	Quadri et al.	N/A	N/A
2010/0094248	12/2009	Nguyen et al.	N/A	N/A
2010/0100167	12/2009	Bortlein et al.	N/A	N/A
2010/0114180	12/2009	Rock	N/A	N/A
2010/0114299	12/2009	Ben-Muvhar et al.	N/A	N/A
2010/0121349	12/2009	Meier	N/A	N/A
2010/0130992	12/2009	Machold et al.	N/A	N/A
2010/0131054	12/2009	Tuval et al.	N/A	N/A
2010/0137979	12/2009	Tuval et al.	N/A	N/A
2010/0152845	12/2009	Bloom et al.	N/A	N/A
2010/0160958	12/2009	Clark	N/A	N/A
2010/0161036	12/2009	Pintor et al.	N/A	N/A
2010/0161041	12/2009	Maisano et al.	N/A	N/A
2010/0161042	12/2009	Maisano et al.	N/A	N/A
2010/0161043	12/2009	Maisano et al.	N/A	N/A
2010/0161047	12/2009	Cabiri	N/A	N/A
2010/0168845	12/2009	Wright	N/A	N/A
2010/0174358	12/2009	Rabkin et al.	N/A	N/A
2010/0174363	12/2009	Castro	N/A	N/A
2010/0179574	12/2009	Longoria et al.	N/A	N/A
2010/0179643	12/2009	Shalev	N/A	N/A
2010/0179648	12/2009	Richter et al.	N/A	N/A
2010/0179649	12/2009	Richter et al.	N/A	N/A
2010/0185277	12/2009	Braido et al.	N/A	N/A
2010/0198347	12/2009	Zakay et al.	N/A	N/A
2010/0217382	12/2009	Chau et al.	N/A	N/A
2010/0222810	12/2009	DeBeer et al.	N/A	N/A
2010/0228285	12/2009	Miles et al.	N/A	N/A
2010/0234935	12/2009	Bashiri et al.	N/A	N/A
2010/0234940	12/2009	Dolan	N/A	N/A
2010/0249908	12/2009	Chau et al.	N/A	N/A
2010/0249915	12/2009	Zhang	N/A	N/A
2010/0249917	12/2009	Zhang	N/A	N/A
2010/0249920	12/2009	Bolling et al.	N/A	N/A
2010/0256737	12/2009	Pollock et al.	N/A	N/A
2010/0262232	12/2009	Annest	N/A	N/A
2010/0262233	12/2009	He	N/A	N/A
2010/0280603	12/2009	Maisano et al.	N/A	N/A
2010/0280604	12/2009	Zipory et al.	N/A	N/A
2010/0280605	12/2009	Hammer et al.	N/A	N/A
2010/0280606	12/2009	Naor	N/A	N/A
2010/0286628	12/2009	Gross	N/A	N/A
2010/0286767	12/2009	Zipory et al.	N/A	N/A
2010/0305475	12/2009	Hinchliffe et al.	N/A	N/A
2010/0312333	12/2009	Navia et al.	N/A	N/A
2010/0324595	12/2009	Linder et al.	N/A	N/A
2010/0331971	12/2009	Keränen et al.	N/A	N/A

2011/0004210	12/2010	Johnson et al.	N/A	N/A
2011/0004227	12/2010	Goldfarb et al.	N/A	N/A
2011/0004296	12/2010	Lutter et al.	N/A	N/A
2011/0004298	12/2010	Lee et al.	N/A	N/A
2011/0004299	12/2010	Navia et al.	N/A	N/A
2011/0011917	12/2010	Loulmet	N/A	N/A
2011/0015729	12/2010	Jimenez et al.	N/A	N/A
2011/0015731	12/2010	Carpentier et al.	N/A	N/A
2011/0015739	12/2010	Cheung et al.	N/A	N/A
2011/0021985	12/2010	Spargias	N/A	N/A
2011/0022165	12/2010	Oba et al.	N/A	N/A
2011/0026208	12/2010	Otsuro et al.	N/A	N/A
2011/0029066	12/2010	Gilad et al.	N/A	N/A
2011/0029067	12/2010	Mcguckin, Jr. et al.	N/A	N/A
2011/0029072	12/2010	Gabbay	N/A	N/A
2011/0035000	12/2010	Nieminen et al.	N/A	N/A
2011/0040374	12/2010	Goetz et al.	N/A	N/A
2011/0040375	12/2010	Letac et al.	N/A	N/A
2011/0046662	12/2010	Moszner et al.	N/A	N/A
2011/0054466	12/2010	Rothstein et al.	N/A	N/A
2011/0054596	12/2010	Taylor	N/A	N/A
2011/0054598	12/2010	Johnson	N/A	N/A
2011/0066231	12/2010	Cartledge et al.	N/A	N/A
2011/0066233	12/2010	Thornton et al.	N/A	N/A
2011/0067770	12/2010	Pederson et al.	N/A	N/A
2011/0071626	12/2010	Wright et al.	N/A	N/A
2011/0077730	12/2010	Fenster	N/A	N/A
2011/0082538	12/2010	Dahlgren et al.	N/A	N/A
2011/0087146	12/2010	Ryan et al.	N/A	N/A
2011/0087322	12/2010	Letac et al.	N/A	N/A
2011/0093002	12/2010	Rucker et al.	N/A	N/A
2011/0093063	12/2010	Schreck	N/A	N/A
2011/0098525	12/2010	Kermode et al.	N/A	N/A
2011/0098802	12/2010	Braido et al.	N/A	N/A
2011/0098805	12/2010	Dwork et al.	N/A	N/A
2011/0106245	12/2010	Miller et al.	N/A	N/A
2011/0106247	12/2010	Miller et al.	N/A	N/A
2011/0112625	12/2010	Ben-Muvhar et al.	N/A	N/A
2011/0112632	12/2010	Chau et al.	N/A	N/A
2011/0113768	12/2010	Bauer et al.	N/A	N/A
2011/0118830	12/2010	Liddicoat et al.	N/A	N/A
2011/0118832	12/2010	Punjabi	N/A	N/A
2011/0125257	12/2010	Seguin et al.	N/A	N/A
2011/0125258	12/2010	Centola	N/A	N/A
2011/0137326	12/2010	Bachman	N/A	N/A
2011/0137397	12/2010	Chau et al.	N/A	N/A
2011/0137409	12/2010	Yang et al.	N/A	N/A
2011/0137410	12/2010	Hacohen	N/A	N/A
2011/0144703	12/2010	Krause et al.	N/A	N/A
2011/0144742	12/2010	Madrid et al.	N/A	N/A
2011/0166636	12/2010	Rowe	N/A	N/A
2011/0166649	12/2010	Gross et al.	N/A	N/A
2011/0172784	12/2010	Richter	N/A	N/A
2011/0178597	12/2010	Navia et al.	N/A	N/A
2011/0184510	12/2010	Maisano et al.	N/A	N/A

2011/0190877	12/2010	Lane et al.	N/A	N/A
2011/0190879	12/2010	Bobo et al.	N/A	N/A
2011/0202076	12/2010	Richter	N/A	N/A
2011/0202130	12/2010	Cartledge et al.	N/A	N/A
2011/0208283	12/2010	Rust	N/A	N/A
2011/0208293	12/2010	Tabor	N/A	N/A
2011/0208298	12/2010	Tuval et al.	N/A	N/A
2011/0213459	12/2010	Garrison et al.	N/A	N/A
2011/0213461	12/2010	Seguin et al.	N/A	N/A
2011/0218619	12/2010	Benichou et al.	N/A	N/A
2011/0218620	12/2010	Meiri et al.	N/A	N/A
2011/0224785	12/2010	Hacohen	N/A	N/A
2011/0230941	12/2010	Markus	N/A	N/A
2011/0230961	12/2010	Langer et al.	N/A	N/A
2011/0238088	12/2010	Bodluc et al.	N/A	N/A
2011/0238094	12/2010	Thomas et al.	N/A	N/A
2011/0238159	12/2010	Guyenot et al.	N/A	N/A
2011/0245911	12/2010	Quill et al.	N/A	N/A
2011/0245917	12/2010	Savage et al.	N/A	N/A
2011/0251675	12/2010	Dwork	N/A	N/A
2011/0251676	12/2010	Sweeney et al.	N/A	N/A
2011/0251678	12/2010	Eidenschink et al.	N/A	N/A
2011/0251679	12/2010	Weimeyer et al.	N/A	N/A
2011/0251680	12/2010	Tran et al.	N/A	N/A
2011/0251682	12/2010	Murray, III et al.	N/A	N/A
2011/0251683	12/2010	Tabor	N/A	N/A
2011/0257433	12/2010	Walker	N/A	N/A
2011/0257633	12/2010	Cartledge et al.	N/A	N/A
2011/0257721	12/2010	Tabor	N/A	N/A
2011/0257728	12/2010	Kuehn	N/A	N/A
2011/0257729	12/2010	Spenser et al.	N/A	N/A
2011/0257736	12/2010	Marquez et al.	N/A	N/A
2011/0257737	12/2010	Fogarty et al.	N/A	N/A
2011/0264191	12/2010	Rothstein	N/A	N/A
2011/0264196	12/2010	Savage et al.	N/A	N/A
2011/0264198	12/2010	Murray, III et al.	N/A	N/A
2011/0264199	12/2010	Tran et al.	N/A	N/A
2011/0264200	12/2010	Tran et al.	N/A	N/A
2011/0264201	12/2010	Yeung	N/A	N/A
2011/0264202	12/2010	Murray, III et al.	N/A	N/A
2011/0264203	12/2010	Dwork et al.	N/A	N/A
2011/0264206	12/2010	Tabor	N/A	N/A
2011/0264208	12/2010	Duffy	N/A	N/A
2011/0270276	12/2010	Rothstein et al.	N/A	N/A
2011/0271967	12/2010	Mortier et al.	N/A	N/A
2011/0276062	12/2010	Bolduc	N/A	N/A
2011/0276128	12/2010	Cao et al.	N/A	N/A
2011/0282361	12/2010	Miller et al.	N/A	N/A
2011/0282438	12/2010	Drews et al.	N/A	N/A
2011/0282439	12/2010	Thill et al.	N/A	N/A
2011/0282440	12/2010	Cao	N/A	N/A
2011/0283514	12/2010	Fogarty et al.	N/A	N/A
2011/0288435	12/2010	Christy et al.	N/A	N/A
2011/0288632	12/2010	White	N/A	N/A
2011/0288634	12/2010	Tuval et al.	N/A	N/A

2011/0288635	12/2010	Miller et al.	N/A	N/A
2011/0295354	12/2010	Bueche et al.	N/A	N/A
2011/0295363	12/2010	Girard et al.	N/A	N/A
2011/0301498	12/2010	Maenhout et al.	N/A	N/A
2011/0301688	12/2010	Dolan	N/A	N/A
2011/0301698	12/2010	Miller et al.	N/A	N/A
2011/0301701	12/2010	Padala et al.	N/A	N/A
2011/0301702	12/2010	Rust et al.	N/A	N/A
2011/0306916	12/2010	Nitzan et al.	N/A	N/A
2011/0307049	12/2010	Kao	N/A	N/A
2011/0313452	12/2010	Carley et al.	N/A	N/A
2011/0313515	12/2010	Quadri et al.	N/A	N/A
2011/0319988	12/2010	Schankereli et al.	N/A	N/A
2011/0319989	12/2010	Lane et al.	N/A	N/A
2011/0319991	12/2010	Hariton et al.	N/A	N/A
2012/0010694	12/2011	Lutter et al.	N/A	N/A
2012/0016468	12/2011	Robin et al.	N/A	N/A
2012/0022557	12/2011	Cabiri et al.	N/A	N/A
2012/0022629	12/2011	Perera et al.	N/A	N/A
2012/0022633	12/2011	Olson et al.	N/A	N/A
2012/0022637	12/2011	Ben-Movhar et al.	N/A	N/A
2012/0022639	12/2011	Hacohen et al.	N/A	N/A
2012/0022640	12/2011	Gross et al.	N/A	N/A
2012/0022644	12/2011	Reich et al.	N/A	N/A
2012/0035703	12/2011	Lutter et al.	N/A	N/A
2012/0035712	12/2011	Maisano et al.	N/A	N/A
2012/0035713	12/2011	Lutter et al.	N/A	N/A
2012/0035722	12/2011	Tuval et al.	N/A	N/A
2012/0041547	12/2011	Duffy et al.	N/A	N/A
2012/0041551	12/2011	Spenser et al.	N/A	N/A
2012/0046738	12/2011	Lau et al.	N/A	N/A
2012/0046742	12/2011	Tuval et al.	N/A	N/A
2012/0053676	12/2011	Ku et al.	N/A	N/A
2012/0053680	12/2011	Bolling et al.	N/A	N/A
2012/0053682	12/2011	Kovalsky et al.	N/A	N/A
2012/0053688	12/2011	Fogarty et al.	N/A	N/A
2012/0059337	12/2011	Eilat	N/A	N/A
2012/0059454	12/2011	Millwee et al.	N/A	N/A
2012/0059458	12/2011	Buchbinder et al.	N/A	N/A
2012/0065464	12/2011	Ellis et al.	N/A	N/A
2012/0078237	12/2011	Wang et al.	N/A	N/A
2012/0078353	12/2011	Quadri et al.	N/A	N/A
2012/0078355	12/2011	Zipory et al.	N/A	N/A
2012/0078357	12/2011	Conklin	N/A	N/A
2012/0078359	12/2011	Li et al.	N/A	N/A
2012/0083832	12/2011	Delaloye et al.	N/A	N/A
2012/0083839	12/2011	Letac et al.	N/A	N/A
2012/0083874	12/2011	Dale et al.	N/A	N/A
2012/0083879	12/2011	Eberhardt et al.	N/A	N/A
2012/0089022	12/2011	House et al.	N/A	N/A
2012/0089223	12/2011	Nguyen et al.	N/A	N/A
2012/0095552	12/2011	Spence et al.	N/A	N/A
2012/0101570	12/2011	Tuval et al.	N/A	N/A
2012/0101571	12/2011	Thambar et al.	N/A	N/A
2012/0101572	12/2011	Kovalsky et al.	N/A	N/A

2012/0109155	12/2011	Robinson et al.	N/A	N/A
2012/0123511	12/2011	Brown	N/A	N/A
2012/0123529	12/2011	Levi et al.	N/A	N/A
2012/0123530	12/2011	Carpentier et al.	N/A	N/A
2012/0123531	12/2011	Tsukashima et al.	N/A	N/A
2012/0130301	12/2011	McNamara et al.	N/A	N/A
2012/0130473	12/2011	Norris et al.	N/A	N/A
2012/0130474	12/2011	Buckley	N/A	N/A
2012/0130475	12/2011	Shaw	N/A	N/A
2012/0136434	12/2011	Carpentier et al.	N/A	N/A
2012/0136436	12/2011	Cabiri et al.	N/A	N/A
2012/0143323	12/2011	Hasenkam et al.	N/A	N/A
2012/0150218	12/2011	Sandgren et al.	N/A	N/A
2012/0150290	12/2011	Gabbay	N/A	N/A
2012/0158021	12/2011	Morrill	N/A	N/A
2012/0165915	12/2011	Melsheimer et al.	N/A	N/A
2012/0165930	12/2011	Gifford, III et al.	N/A	N/A
2012/0179086	12/2011	Shank et al.	N/A	N/A
2012/0179244	12/2011	Schankereli et al.	N/A	N/A
2012/0191182	12/2011	Hauser et al.	N/A	N/A
2012/0197292	12/2011	Chin-Chen et al.	N/A	N/A
2012/0197388	12/2011	Khairkhahan et al.	N/A	N/A
2012/0215303	12/2011	Quadri et al.	N/A	N/A
2012/0239142	12/2011	Liu et al.	N/A	N/A
2012/0245604	12/2011	Tegzes	N/A	N/A
2012/0265296	12/2011	McNamara et al.	N/A	N/A
2012/0271198	12/2011	Whittaker et al.	N/A	N/A
2012/0277845	12/2011	Bowe	N/A	N/A
2012/0283757	12/2011	Miller et al.	N/A	N/A
2012/0283824	12/2011	Lutter et al.	N/A	N/A
2012/0290062	12/2011	McNamara et al.	N/A	N/A
2012/0296349	12/2011	Smith et al.	N/A	N/A
2012/0296360	12/2011	Norris et al.	N/A	N/A
2012/0296417	12/2011	Hill et al.	N/A	N/A
2012/0296418	12/2011	Bonyuet et al.	N/A	N/A
2012/0296419	12/2011	Richardson	N/A	N/A
2012/0300063	12/2011	Majkrzak et al.	N/A	N/A
2012/0303116	12/2011	Gorman, III et al.	N/A	N/A
2012/0310328	12/2011	Olson et al.	N/A	N/A
2012/0310330	12/2011	Buchbinder et al.	N/A	N/A
2012/0323313	12/2011	Seguin	N/A	N/A
2012/0323316	12/2011	Chau et al.	N/A	N/A
2012/0323317	12/2011	Karapetian et al.	N/A	N/A
2012/0330408	12/2011	Hillukka et al.	N/A	N/A
2012/0330410	12/2011	Hammer et al.	N/A	N/A
2012/0330411	12/2011	Gross et al.	N/A	N/A
2013/0006347	12/2012	McHugo	N/A	N/A
2013/0018450	12/2012	Hunt	N/A	N/A
2013/0018458	12/2012	Yohanan et al.	N/A	N/A
2013/0023758	12/2012	Fabro	N/A	N/A
2013/0030519	12/2012	Tran et al.	N/A	N/A
2013/0030522	12/2012	Rowe et al.	N/A	N/A
2013/0035759	12/2012	Gross et al.	N/A	N/A
2013/0041204	12/2012	Heilman et al.	N/A	N/A
2013/0041451	12/2012	Patterson et al.	N/A	N/A

2013/0046373	12/2012	Cartledge et al.	N/A	N/A
2013/0066341	12/2012	Ketai et al.	N/A	N/A
2013/0066342	12/2012	Dell et al.	N/A	N/A
2013/0079872	12/2012	Gallagher	N/A	N/A
2013/0079873	12/2012	Migliazza et al.	N/A	N/A
2013/0085529	12/2012	Housman	N/A	N/A
2013/0090724	12/2012	Subramanian et al.	N/A	N/A
2013/0096673	12/2012	Hill et al.	N/A	N/A
2013/0116776	12/2012	Gross et al.	N/A	N/A
2013/0116779	12/2012	Weber	N/A	N/A
2013/0116780	12/2012	Miller et al.	N/A	N/A
2013/0123896	12/2012	Bloss et al.	N/A	N/A
2013/0123900	12/2012	Eblacas et al.	N/A	N/A
2013/0123910	12/2012	Cartledge et al.	N/A	N/A
2013/0131791	12/2012	Hlavka et al.	N/A	N/A
2013/0131792	12/2012	Miller et al.	N/A	N/A
2013/0138068	12/2012	Hu et al.	N/A	N/A
2013/0144381	12/2012	Quadri et al.	N/A	N/A
2013/0150945	12/2012	Crawford et al.	N/A	N/A
2013/0150956	12/2012	Yohanan et al.	N/A	N/A
2013/0158647	12/2012	Norris et al.	N/A	N/A
2013/0166017	12/2012	Cartledge et al.	N/A	N/A
2013/0166022	12/2012	Conklin	N/A	N/A
2013/0172978	12/2012	Vidlund et al.	N/A	N/A
2013/0172992	12/2012	Gross et al.	N/A	N/A
2013/0178930	12/2012	Straubinger et al.	N/A	N/A
2013/0190857	12/2012	Mitra et al.	N/A	N/A
2013/0190861	12/2012	Chau et al.	N/A	N/A
2013/0190863	12/2012	Call et al.	N/A	N/A
2013/0190866	12/2012	Zipory et al.	N/A	N/A
2013/0197622	12/2012	Mitra et al.	N/A	N/A
2013/0197632	12/2012	Kovach et al.	N/A	N/A
2013/0204361	12/2012	Adams et al.	N/A	N/A
2013/0211501	12/2012	Buckley et al.	N/A	N/A
2013/0211508	12/2012	Lane et al.	N/A	N/A
2013/0226289	12/2012	Shaolian et al.	N/A	N/A
2013/0226290	12/2012	Yellin et al.	N/A	N/A
2013/0231735	12/2012	Deem et al.	N/A	N/A
2013/0245742	12/2012	Norris	N/A	N/A
2013/0253634	12/2012	Wilson et al.	N/A	N/A
2013/0253643	12/2012	Rolando et al.	N/A	N/A
2013/0261737	12/2012	Costello	N/A	N/A
2013/0261738	12/2012	Clague et al.	N/A	N/A
2013/0268069	12/2012	Zakai et al.	N/A	N/A
2013/0274870	12/2012	Lombardi et al.	N/A	N/A
2013/0274873	12/2012	Delaloye et al.	N/A	N/A
2013/0281988	12/2012	Magnin et al.	N/A	N/A
2013/0282059	12/2012	Ketai et al.	N/A	N/A
2013/0289711	12/2012	Liddy et al.	N/A	N/A
2013/0289718	12/2012	Tsukashima et al.	N/A	N/A
2013/0289740	12/2012	Liddy et al.	N/A	N/A
2013/0297013	12/2012	Klima et al.	N/A	N/A
2013/0304093	12/2012	Serina et al.	N/A	N/A
2013/0304197	12/2012	Buchbinder et al.	N/A	N/A
2013/0304200	12/2012	McLean et al.	N/A	N/A

2013/0310928	12/2012	Morriss et al.	N/A	N/A
2013/0325114	12/2012	McLean et al.	N/A	N/A
2013/0325118	12/2012	Cartledge	N/A	N/A
2013/0331929	12/2012	Mitra et al.	N/A	N/A
2014/0000112	12/2013	Braido et al.	N/A	N/A
2014/0005767	12/2013	Glazier et al.	N/A	N/A
2014/0005778	12/2013	Buchbinder et al.	N/A	N/A
2014/0018911	12/2013	Zhou et al.	N/A	N/A
2014/0018914	12/2013	Zipory et al.	N/A	N/A
2014/0018915	12/2013	Biadillah et al.	N/A	N/A
2014/0031928	12/2013	Murphy et al.	N/A	N/A
2014/0046430	12/2013	Shaw	N/A	N/A
2014/0052237	12/2013	Lane et al.	N/A	N/A
2014/0067050	12/2013	Costello et al.	N/A	N/A
2014/0067054	12/2013	Chau et al.	N/A	N/A
2014/0081376	12/2013	Burkart et al.	N/A	N/A
2014/0088368	12/2013	Park	N/A	N/A
2014/0094826	12/2013	Sutherland et al.	N/A	N/A
2014/0094903	12/2013	Miller et al.	N/A	N/A
2014/0094906	12/2013	Spence et al.	N/A	N/A
2014/0099726	12/2013	Heller	N/A	N/A
2014/0100653	12/2013	Savage et al.	N/A	N/A
2014/0106951	12/2013	Brandon	N/A	N/A
2014/0120287	12/2013	Jacoby et al.	N/A	N/A
2014/0121749	12/2013	Roeder	N/A	N/A
2014/0121763	12/2013	Duffy et al.	N/A	N/A
2014/0135799	12/2013	Henderson	N/A	N/A
2014/0135894	12/2013	Norris et al.	N/A	N/A
2014/0135895	12/2013	Andress et al.	N/A	N/A
2014/0142619	12/2013	Serina et al.	N/A	N/A
2014/0142681	12/2013	Norris	N/A	N/A
2014/0142688	12/2013	Duffy et al.	N/A	N/A
2014/0142695	12/2013	Gross et al.	N/A	N/A
2014/0148849	12/2013	Serina et al.	N/A	N/A
2014/0148891	12/2013	Johnson	N/A	N/A
2014/0148898	12/2013	Gross et al.	N/A	N/A
2014/0155783	12/2013	Starksen et al.	N/A	N/A
2014/0163668	12/2013	Rafiee	N/A	N/A
2014/0163670	12/2013	Alon et al.	N/A	N/A
2014/0163690	12/2013	White	N/A	N/A
2014/0172069	12/2013	Roeder et al.	N/A	N/A
2014/0172077	12/2013	Bruchman et al.	N/A	N/A
2014/0172082	12/2013	Bruchman	156/185	A61F 2/2418
2014/0188108	12/2013	Goodine et al.	N/A	N/A
2014/0188140	12/2013	Meier et al.	N/A	N/A
2014/0188210	12/2013	Beard et al.	N/A	N/A
2014/0188215	12/2013	Hlavka et al.	N/A	N/A
2014/0188221	12/2013	Chung et al.	N/A	N/A
2014/0194970	12/2013	Chobotov	N/A	N/A
2014/0194971	12/2013	McNamara	N/A	N/A
2014/0194976	12/2013	Starksen et al.	N/A	N/A
2014/0194981	12/2013	Menk et al.	N/A	N/A
2014/0194983	12/2013	Kovalsky et al.	N/A	N/A
2014/0200649	12/2013	Essinger et al.	N/A	N/A
2014/0207175	12/2013	Aggerholm	N/A	N/A

2014/0207231	12/2013	Hacohen et al.	N/A	N/A
2014/0214157	12/2013	Börtlein et al.	N/A	N/A
2014/0214159	12/2013	Vidlund et al.	N/A	N/A
2014/0222136	12/2013	Geist et al.	N/A	N/A
2014/0222137	12/2013	Miller et al.	N/A	N/A
2014/0222142	12/2013	Kovalsky et al.	N/A	N/A
2014/0236287	12/2013	Clague et al.	N/A	N/A
2014/0236289	12/2013	Alkhatib	N/A	N/A
2014/0243859	12/2013	Robinson	N/A	N/A
2014/0243894	12/2013	Groothuis et al.	N/A	N/A
2014/0243963	12/2013	Sheps et al.	N/A	N/A
2014/0249622	12/2013	Carmi	623/2.11	A61F 2/2418
2014/0257461	12/2013	Robinson et al.	N/A	N/A
2014/0257467	12/2013	Lane et al.	N/A	N/A
2014/0257475	12/2013	Gross et al.	N/A	N/A
2014/0257476	12/2013	Montorfano et al.	N/A	N/A
2014/0275757	12/2013	Goodwin et al.	N/A	N/A
2014/0276648	12/2013	Hammer et al.	N/A	N/A
2014/0277045	12/2013	Fazio et al.	N/A	N/A
2014/0277054	12/2013	McNamara et al.	N/A	N/A
2014/0277358	12/2013	Slazas	N/A	N/A
2014/0277409	12/2013	Börtlein et al.	N/A	N/A
2014/0277411	12/2013	Bortlein	623/2.11	A61F 2/243
2014/0277412	12/2013	Börtlein et al.	N/A	N/A
2014/0277413	12/2013	Richter et al.	N/A	N/A
2014/0277417	12/2013	Schraut et al.	N/A	N/A
2014/0277418	12/2013	Miller	N/A	N/A
2014/0277419	12/2013	Garde et al.	N/A	N/A
2014/0277422	12/2013	Ratz et al.	N/A	N/A
2014/0277427	12/2013	Ratz et al.	N/A	N/A
2014/0296962	12/2013	Cartledge et al.	N/A	N/A
2014/0296969	12/2013	Tegels et al.	N/A	N/A
2014/0303649	12/2013	Nguyen et al.	N/A	N/A
2014/0303720	12/2013	Sugimoto et al.	N/A	N/A
2014/0309661	12/2013	Sheps et al.	N/A	N/A
2014/0309730	12/2013	Alon et al.	N/A	N/A
2014/0324164	12/2013	Gross et al.	N/A	N/A
2014/0329225	12/2013	Morin	N/A	N/A
2014/0330371	12/2013	Gloss et al.	N/A	N/A
2014/0331475	12/2013	Duffy et al.	N/A	N/A
2014/0336744	12/2013	Tani et al.	N/A	N/A
2014/0343668	12/2013	Zipory et al.	N/A	N/A
2014/0343670	12/2013	Bakis et al.	N/A	N/A
2014/0350565	12/2013	Yacoby et al.	N/A	N/A
2014/0350662	12/2013	Vaturi	N/A	N/A
2014/0350670	12/2013	Keränen	N/A	N/A
2014/0358222	12/2013	Gorman, III et al.	N/A	N/A
2014/0358224	12/2013	Tegels et al.	N/A	N/A
2014/0378331	12/2013	Morin	N/A	N/A
2014/0379006	12/2013	Sutherland et al.	N/A	N/A
2014/0379065	12/2013	Johnson et al.	N/A	N/A
2014/0379074	12/2013	Spence et al.	N/A	N/A
2014/0379076	12/2013	Vidlund et al.	N/A	N/A
2015/0012087	12/2014	Miller et al.	N/A	N/A
2015/0018940	12/2014	Quill et al.	N/A	N/A

2015/0018944	12/2014	O'Connor et al.	N/A	N/A
2015/0032205	12/2014	Matheny	N/A	N/A
2015/0045880	12/2014	Hacohen	N/A	N/A
2015/0045881	12/2014	Lim	N/A	N/A
2015/0051697	12/2014	Spence et al.	N/A	N/A
2015/0073544	12/2014	Gorman, III et al.	N/A	N/A
2015/0081011	12/2014	Young et al.	N/A	N/A
2015/0081014	12/2014	Gross et al.	N/A	N/A
2015/0094802	12/2014	Buchbinder et al.	N/A	N/A
2015/0105855	12/2014	Cabiri et al.	N/A	N/A
2015/0119796	12/2014	Finch	N/A	N/A
2015/0119970	12/2014	Nakayama et al.	N/A	N/A
2015/0127097	12/2014	Neumann et al.	N/A	N/A
2015/0142100	12/2014	Morriss et al.	N/A	N/A
2015/0142103	12/2014	Vidlund	N/A	N/A
2015/0148731	12/2014	McNamara et al.	N/A	N/A
2015/0148894	12/2014	Damm et al.	N/A	N/A
2015/0148896	12/2014	Karapetian et al.	N/A	N/A
2015/0157457	12/2014	Hacohen	N/A	N/A
2015/0157458	12/2014	Thambar et al.	N/A	N/A
2015/0164640	12/2014	McLean et al.	N/A	N/A
2015/0173896	12/2014	Richter et al.	N/A	N/A
2015/0173897	12/2014	Raanani et al.	N/A	N/A
2015/0182336	12/2014	Zipory et al.	N/A	N/A
2015/0196390	12/2014	Ma et al.	N/A	N/A
2015/0196393	12/2014	Vidlund et al.	N/A	N/A
2015/0216661	12/2014	Hacohen et al.	N/A	N/A
2015/0230923	12/2014	Levi	N/A	N/A
2015/0230924	12/2014	Miller et al.	N/A	N/A
2015/0238313	12/2014	Spence et al.	N/A	N/A
2015/0238315	12/2014	Rabito et al.	N/A	N/A
2015/0245908	12/2014	Nitzan et al.	N/A	N/A
2015/0245934	12/2014	Lombardi et al.	N/A	N/A
2015/0250588	12/2014	Yang et al.	N/A	N/A
2015/0272730	12/2014	Melnick et al.	N/A	N/A
2015/0272731	12/2014	Racchini et al.	N/A	N/A
2015/0272734	12/2014	Sheps et al.	N/A	N/A
2015/0282964	12/2014	Beard et al.	N/A	N/A
2015/0305903	12/2014	Kitaoka	N/A	N/A
2015/0320556	12/2014	Levi et al.	N/A	N/A
2015/0327994	12/2014	Morriss et al.	N/A	N/A
2015/0328000	12/2014	Ratz et al.	N/A	N/A
2015/0335429	12/2014	Morriss et al.	N/A	N/A
2015/0342736	12/2014	Rabito et al.	N/A	N/A
2015/0351903	12/2014	Morriss et al.	N/A	N/A
2015/0351904	12/2014	Cooper	623/2.1	A61F 2/2418
2015/0351906	12/2014	Hammer et al.	N/A	N/A
2015/0359629	12/2014	Ganesan et al.	N/A	N/A
2015/0359631	12/2014	Sheahan et al.	N/A	N/A
2016/0008129	12/2015	Siegel	N/A	N/A
2016/0022423	12/2015	McNamara et al.	N/A	N/A
2016/0022970	12/2015	Forcucci et al.	N/A	N/A
2016/0030165	12/2015	Mitra et al.	N/A	N/A
2016/0030169	12/2015	Shahriari	N/A	N/A
2016/0030170	12/2015	Alkhatib et al.	N/A	N/A

2016/0030171	12/2015	Quijano et al.	N/A	N/A
2016/0038281	12/2015	Delaloye et al.	N/A	N/A
2016/0089482	12/2015	Siegenthaler	N/A	N/A
2016/0095700	12/2015	Righini	N/A	N/A
2016/0100939	12/2015	Armstrong et al.	N/A	N/A
2016/0106537	12/2015	Christianson et al.	N/A	N/A
2016/0106539	12/2015	Buchbinder et al.	N/A	N/A
2016/0113765	12/2015	Ganesan et al.	N/A	N/A
2016/0113766	12/2015	Ganesan et al.	N/A	N/A
2016/0113768	12/2015	Ganesan et al.	N/A	N/A
2016/0120550	12/2015	McNamara et al.	N/A	N/A
2016/0125160	12/2015	Heneghan et al.	N/A	N/A
2016/0157862	12/2015	Hernandez et al.	N/A	N/A
2016/0166381	12/2015	Sugimoto et al.	N/A	N/A
2016/0175095	12/2015	Dienno et al.	N/A	N/A
2016/0184098	12/2015	Vaturi	N/A	N/A
2016/0200773	12/2015	Morin	N/A	N/A
2016/0213473	12/2015	Hacohen et al.	N/A	N/A
2016/0220367	12/2015	Barrett	N/A	N/A
2016/0228244	12/2015	Cerf et al.	N/A	N/A
2016/0228247	12/2015	Maimon et al.	N/A	N/A
2016/0228249	12/2015	Mantanus et al.	N/A	N/A
2016/0242902	12/2015	Morriss et al.	N/A	N/A
2016/0245802	12/2015	Morin et al.	N/A	N/A
2016/0258939	12/2015	Morin et al.	N/A	N/A
2016/0262885	12/2015	Sandstrom et al.	N/A	N/A
2016/0266089	12/2015	Morin et al.	N/A	N/A
2016/0270911	12/2015	Ganesan et al.	N/A	N/A
2016/0296328	12/2015	Tabor et al.	N/A	N/A
2016/0296330	12/2015	Hacohen	N/A	N/A
2016/0296332	12/2015	Zhou et al.	N/A	N/A
2016/0310268	12/2015	Oba et al.	N/A	N/A
2016/0310274	12/2015	Gross et al.	N/A	N/A
2016/0317301	12/2015	Quadri et al.	N/A	N/A
2016/0317305	12/2015	Pelled et al.	N/A	N/A
2016/0324633	12/2015	Gross et al.	N/A	N/A
2016/0324635	12/2015	Vidlund et al.	N/A	N/A
2016/0324640	12/2015	Gifford et al.	N/A	N/A
2016/0331526	12/2015	Schweich et al.	N/A	N/A
2016/0331527	12/2015	Vidlund et al.	N/A	N/A
2016/0338706	12/2015	Rowe	N/A	N/A
2016/0367360	12/2015	Cartledge et al.	N/A	N/A
2016/0367368	12/2015	Vidlund et al.	N/A	N/A
2016/0374801	12/2015	Jimenez et al.	N/A	N/A
2016/0374802	12/2015	Levi et al.	N/A	N/A
2017/0042678	12/2016	Ganesan et al.	N/A	N/A
2017/0049435	12/2016	Sauer et al.	N/A	N/A
2017/0056166	12/2016	Ratz et al.	N/A	N/A
2017/0056169	12/2016	Johnson et al.	N/A	N/A
2017/0056171	12/2016	Cooper et al.	N/A	N/A
2017/0065407	12/2016	Hacohen et al.	N/A	N/A
2017/0065411	12/2016	Grundeman et al.	N/A	N/A
2017/0074855	12/2016	Morin et al.	N/A	N/A
2017/0100236	12/2016	Robertson et al.	N/A	N/A
2017/0113026	12/2016	Finch et al.	N/A	N/A

2017/0128205	12/2016	Tamir et al.	N/A	N/A
2017/0128705	12/2016	Forcucci et al.	N/A	N/A
2017/0135816	12/2016	Lashinski et al.	N/A	N/A
2017/0143938	12/2016	Ogle et al.	N/A	N/A
2017/0165054	12/2016	Benson et al.	N/A	N/A
2017/0165063	12/2016	Anderson et al.	N/A	N/A
2017/0189174	12/2016	Braido et al.	N/A	N/A
2017/0196688	12/2016	Christianson et al.	N/A	N/A
2017/0196692	12/2016	Kirk et al.	N/A	N/A
2017/0209264	12/2016	Chau et al.	N/A	N/A
2017/0216026	12/2016	Quill et al.	N/A	N/A
2017/0224323	12/2016	Rowe et al.	N/A	N/A
2017/0231757	12/2016	Gassler	N/A	N/A
2017/0231759	12/2016	Geist et al.	N/A	N/A
2017/0231760	12/2016	Lane et al.	N/A	N/A
2017/0231766	12/2016	Hariton et al.	N/A	N/A
2017/0234850	12/2016	Morin	N/A	N/A
2017/0239048	12/2016	Goldfarb et al.	N/A	N/A
2017/0252159	12/2016	Hacohen et al.	N/A	N/A
2017/0266003	12/2016	Hammer et al.	N/A	N/A
2017/0281337	12/2016	Campbell	N/A	N/A
2017/0325948	12/2016	Wallace et al.	N/A	N/A
2017/0333183	12/2016	Backus	N/A	N/A
2017/0333187	12/2016	Hariton et al.	N/A	N/A
2017/0349940	12/2016	Morin et al.	N/A	N/A
2017/0360426	12/2016	Hacohen et al.	N/A	N/A
2017/0367823	12/2016	Hariton et al.	N/A	N/A
2018/0000580	12/2017	Wallace et al.	N/A	N/A
2018/0014930	12/2017	Hariton et al.	N/A	N/A
2018/0014932	12/2017	Hammer et al.	N/A	N/A
2018/0021129	12/2017	Peterson et al.	N/A	N/A
2018/0023114	12/2017	Morin et al.	N/A	N/A
2018/0023115	12/2017	Morin et al.	N/A	N/A
2018/0028215	12/2017	Cohen	N/A	N/A
2018/0028311	12/2017	Hacohen	N/A	N/A
2018/0049873	12/2017	Manash et al.	N/A	N/A
2018/0055628	12/2017	Patel et al.	N/A	N/A
2018/0055629	12/2017	Oba et al.	N/A	N/A
2018/0055630	12/2017	Patel et al.	N/A	N/A
2018/0098850	12/2017	Rafiee et al.	N/A	N/A
2018/0116790	12/2017	Ratz et al.	N/A	N/A
2018/0116843	12/2017	Schreck et al.	N/A	N/A
2018/0125644	12/2017	Conklin	N/A	N/A
2018/0132999	12/2017	Perouse	N/A	N/A
2018/0133003	12/2017	Levi	N/A	N/A
2018/0147059	12/2017	Hammer et al.	N/A	N/A
2018/0153687	12/2017	Hariton et al.	N/A	N/A
2018/0153689	12/2017	Maimon et al.	N/A	N/A
2018/0153695	12/2017	Cunningham et al.	N/A	N/A
2018/0153696	12/2017	Albitov et al.	N/A	N/A
2018/0161159	12/2017	Lee et al.	N/A	N/A
2018/0177593	12/2017	Hariton et al.	N/A	N/A
2018/0177594	12/2017	Patel et al.	N/A	N/A
2018/0185148	12/2017	Hariton et al.	N/A	N/A
2018/0206982	12/2017	Haivatov et al.	N/A	N/A

2018/0206983	12/2017	Noe et al.	N/A	N/A
2018/0214263	12/2017	Rolando et al.	N/A	N/A
2018/0243086	12/2017	Barbarino et al.	N/A	N/A
2018/0250126	12/2017	O'Connor et al.	N/A	N/A
2018/0250130	12/2017	Hariton et al.	N/A	N/A
2018/0250147	12/2017	Syed	N/A	N/A
2018/0256323	12/2017	Hariton et al.	N/A	N/A
2018/0256325	12/2017	Hariton et al.	N/A	N/A
2018/0271654	12/2017	Hariton et al.	N/A	N/A
2018/0271655	12/2017	Hariton et al.	N/A	N/A
2018/0280136	12/2017	Hariton et al.	N/A	N/A
2018/0289479	12/2017	Hariton et al.	N/A	N/A
2018/0296333	12/2017	Dixon et al.	N/A	N/A
2018/0296336	12/2017	Cooper et al.	N/A	N/A
2018/0296341	12/2017	Noe et al.	N/A	N/A
2018/0325671	12/2017	Abunassar et al.	N/A	N/A
2018/0338829	12/2017	Hariton et al.	N/A	N/A
2018/0338830	12/2017	Hariton et al.	N/A	N/A
2018/0338831	12/2017	Hariton et al.	N/A	N/A
2018/0344457	12/2017	Gross et al.	N/A	N/A
2018/0344490	12/2017	Fox et al.	N/A	N/A
2018/0353294	12/2017	Calomeni et al.	N/A	N/A
2018/0360457	12/2017	Ellis et al.	N/A	N/A
2019/0000613	12/2018	Delgado et al.	N/A	N/A
2019/0008640	12/2018	Cooper et al.	N/A	N/A
2019/0015093	12/2018	Hacohen et al.	N/A	N/A
2019/0015200	12/2018	Delgado et al.	N/A	N/A
2019/0021852	12/2018	Delgado et al.	N/A	N/A
2019/0021857	12/2018	Hacohen et al.	N/A	N/A
2019/0038404	12/2018	Iamberger et al.	N/A	N/A
2019/0038405	12/2018	Iamberger et al.	N/A	N/A
2019/0046314	12/2018	Levi et al.	N/A	N/A
2019/0053895	12/2018	Levi	N/A	N/A
2019/0053896	12/2018	Adamek-bowers et al.	N/A	N/A
2019/0060060	12/2018	Chau et al.	N/A	N/A
2019/0060068	12/2018	Cope et al.	N/A	N/A
2019/0060070	12/2018	Groothuis et al.	N/A	N/A
2019/0069997	12/2018	Ratz et al.	N/A	N/A
2019/0069998	12/2018	Hacohen	N/A	N/A
2019/0076244	12/2018	Yohanan et al.	N/A	N/A
2019/0076245	12/2018	Arcaro et al.	N/A	N/A
2019/0083242	12/2018	Hariton et al.	N/A	N/A
2019/0083243	12/2018	Hariton et al.	N/A	N/A
2019/0083244	12/2018	Hariton et al.	N/A	N/A
2019/0083245	12/2018	Hariton et al.	N/A	N/A
2019/0083246	12/2018	Hariton et al.	N/A	N/A
2019/0083247	12/2018	Hariton et al.	N/A	N/A
2019/0083248	12/2018	Hariton et al.	N/A	N/A
2019/0083249	12/2018	Hariton et al.	N/A	N/A
2019/0083250	12/2018	Hariton et al.	N/A	N/A
2019/0083251	12/2018	Hariton et al.	N/A	N/A
2019/0083252	12/2018	Hariton et al.	N/A	N/A
2019/0083253	12/2018	Hariton et al.	N/A	N/A
2019/0083254	12/2018	Hariton et al.	N/A	N/A
2019/0083261	12/2018	Perszyk et al.	N/A	N/A

2019/0083262	12/2018	Hariton et al.	N/A	N/A
2019/0083263	12/2018	Hariton et al.	N/A	N/A
2019/0105153	12/2018	Barash et al.	N/A	N/A
2019/0117391	12/2018	Humair	N/A	N/A
2019/0167423	12/2018	Hariton et al.	N/A	N/A
2019/0175339	12/2018	Vidlund	N/A	N/A
2019/0175342	12/2018	Hariton et al.	N/A	N/A
2019/0183639	12/2018	Moore	N/A	N/A
2019/0183644	12/2018	Hacohen	N/A	N/A
2019/0192295	12/2018	Spence et al.	N/A	N/A
2019/0216602	12/2018	Lozonschi	N/A	N/A
2019/0224008	12/2018	Bressloff et al.	N/A	N/A
2019/0231525	12/2018	Hariton et al.	N/A	N/A
2019/0240010	12/2018	Hacohen	N/A	N/A
2019/0254818	12/2018	Quill et al.	N/A	N/A
2019/0262507	12/2018	Adamek-bowers et al.	N/A	N/A
2019/0321172	12/2018	Gross et al.	N/A	N/A
2019/0328519	12/2018	Hariton et al.	N/A	N/A
2019/0336280	12/2018	Naor	N/A	N/A
2019/0343627	12/2018	Hariton et al.	N/A	N/A
2019/0350701	12/2018	Adamek-Bowers et al.	N/A	N/A
2019/0365530	12/2018	Hoang et al.	N/A	N/A
2019/0388218	12/2018	Vidlund et al.	N/A	N/A
2019/0388220	12/2018	Vidlund et al.	N/A	N/A
2019/0388223	12/2018	Hariton et al.	N/A	N/A
2020/0000449	12/2019	Goldfarb et al.	N/A	N/A
2020/0000579	12/2019	Manash et al.	N/A	N/A
2020/0000580	12/2019	Hacohen	N/A	N/A
2020/0015964	12/2019	Noe et al.	N/A	N/A
2020/0030098	12/2019	Delgado et al.	N/A	N/A
2020/0038181	12/2019	Hariton et al.	N/A	N/A
2020/0046496	12/2019	Hammer et al.	N/A	N/A
2020/0046497	12/2019	Hariton et al.	N/A	N/A
2020/0054335	12/2019	Hernandez et al.	N/A	N/A
2020/0054451	12/2019	Hariton et al.	N/A	N/A
2020/0060818	12/2019	Geist et al.	N/A	N/A
2020/0069417	12/2019	Morin et al.	N/A	N/A
2020/0069424	12/2019	Hariton et al.	N/A	N/A
2020/0078002	12/2019	Hacohen et al.	N/A	N/A
2020/0113677	12/2019	McCann et al.	N/A	N/A
2020/0113689	12/2019	McCann et al.	N/A	N/A
2020/0113692	12/2019	McCann et al.	N/A	N/A
2020/0129294	12/2019	Hariton et al.	N/A	N/A
2020/0138567	12/2019	Marr et al.	N/A	N/A
2020/0146671	12/2019	Hacohen et al.	N/A	N/A
2020/0146824	12/2019	Hammer et al.	N/A	N/A
2020/0163760	12/2019	Hariton et al.	N/A	N/A
2020/0163761	12/2019	Hariton et al.	N/A	N/A
2020/0205969	12/2019	Hacohen	N/A	N/A
2020/0214832	12/2019	Metchik et al.	N/A	N/A
2020/0237512	12/2019	McCann et al.	N/A	N/A
2020/0246136	12/2019	Marr et al.	N/A	N/A
2020/0246140	12/2019	Hariton et al.	N/A	N/A
2020/0253600	12/2019	Darabian	N/A	N/A
2020/0261094	12/2019	Goldfarb et al.	N/A	N/A

2020/0281721	12/2019	Hariton et al.	N/A	N/A
2020/0297486	12/2019	Hariton et al.	N/A	N/A
2020/0306037	12/2019	Siegel et al.	N/A	N/A
2020/0315786	12/2019	Metchnik et al.	N/A	N/A
2020/0315789	12/2019	Hariton et al.	N/A	N/A
2020/0315797	12/2019	Hariton et al.	N/A	N/A
2020/0330221	12/2019	Hacohen	N/A	N/A
2020/0330227	12/2019	Hacohen	N/A	N/A
2020/0337842	12/2019	Metchnik et al.	N/A	N/A
2020/0360139	12/2019	Hammer et al.	N/A	N/A
2020/0390546	12/2019	Hariton et al.	N/A	N/A
2020/0390548	12/2019	Hariton et al.	N/A	N/A
2020/0397573	12/2019	Hariton et al.	N/A	N/A
2021/0085455	12/2020	Bateman et al.	N/A	N/A
2021/0085457	12/2020	Hariton et al.	N/A	N/A
2021/0093449	12/2020	Hariton et al.	N/A	N/A
2021/0106419	12/2020	Abunassar	N/A	N/A
2021/0113331	12/2020	Quadri et al.	N/A	N/A
2021/0137680	12/2020	Kizuka et al.	N/A	N/A
2021/0145578	12/2020	Hariton et al.	N/A	N/A
2021/0169467	12/2020	Hacohen et al.	N/A	N/A
2021/0196461	12/2020	Hariton et al.	N/A	N/A
2021/0259835	12/2020	Tyler, II et al.	N/A	N/A
2021/0322167	12/2020	Hariton et al.	N/A	N/A
2021/0330456	12/2020	Hacohen et al.	N/A	N/A
2021/0361422	12/2020	Gross et al.	N/A	N/A
2021/0361426	12/2020	Hacohen	N/A	N/A
2021/0393402	12/2020	Hammer et al.	N/A	N/A
2021/0401573	12/2020	Gross et al.	N/A	N/A
2022/0000612	12/2021	Hacohen	N/A	N/A
2022/0023036	12/2021	Levi et al.	N/A	N/A
2022/0061984	12/2021	Humair et al.	N/A	N/A
2022/0105238	12/2021	Reimer et al.	N/A	N/A
2022/0151779	12/2021	Pintor	N/A	N/A
2022/0175526	12/2021	Albitov et al.	N/A	N/A
2022/0338985	12/2021	Hariton et al.	N/A	N/A
2023/0049163	12/2022	Hariton et al.	N/A	N/A
2023/0137909	12/2022	Hariton et al.	N/A	N/A
2023/0201015	12/2022	Gurovich et al.	N/A	N/A
2023/0255764	12/2022	Hariton et al.	N/A	N/A
2023/0346549	12/2022	Hariton et al.	N/A	N/A
2023/0346550	12/2022	Gross et al.	N/A	N/A
2024/0050227	12/2023	Hacohen	N/A	N/A
2024/0081983	12/2023	Hariton et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2822801	12/2005	CA	N/A
2671966	12/2007	CA	N/A
101653365	12/2009	CN	N/A
202313807	12/2011	CN	N/A
103974674	12/2013	CN	N/A
103997990	12/2013	CN	N/A
105324091	12/2015	CN	N/A
112603598	12/2020	CN	N/A

0170262	12/1985	EP	N/A
06/14342	12/1993	EP	N/A
10/06905	12/1999	EP	N/A
0954257	12/1999	EP	N/A
1258437	12/2001	EP	N/A
1264582	12/2001	EP	N/A
0871417	12/2002	EP	N/A
1266641	12/2003	EP	N/A
1034753	12/2004	EP	N/A
1258232	12/2005	EP	N/A
1637092	12/2005	EP	N/A
1990014	12/2007	EP	N/A
1562522	12/2007	EP	N/A
1420723	12/2008	EP	N/A
1903991	12/2008	EP	N/A
1418865	12/2008	EP	N/A
2119399	12/2008	EP	N/A
1531762	12/2009	EP	N/A
1450733	12/2010	EP	N/A
2446915	12/2011	EP	N/A
2088965	12/2011	EP	N/A
2641569	12/2012	EP	N/A
1768630	12/2014	EP	N/A
1861045	12/2014	EP	N/A
1465555	12/2014	EP	N/A
2219558	12/2014	EP	N/A
2349124	12/2017	EP	N/A
2739214	12/2017	EP	N/A
3417813	12/2017	EP	N/A
3583922	12/2018	EP	N/A
3270825	12/2019	EP	N/A
2485795	12/2019	EP	N/A
223448	12/2011	IL	N/A
S53152790	12/1977	JP	N/A
20010046894	12/2000	KR	N/A
92/05093	12/1991	WO	N/A
93/10714	12/1992	WO	N/A
96/39963	12/1995	WO	N/A
96/40344	12/1995	WO	N/A
97/01369	12/1996	WO	N/A
98/46149	12/1997	WO	N/A
1998/043557	12/1997	WO	N/A
1999/030647	12/1998	WO	N/A
00/22981	12/1999	WO	N/A
2000-047139	12/1999	WO	N/A
01/26586	12/2000	WO	N/A
01/56457	12/2000	WO	N/A
2001-062189	12/2000	WO	N/A
01/82832	12/2000	WO	N/A
02/085250	12/2001	WO	N/A
02/085251	12/2001	WO	N/A
02/085252	12/2001	WO	N/A
2003/020179	12/2002	WO	N/A
2003/028558	12/2002	WO	N/A
03/047467	12/2002	WO	N/A

2003/049647	12/2002	WO	N/A
2003/105667	12/2002	WO	N/A
2004/028399	12/2003	WO	N/A
04/103434	12/2003	WO	N/A
2004/108191	12/2003	WO	N/A
05/021063	12/2004	WO	N/A
05/046488	12/2004	WO	N/A
2005/062931	12/2004	WO	N/A
2005/107650	12/2004	WO	N/A
2006/007389	12/2005	WO	N/A
2006/007401	12/2005	WO	N/A
06/012013	12/2005	WO	N/A
06/012038	12/2005	WO	N/A
06/054930	12/2005	WO	N/A
2006/065212	12/2005	WO	N/A
2006/070372	12/2005	WO	N/A
06/086434	12/2005	WO	N/A
2006/089236	12/2005	WO	N/A
2006/091163	12/2005	WO	N/A
06/097931	12/2005	WO	N/A
06/105084	12/2005	WO	N/A
2006/113906	12/2005	WO	N/A
06/116558	12/2005	WO	N/A
2006/128193	12/2005	WO	N/A
07/011799	12/2006	WO	N/A
2007/030063	12/2006	WO	N/A
2007/047488	12/2006	WO	N/A
2007/059252	12/2006	WO	N/A
07/121314	12/2006	WO	N/A
07/136783	12/2006	WO	N/A
07/136981	12/2006	WO	N/A
08/013915	12/2007	WO	N/A
2008/014144	12/2007	WO	N/A
2008/029296	12/2007	WO	N/A
2008/031103	12/2007	WO	N/A
2008/058940	12/2007	WO	N/A
08/068756	12/2007	WO	N/A
2008/070797	12/2007	WO	N/A
2008/103722	12/2007	WO	N/A
2009/026563	12/2008	WO	N/A
09/033469	12/2008	WO	N/A
09/053497	12/2008	WO	N/A
2009/080801	12/2008	WO	N/A
2009/091509	12/2008	WO	N/A
2009/108942	12/2008	WO	N/A
2009/130631	12/2008	WO	N/A
10/004546	12/2009	WO	N/A
2010/000454	12/2009	WO	N/A
2010/005827	12/2009	WO	N/A
2010/006627	12/2009	WO	N/A
2010/006905	12/2009	WO	N/A
2010/027485	12/2009	WO	N/A
2010/037141	12/2009	WO	N/A
2010/044851	12/2009	WO	N/A
2010/045297	12/2009	WO	N/A

2010/057262	12/2009	WO	N/A
2010/073246	12/2009	WO	N/A
2010/081033	12/2009	WO	N/A
2010/085649	12/2009	WO	N/A
2010/121076	12/2009	WO	N/A
2010/128502	12/2009	WO	N/A
2010/128503	12/2009	WO	N/A
2010/150178	12/2009	WO	N/A
2011/025972	12/2010	WO	N/A
2011/051942	12/2010	WO	N/A
2011/057087	12/2010	WO	N/A
2011/067770	12/2010	WO	N/A
2011/069048	12/2010	WO	N/A
2011/072084	12/2010	WO	N/A
2011/089401	12/2010	WO	N/A
2011/089601	12/2010	WO	N/A
2011/106137	12/2010	WO	N/A
2011/111047	12/2010	WO	N/A
01/87190	12/2010	WO	N/A
2011/137531	12/2010	WO	N/A
2011-143263	12/2010	WO	N/A
2011/144351	12/2010	WO	N/A
2011/148374	12/2010	WO	N/A
2011/154942	12/2010	WO	N/A
2012/011108	12/2011	WO	N/A
2012/014201	12/2011	WO	N/A
2012/024428	12/2011	WO	N/A
2012/036740	12/2011	WO	N/A
2012/048035	12/2011	WO	N/A
2012/068541	12/2011	WO	N/A
2012/127309	12/2011	WO	N/A
2012/176195	12/2011	WO	N/A
2012/177942	12/2011	WO	N/A
2012/178115	12/2011	WO	N/A
2013/021374	12/2012	WO	N/A
2013/021375	12/2012	WO	N/A
2013/021384	12/2012	WO	N/A
2013/028387	12/2012	WO	N/A
2013/059743	12/2012	WO	N/A
2013/059747	12/2012	WO	N/A
2013/069019	12/2012	WO	N/A
2013/072496	12/2012	WO	N/A
2013/078497	12/2012	WO	N/A
2013/088327	12/2012	WO	N/A
2013/114214	12/2012	WO	N/A
2013/128436	12/2012	WO	N/A
2013/175468	12/2012	WO	N/A
2014/022124	12/2013	WO	N/A
2014/064694	12/2013	WO	N/A
2014/064695	12/2013	WO	N/A
2014/076696	12/2013	WO	N/A
2014/087402	12/2013	WO	N/A
2014/099655	12/2013	WO	N/A
2014/115149	12/2013	WO	N/A
2014/121275	12/2013	WO	N/A

2014/121280	12/2013	WO	N/A
2014/144937	12/2013	WO	N/A
2014/145338	12/2013	WO	N/A
2014/164151	12/2013	WO	N/A
2014/164364	12/2013	WO	N/A
2014/194178	12/2013	WO	N/A
2014/195786	12/2013	WO	N/A
2014/210124	12/2013	WO	N/A
2015/059699	12/2014	WO	N/A
2015/173794	12/2014	WO	N/A
2015/191923	12/2014	WO	N/A
2016/016899	12/2015	WO	N/A
2016/093877	12/2015	WO	N/A
2016/098104	12/2015	WO	N/A
2016/113743	12/2015	WO	N/A
2016/125160	12/2015	WO	N/A
2016/150806	12/2015	WO	N/A
2016/183526	12/2015	WO	N/A
2017/223486	12/2016	WO	N/A
2018/025260	12/2017	WO	N/A
2018/025263	12/2017	WO	N/A
2018/029680	12/2017	WO	N/A
2018/039631	12/2017	WO	N/A
2018/106837	12/2017	WO	N/A
2018/108837	12/2017	WO	N/A
2018/112429	12/2017	WO	N/A
2018/118717	12/2017	WO	N/A
2018/131042	12/2017	WO	N/A
2018/131043	12/2017	WO	N/A
2019/026059	12/2018	WO	N/A
2019/027507	12/2018	WO	N/A
2019/030753	12/2018	WO	N/A
2019/077595	12/2018	WO	N/A
2019/086958	12/2018	WO	N/A
2019/116369	12/2018	WO	N/A
2019/138400	12/2018	WO	N/A
2019/195860	12/2018	WO	N/A
2019/202579	12/2018	WO	N/A
2020/058972	12/2019	WO	N/A
2020/167677	12/2019	WO	N/A
2021/156866	12/2020	WO	N/A
2021/178400	12/2020	WO	N/A
2021/186424	12/2020	WO	N/A
2022/015910	12/2021	WO	N/A
2022/046568	12/2021	WO	N/A
2022/061017	12/2021	WO	N/A
2022/118316	12/2021	WO	N/A
2023/009379	12/2022	WO	N/A

OTHER PUBLICATIONS

An Office Action dated Nov. 23, 2012, which issued during the prosecution of U.S. Appl. No. 13/033,852. cited by applicant

An Office Action dated Dec. 31, 2012, which issued during the prosecution of U.S. Appl. No. 13/044,694. cited by applicant

An Office Action dated Feb. 6, 2013, which issued during the prosecution of U.S. Appl. No. 13/412,814. cited by

applicant

Langer F et al., "Ring plus String: Papillary muscle repositioning as an adjunctive repair technique for ischemic mitral regurgitation," J Thorac Cardiovasc Surg 133:247-9, Jan. 2007. cited by applicant

Langer F et al., "Ring+String: Successful repair technique for ischemic mitral regurgitation with severe leaflet tethering," Circulation 120[suppl 1]: S85-S91, Sep. 2009. cited by applicant

"Transcatheter Valve-in-Valve Implantation for Failed Bioprosthetic Heart Valves", J Webb et al., Circulation. Apr. 2010; 121: 1848-1857. cited by applicant

Jansen, J., Willeke, S., Reul, H. and Rum, G. (1992), Detachable Shape-Memory Sewing Ring for Heart Valves. Artificial Organs, 16:294-297. 1992 (an abstract). cited by applicant

Alexander S. Geha, et al., Replacement of degenerated mitral and aortic bioprostheses without explantation Ann Thorac Surg. Jun. 2001; 72:1509-1514. cited by applicant

An International Search Report and a Written Opinion both dated Oct. 13, 2011 which issued during the prosecution of Applicant's PCT/IL11/00231. cited by applicant

An Office Action dated Jul. 1, 2016, which issued during the prosecution of U.S. Appl. No. 14/161,921. cited by applicant

An International Search Report and a Written Opinion both dated Dec. 5, 2011, which issued during the prosecution of Applicant's PCT/IL11/00582. cited by applicant

An Office Action dated May 29, 2012, which issued during the prosecution of U.S. Appl. No. 12/840,463. cited by applicant

U.S. Appl. No. 61/555,160, filed Nov. 3, 2011. cited by applicant

U.S. Appl. No. 61/525,281, filed Aug. 19, 2011. cited by applicant

U.S. Appl. No. 61/537,276, filed Sep. 21, 2011. cited by applicant

U.S. Appl. No. 61/515,372, filed Aug. 5, 2011. cited by applicant

U.S. Appl. No. 61/492,449, filed Jun. 2, 2011. cited by applicant

U.S. Appl. No. 61/588,892, filed Jan. 20, 2012. cited by applicant

An International Search Report and a Written Opinion both dated Feb. 6, 2013, which issued during the prosecution of Applicant's PCT/IL12/00292. cited by applicant

An International Search Report and a Written Opinion both dated Feb. 6, 2013, which issued during the prosecution of Applicant's PCT/IL12/00293. cited by applicant

An Office Action dated Nov. 18, 2012, which issued during the prosecution of U.S. Appl. No. 12/961,721. cited by applicant

An Office Action dated Feb. 15, 2013, which issued during the prosecution of U.S. Appl. No. 12/840,463. cited by applicant

An Office Action dated Feb. 10, 2014, which issued during the prosecution of U.S. Appl. No. 13/033,852. cited by applicant

An Office Action dated Sep. 19, 2014, which issued during the prosecution of U.S. Appl. No. 13/044,694. cited by applicant

An International Search Report and a Written Opinion both dated Sep. 4, 2014 which issued during the prosecution of Applicant's PCT/IL2014/050087. cited by applicant

Invitation to Pay Additional Fees dated Jun. 12, 2014 PCT/IL2014/050087. cited by applicant

An Office Action dated Jun. 17, 2014, which issued during the prosecution of U.S. Appl. No. 12/961,721. cited by applicant

An Office Action dated Jul. 3, 2014, which issued during the prosecution of U.S. Appl. No. 13/033,852. cited by applicant

An Office Action dated May 23, 2014, which issued during the prosecution of U.S. Appl. No. 13/412,814. cited by applicant

Dominique Himbert; Mitral Regurgitation and Stenosis from Bioprosthesis and Annuloplasty Failure: Transcatheter approaches and outcomes, 24 pages Oct. 28, 2013. cited by applicant

An International Search Report and a Written Opinion both dated Mar. 17, 2014 which issued during the prosecution of Applicant's PCT/IL2013/050937. cited by applicant

An International Preliminary Report on patentability dated Dec. 2, 2013, which issued during the prosecution of Applicant's PCT/IL11/00582. cited by applicant

An Office Action dated Sep. 12, 2013, which issued during the prosecution of U.S. Appl. No. 13/412,814. cited by applicant

An Office Action dated Aug. 2, 2013, which issued during the prosecution of U.S. Appl. No. 13/033,852. cited by applicant

applicant

An International Preliminary Report on patentability dated Sep. 11, 2012, which issued during the prosecution of Applicant's PCT/IL2011/000231. cited by applicant

An Office Action dated Jul. 2, 2014, which issued during the prosecution of U.S. Appl. No. 13/811,308. cited by applicant

An Office Action dated Jan. 20, 2016, which issued during the prosecution of U.S. Appl. No. 14/161,921. cited by applicant

An Office Action dated Jul. 23, 2013, which issued during the prosecution of U.S. Appl. No. 12/961,721. cited by applicant

An Office Action dated Jul. 18, 2013, which issued during the prosecution of U.S. Appl. No. 13/044,694. cited by applicant

An Office Action dated Nov. 8, 2013, which issued during the prosecution of U.S. Appl. No. 12/840,463. cited by applicant

An Office Action dated Jun. 4, 2014, which issued during the prosecution of U.S. Appl. No. 12/840,463. cited by applicant

An Office Action dated Aug. 13, 2012, which issued during the prosecution of U.S. Appl. No. 13/044,694. cited by applicant

An Office Action dated Jul. 2, 2012, which issued during the prosecution of U.S. Appl. No. 13/033,852. cited by applicant

An Office Action dated Feb. 3, 2014, which issued during the prosecution of U.S. Appl. No. 13/811,308. cited by applicant

An International Preliminary Report on patentability dated Feb. 11, 2014, which issued during the prosecution of Applicant's PCT/IL12/00292. cited by applicant

An International Preliminary Report on patentability dated Feb. 11, 2014, which issued during the prosecution of Applicant's PCT/IL12/00293. cited by applicant

A Notice of Allowance dated Aug. 15, 2014, which issued during the prosecution of U.S. Appl. No. 13/412,814. cited by applicant

An Office Action dated Aug. 14, 2012, which issued during the prosecution of U.S. Appl. No. 12/961,721. cited by applicant

U.S. Appl. No. 61/283,819, filed Dec. 8, 2009. cited by applicant

Notice of Allowance dated Apr. 8, 2016, which issued during the prosecution of U.S. Appl. No. 14/237,258. cited by applicant

U.S. Appl. No. 61/756,034, filed Jan. 24, 2013. cited by applicant

U.S. Appl. No. 61/756,049, filed Jan. 24, 2013. cited by applicant

An International Preliminary Report on Patentability dated Jan. 31, 2017, which issued during the prosecution of Applicant's PCT/IL2015/050792. cited by applicant

U.S. Appl. No. 62/372,861, filed Aug. 10, 2016. cited by applicant

Notice of Allowance dated Aug. 13, 2018, which issued during the prosecution of U.S. Appl. No. 15/995,597. cited by applicant

Notice of Allowance dated Apr. 20, 2018, which issued during the prosecution of U.S. Appl. No. 15/878,206. cited by applicant

An Office Action dated Dec. 10, 2015, which issued during the prosecution of U.S. Appl. No. 14/237,258. cited by applicant

An International Preliminary Report on Patentability dated Jul. 28, 2015, which issued during the prosecution of Applicant's PCT/IL2014/050087. cited by applicant

An Office Action dated Nov. 27, 2015, which issued during the prosecution of U.S. Appl. No. 14/626,267. cited by applicant

An Office Action dated Jan. 21, 2016, which issued during the prosecution of U.S. Appl. No. 14/237,264. cited by applicant

An Office Action dated Jan. 30, 2015, which issued during the prosecution of UK Patent Application No. 1413474.6. cited by applicant

An International Search Report and a Written Opinion both dated May 30, 2016, which issued during the prosecution of Applicant's PCT/IL2016/050125. cited by applicant

An Office Action dated Sep. 26 2016, which issued during the prosecution of U.S. Appl. No. 14/763,004. cited by applicant

An Office Action dated Jan. 18, 2017, which issued during the prosecution of U.S. Appl. No. 14/626,267. cited by applicant

An Office Action dated Feb. 7, 2017, which issued during the prosecution of U.S. Appl. No. 14/689,608. cited by applicant

An Office Action dated Feb. 8, 2017, which issued during the prosecution of UK Patent Application No. 1613219.3. cited by applicant

An Office Action together dated Feb. 10, 2017, which issued during the prosecution of European Patent Application No. 12821522.5. cited by applicant

An International Search Report and a Written Opinion both dated Oct. 27, 2015, which issued during the prosecution of Applicant's PCT/IL2015/050792. cited by applicant

European Search Report dated Feb. 18, 2015, which issued during the prosecution of Applicant's European App No. 12821522.5. cited by applicant

Saturn Project—a novel solution for transcatheter heart valve replacement specifically designed to address clinical therapeutic needs on mitral valve: Dec. 2016. cited by applicant

Righini presentation EuroPCR May 2015 (Saturn)—(downloaded from: <https://www.pconline.com/Cases-resourcesimages/Resources/Course-videos-slides/2015/Cardiovascularinnovation-pipeline-Mitral-and-tricuspid-valve-interventions>). cited by applicant

An Advisory Action dated Apr. 2, 2018, which issued during the prosecution of U.S. Appl. No. 14/763,004. cited by applicant

An Office Action dated Jul. 26, 2018, which issued during the prosecution of U.S. Appl. No. 15/872,501. cited by applicant

An Office Action dated May 4, 2018, which issued during the prosecution of U.S. Appl. No. 15/872,501. cited by applicant

An Office Action dated Apr. 20, 2018, which issued during the prosecution of U.S. Appl. No. 15/886,517. cited by applicant

An Office Action dated Aug. 9, 2018, which issued during the prosecution of U.S. Appl. No. 15/899,858. cited by applicant

An Office Action dated Aug. 9, 2018, which issued during the prosecution of U.S. Appl. No. 15/902,403. cited by applicant

An Office Action dated Jun. 28, 2018, which issued during the prosecution of Design U.S. Appl. No. 29/635,658. cited by applicant

An Office Action dated Jun. 28, 2018, which issued during the prosecution of Design U.S. Appl. No. 29/635,661. cited by applicant

Georg Lutter, MD, et al; “Percutaneous Valve Replacement: Current State and Future Prospects”, The Annals of Thoracic Surgery ; vol. 78, pp. 2199-2206; Dec. 2004. cited by applicant

An Office Action dated Jun. 6, 2018, which issued during the prosecution of UK Patent Application No. 1720803.4. cited by applicant

An International Search Report and a Written Opinion both dated Jun. 20, 2018, which issued during the prosecution of Applicant's PCT/IL2018/050024. cited by applicant

An Office Action dated Jun. 18, 2018, which issued during the prosecution of UK Patent Application No. 1800399.6. cited by applicant

An Office Action dated Oct. 23, 2017, which issued during the prosecution of U.S. Appl. No. 14/763,004. cited by applicant

An Office Action dated Dec. 7, 2017, which issued during the prosecution of U.S. Appl. No. 15/213,791. cited by applicant

Interview Summary dated Feb. 8, 2018, which issued during the prosecution of U.S. Appl. No. 15/213,791. cited by applicant

An Office Action dated Feb. 7, 2018, which issued during the prosecution of U.S. Appl. No. 15/197,069. cited by applicant

An International Search Report and a Written Opinion both dated Nov. 24, 2017, which issued during the prosecution of Applicant's PCT/IL2017/050873. cited by applicant

An Office Action dated Jan. 5, 2018, which issued during the prosecution of U.S. Appl. No. 15/541,783. cited by applicant

An Office Action dated Feb. 2, 2018, which issued during the prosecution of U.S. Appl. No. 15/329,920. cited by applicant

An Invitation to pay additional fees dated Jan. 2, 2018, which issued during the prosecution of Applicant's PCT/IL2017/050849. cited by applicant

An Invitation to pay additional fees dated Sep. 29, 2017, which issued during the prosecution of Applicant's PCT/IL2017/050873. cited by applicant

European Search Report dated Jun. 29, 2017, which issued during the prosecution of Applicant's European App No. 11809374.9. cited by applicant

An Invitation to pay additional fees dated Oct. 11, 2018, which issued during the prosecution of Applicant's PCT/IL2018/050725. cited by applicant

An Office Action dated Dec. 4, 2018, which issued during the prosecution of U.S. Appl. No. 16/045,059. cited by applicant

An Office Action together with the English translation dated Nov. 5, 2018 which issued during the prosecution of Chinese Patent Application No. 201680008328.5. cited by applicant

Notice of Allowance dated Sep. 25, 2018, which issued during the prosecution of U.S. Appl. No. 15/188,507. cited by applicant

European Search Report dated Sep. 26, 2018 which issued during the prosecution of Applicant's European App No. 18186784.7. cited by applicant

An Office Action dated Jun. 30, 2015, which issued during the prosecution of U.S. Appl. No. 14/522,987. cited by applicant

Notice of Allowance dated Dec. 13, 2013, which issued during the prosecution of U.S. Appl. No. 13/675,119. cited by applicant

An International Preliminary Report on Patentability dated Aug. 8, 2017, which issued during the prosecution of Applicant's PCT/IL2016/050125. cited by applicant

An Office Action dated Jan. 17, 2018, which issued during the prosecution of U.S. Appl. No. 14/763,004. cited by applicant

An Office Action dated Mar. 25, 2015, which issued during the prosecution of U.S. Appl. No. 12/840,463. cited by applicant

An Office Action dated Feb. 25, 2016, which issued during the prosecution of U.S. Appl. No. 14/522,987. cited by applicant

An Office Action dated Apr. 13, 2016, which issued during the prosecution of U.S. Appl. No. 14/626,267. cited by applicant

An Office Action dated Aug. 28, 2015, which issued during the prosecution of U.S. Appl. No. 14/237,264. cited by applicant

Maisano (2015) TCR presentation re Cardiovalve. cited by applicant

Notice of Allowance dated Sep. 29, 2016, which issued during the prosecution of U.S. Appl. No. 14/442,541. cited by applicant

Notice of Allowance dated May 10, 2016, which issued during the prosecution of U.S. Appl. No. 14/237,258. cited by applicant

Notice of Allowance dated May 20, 2016, which issued during the prosecution of U.S. Appl. No. 14/237,258. cited by applicant

An International Preliminary Report on Patentability dated May 19, 2015, which issued during the prosecution of Applicant's PCT/IL2013/050937. cited by applicant

Dusan Pavcnik, MD, PhD², et al; "Development and Initial Experimental Evaluation of a Prosthetic Aortic Valve for Transcatheter Placement", Cardiovascular Radiology. Radiology Apr. 1992, vol. 183, pp. 151-154. cited by applicant

Notice of Allowance dated Oct. 16, 2013, which issued during the prosecution of U.S. Appl. No. 13/675,119. cited by applicant

Notice of Allowance dated Feb. 11, 2015, which issued during the prosecution of U.S. Appl. No. 13/033,852. cited by applicant

Notice of Allowance dated May 5, 2015, which issued during the prosecution of U.S. Appl. No. 12/840,463. cited by applicant

Notice of Allowance dated Mar. 10, 2015, which issued during the prosecution of U.S. Appl. No. 13/811,308. cited by applicant

Notice of Allowance dated Jul. 1, 2016, which issued during the prosecution of U.S. Appl. No. 14/442,541. cited by applicant

An Office Action dated Mar. 25, 2019, which issued during the prosecution of European Patent Application No.

14710060.6. cited by applicant

An International Search Report and a Written Opinion both dated Nov. 9, 2018, which issued during the prosecution of Applicant's PCT/IL2018/050869. cited by applicant

An International Search Report and a Written Opinion both dated Dec. 5, 2018, which issued during the prosecution of Applicant's PCT/IL2018/050725. cited by applicant

An International Search Report and a Written Opinion both dated Apr. 25, 2019, which issued during the prosecution of Applicant's PCT/IL2019/050142. cited by applicant

An International Preliminary Report on Patentability dated Feb. 12, 2019, which issued during the prosecution of Applicant's PCT/IL2017/050873. cited by applicant

An Office Action dated Sep. 13, 2019, which issued during the prosecution of U.S. Appl. No. 16/460,313. cited by applicant

An Office Action dated Nov. 26, 2019, which issued during the prosecution of U.S. Appl. No. 16/532,945. cited by applicant

An Office Action dated Aug. 16, 2019, which issued during the prosecution of U.S. Appl. No. 15/668,659. cited by applicant

An Office Action dated Nov. 1, 2019, which issued during the prosecution of U.S. Appl. No. 15/872,501. cited by applicant

An Office Action dated Jun. 14, 2019, which issued during the prosecution of U.S. Appl. No. 15/703,385. cited by applicant

An Office Action dated Oct. 4, 2019, which issued during the prosecution of U.S. Appl. No. 16/183,140. cited by applicant

An Office Action dated Jun. 13, 2019, which issued during the prosecution of U.S. Appl. No. 16/388,038. cited by applicant

An International Preliminary Report on Patentability dated Feb. 4, 2020, which issued during the prosecution of Applicant's PCT/IL2018/050725. cited by applicant

An International Search Report and a Written Opinion both dated Jan. 25, 2019, which issued during the prosecution of Applicant's PCT/IL2018/051122. cited by applicant

An International Search Report and a Written Opinion both dated May 13, 2019, which issued during the prosecution of Applicant's PCT/IL2018/051350. cited by applicant

An International Preliminary Report on Patentability dated Feb. 5, 2019, which issued during the prosecution of Applicant's PCT/IL2017/050849. cited by applicant

An Office Action dated Oct. 25, 2018, which issued during the prosecution of U.S. Appl. No. 14/763,004. cited by applicant

An Office Action dated Mar. 4, 2019, which issued during the prosecution of U.S. Appl. No. 14/763,004. cited by applicant

An Office Action dated Jan. 9, 2019, which issued during the prosecution of U.S. Appl. No. 15/329,920. cited by applicant

An Office Action dated Jan. 30, 2019, which issued during the prosecution of U.S. Appl. No. 15/872,501. cited by applicant

An Office Action dated Feb. 5, 2019, which issued during the prosecution of U.S. Appl. No. 15/899,858. cited by applicant

An Office Action dated May 23, 2019, which issued during the prosecution of U.S. Appl. No. 15/668,659. cited by applicant

An Office Action dated May 1, 2019, which issued during the prosecution of U.S. Appl. No. 15/691,032. cited by applicant

An Office Action dated Aug. 1, 2019, which issued during the prosecution of U.S. Appl. No. 15/668,559. cited by applicant

An Office Action dated Jun. 19, 2019, which issued during the prosecution of U.S. Appl. No. 15/682,789. cited by applicant

Notice of Allowance dated Jan. 13, 2020, which issued during the prosecution of U.S. Appl. No. 15/956,956. cited by applicant

An Office Action dated Jun. 25, 2019, which issued during the prosecution of U.S. Appl. No. 15/329,920. cited by applicant

An Office Action dated May 16, 2019, which issued during the prosecution of U.S. Appl. No. 15/433,547. cited by applicant

U.S. Appl. No. 62/560,384, filed Sep. 19, 2017. cited by applicant

U.S. Appl. No. 62/112,343, filed Feb. 5, 2015. cited by applicant

An International Preliminary Report on Patentability dated Feb. 11, 2020, which issued during the prosecution of Applicant's PCT/IL2018/050869. cited by applicant

An International Preliminary Report on Patentability dated Oct. 20, 2020, which issued during the prosecution of Applicant's PCT/IL2019/050142. cited by applicant

An Office Action dated Jan. 6, 2020, which issued during the prosecution of U.S. Appl. No. 16/660,231. cited by applicant

An Office Action dated Dec. 31, 2019, which issued during the prosecution of U.S. Appl. No. 16/183,140. cited by applicant

Notice of Allowance dated Apr. 24, 2019, which issued during the prosecution of U.S. Appl. No. 16/045,059. cited by applicant

An Office Action dated Jan. 14, 2020, which issued during the prosecution of U.S. Appl. No. 16/284,331. cited by applicant

European Search Report dated Mar. 5, 2020 which issued during the prosecution of Applicant's European App No. 17752184.6. cited by applicant

European Search Report dated Mar. 4, 2020 which issued during the prosecution of Applicant's European App No. 16706913.7. cited by applicant

Notice of Allowance dated Mar. 12, 2020, which issued during the prosecution of U.S. Appl. No. 16/460,313. cited by applicant

An Office Action dated Jan. 9, 2020, which issued during the prosecution of U.S. Appl. No. 15/600,190. cited by applicant

An Office Action dated Jan. 3, 2020, which issued during the prosecution of U.S. Appl. No. 16/678,355. cited by applicant

An Office Action dated Feb. 6, 2020, which issued during the prosecution of U.S. Appl. No. 15/668,659. cited by applicant

Notice of Allowance dated Jan. 16, 2020, which issued during the prosecution of U.S. Appl. No. 16/532,945. cited by applicant

Notice of Allowance dated Aug. 19, 2020, which issued during the prosecution of U.S. Appl. No. 16/637,166. cited by applicant

Notice of Allowance dated Jul. 27, 2020, which issued during the prosecution of U.S. Appl. No. 16/637,166. cited by applicant

Notice of Allowance dated Jun. 23, 2020, which issued during the prosecution of U.S. Appl. No. 16/637,166. cited by applicant

Notice of Allowance dated May 7, 2020, which issued during the prosecution of U.S. Appl. No. 16/637,166. cited by applicant

Sündermann, Simon H., et al. "Feasibility of the Engager™ aortic transcatheter valve system using a flexible over-the-wire design." *European Journal of Cardio-Thoracic Surgery* 42.4 (2012): e48-e52. cited by applicant

An Office Action summarized English translation and Search Report dated Jul. 3, 2020, which issued during the prosecution of Chinese Patent Application No. 201780061210.3. cited by applicant

Serruys, P. W., Piazza, N., Cribier, A., Webb, J., Laborde, J. C., & de Jaegere, P. (Eds.). (2009). *Transcatheter aortic valve implantation: tips and tricks to avoid failure*. CRC Press.—Screenshots from Google Books downloaded from: https://books.google.co.il/books?id=FLzLBQAAQBAJ&lpg=PA198&ots=soqWrDH-y_&dq=%20%22Edwards%20SAPIEN%22&lr&pg=PA20#v=onepage&q=%22Edwards%20SAPIEN%22&f=false; Downloaded on Jun. 18, 2020. cited by applicant

An International Search Report and a Written Opinion both dated Jun. 24, 2020, which issued during the prosecution of Applicant's PCT/IL2019/051398. cited by applicant

An Office Action dated Jul. 14, 2020, which issued during the prosecution of U.S. Appl. No. 16/324,339. cited by applicant

Notice of Allowance dated Aug. 28, 2020, which issued during the prosecution of U.S. Appl. No. 16/324,339. cited by applicant

Notice of Allowance dated Jul. 29, 2020, which issued during the prosecution of U.S. Appl. No. 16/132,937. cited by applicant

An Office Action dated Jul. 29, 2020, which issued during the prosecution of U.S. Appl. No. 16/269,328. cited by applicant

Notice of Allowance dated Aug. 26, 2020, which issued during the prosecution of U.S. Appl. No. 16/269,328. cited by applicant

An Office Action dated Aug. 7, 2020, which issued during the prosecution of U.S. Appl. No. 15/668,659. cited by applicant

Tchetché, D. and Nicolas M. Van Mieghem: “New-generation TAVI devices: description and specifications” EuroIntervention, 2014, No. 10:U90-U100. cited by applicant

An Office Action dated Aug. 23, 2019, which issued during the prosecution of U.S. Appl. No. 15/600,190. cited by applicant

Symetis S.A.: “ACURATE neo™ Aortic Bioprosthesis for Implantation using the ACURATE neo™ TA Transapical Delivery System in Patients with Severe Aortic Stenosis,” Clinical Investigation Plan, Protocol No. 2015-01, Vs. No. 2, 2015:1-76. cited by applicant

Notice of Allowance dated Sep. 10, 2020, which issued during the prosecution of U.S. Appl. No. 15/600,190. cited by applicant

Notice of Allowance dated Sep. 10, 2020, which issued during the prosecution of U.S. Appl. No. 16/324,339. cited by applicant

Notice of Allowance dated Oct. 19, 2020, which issued during the prosecution of U.S. Appl. No. 16/324,339. cited by applicant

Notice of Allowance dated Sep. 21, 2020, which issued during the prosecution of U.S. Appl. No. 16/269,328. cited by applicant

Notice of Allowance dated Oct. 28, 2020, which issued during the prosecution of U.S. Appl. No. 16/269,328. cited by applicant

Notice of Allowance dated Jan. 16, 2020, which issued during the prosecution of U.S. Appl. No. 15/872,501. cited by applicant

An Office Action dated May 11, 2020, which issued during the prosecution of U.S. Appl. No. 16/811,732. cited by applicant

An Office Action dated Sep. 24, 2020, which issued during the prosecution of U.S. Appl. No. 16/811,732. cited by applicant

Notice of Allowance dated Mar. 29, 2017, which issued during the prosecution of U.S. Appl. No. 14/161,921. cited by applicant

Agarwal et al. International Cardiology Perspective Functional Tricuspid Regurgitation, Circ Cardiovasc Interv 2009;2;2:565-573 (2009). cited by applicant

Alfieri et al., “An effective technique to correct anterior mitral leaflet prolapse,” J Card 14(6):468-470 (1999). cited by applicant

Alfieri et al., “The double orifice technique in mitral valve repair: a simple solution for complex problems,” Journal of Thoracic Cardiovascular Surgery 122:674-681 (2001). cited by applicant

Alfieri, “The edge-to-edge repair of the mitral valve,” [Abstract] 6th Annual NewEra Cardiac Care: Innovation & Technology, Heart Surgery Forum pp. 103. (2000). cited by applicant

Alfieri et al. “Novel Suture Device for Beating-Heart Mitral Leaflet Approximation”, Ann Thorac Surg. 2002, 74:1488-1493. cited by applicant

Alfieri et al., “The edge to edge technique,” The European Association for Cardio-Thoracic Surgery 14.SUP.th .Annual Meeting Oct. 7-11, Book of Procees. (2000). cited by applicant

Amplatzer Cardiac Plug brochure (English pages), AGA Medical Corporation (Plymouth, MN) (copyright 2008-2010, downloaded Jan. 11, 2011). cited by applicant

Amplatzer® Cribriform Occluder. A patient guide to Percutaneous, Transcatheter, Atrial Septal Defect Closuer, AGA Medical Corporation, Apr. 2008. cited by applicant

Amplatzer® Septal Occluder. A patient guide to the Non-Surgical Closuer of the Atrial Septal Defect Using the Amplatzer Septal Occluder System, AGA Medical Corporation, Apr. 2008. cited by applicant

Brennan, Jennifer, 510(k) Summary of safety and effectiveness, Jan. 2008. cited by applicant

Dictionary.com definition of “lock”, Jul. 29, 2013. cited by applicant

Dang NC et al. “Simplified Placement of Multiple Artificial Mitral Valve Chords,” The Heart Surgery Forum #2005-1005, 8(3) (2005). cited by applicant

Maisano, The double-orifice technique as a standardized approach to treat mitral . . . , European Journal of Cardio-thoracic Surgery 17 (2000) 201-205. cited by applicant

“Two dimensional real-time ultrasonic imaging of the heart and great vessels”, Mayo Clin Proc. vol. 53:271-303, 1978. cited by applicant

Odell JA et al., "Early Results of a Simplified Method of Mitral Valve Annuloplasty," *Circulation* 92:150-154 (1995). cited by applicant

O'Reilly S et al., "Heart valve surgery pushes the envelope," *Medtech Insight* 8(3): 73, 99-108 (2006). cited by applicant

Swain CP et al., "An endoscopically deliverable tissue-transfixing device for securing biosensors in the gastrointestinal tract," *Gastrointestinal Endoscopy* 40(6): 730-734 (1994). cited by applicant

An Invitation to pay additional fees dated Jan. 31, 2014, which issued during the prosecution of Applicant's PCT/IL2013/050860. cited by applicant

U.S. Appl. No. 62/030,715, filed Jul. 30, 2014. cited by applicant

U.S. Appl. No. 62/139,854, filed Mar. 30, 2015. cited by applicant

U.S. Appl. No. 61/312,412, filed Mar. 10, 2010. cited by applicant

An Invitation to pay additional fees dated Jan. 31, 2014, which issued during the prosecution of Applicant's PCT/IL2013/050861. cited by applicant

An International Preliminary Report on Patentability dated Dec. 23, 2013, which issued during the prosecution of Applicant's PCT/IL2012/000250. cited by applicant

An International Preliminary Report on Patentability dated Sep. 18, 2007, which issued during the prosecution of Applicant's PCT/IL2006/000342. cited by applicant

An International Preliminary Report on Patentability dated Jun. 5, 2012, which issued during the prosecution of Applicant's PCT/IL2010/001024. cited by applicant

An International Preliminary Report on Patentability dated Apr. 28, 2015, which issued during the prosecution of Applicant's PCT/IL2013/050861. cited by applicant

An International Preliminary Report on Patentability dated Apr. 26, 2016, which issued during the prosecution of Applicant's PCT/IL2014/050914. cited by applicant

An International Preliminary Report on Patentability dated Jun. 10, 2009, which issued during the prosecution of Applicant's PCT/IL07/01503. cited by applicant

An International Preliminary Report on Patentability dated Dec. 18, 2010, which issued during the prosecution of Applicant's PCT/IL09/00593. cited by applicant

An International Preliminary Report on Patentability dated Jun. 29, 2011, which issued during the prosecution of Applicant's PCT/IL2009/001209. cited by applicant

Notice of Allowance dated Aug. 18, 2017, which issued during the prosecution of U.S. Appl. No. 14/689,608. cited by applicant

Notice of Allowance dated Jul. 6, 2017, which issued during the prosecution of U.S. Appl. No. 14/689,608. cited by applicant

Notice of Allowance dated May 22, 2017, which issued during the prosecution of U.S. Appl. No. 14/689,608. cited by applicant

An Office Action dated Apr. 21, 2017, which issued during the prosecution of U.S. Appl. No. 15/213,791. cited by applicant

An Office Action dated Sep. 29, 2017, which issued during the prosecution of U.S. Appl. No. 15/197,069. cited by applicant

An International Preliminary Report on Patentability dated Nov. 9, 2011, which issued during the prosecution of Applicant's PCT/IL2010/000357. cited by applicant

An International Preliminary Report on Patentability dated Nov. 9, 2011 which issued during the prosecution of Applicant's PCT/IL2010/000358. cited by applicant

An International Preliminary Report on Patentability dated Nov. 27, 2012, which issued during the prosecution of Applicant's PCT/IL2011/000404. cited by applicant

An International Preliminary Report on Patentability dated Feb. 4, 2014, which issued during the prosecution of Applicant's PCT/IL2011/000446. cited by applicant

An International Preliminary Report on Patentability dated Jan. 29, 2013, which issued during the prosecution of Applicant's PCT/IL2011/000600. cited by applicant

An International Preliminary Report on Patentability dated Dec. 23, 2014, which issued during the prosecution of Applicant's PCT/IL2012/050451. cited by applicant

Notice of Allowance dated Jul. 30, 2015, which issued during the prosecution of U.S. Appl. No. 13/319,007. cited by applicant

An Office Action dated Sep. 29, 2014, which issued during the prosecution of U.S. Appl. No. 13/504,870. cited by applicant

An Office Action dated Jan. 13, 2015, which issued during the prosecution of U.S. Appl. No. 13/707,013. cited by applicant

An Office Action dated Mar. 23, 2015, which issued during the prosecution of U.S. Appl. No. 13/707,013. cited by applicant

Notice of Allowance dated Mar. 25, 2015, which issued during the prosecution of U.S. Appl. No. 13/749,153. cited by applicant

An Office Action dated Oct. 3, 2014, which issued during the prosecution of U.S. Appl. No. 13/749,153. cited by applicant

Notice of Allowance dated May 22, 2015, which issued during the prosecution of U.S. Appl. No. 13/749,153. cited by applicant

Notice of Allowance dated Aug. 3, 2015, which issued during the prosecution of U.S. Appl. No. 13/749,153. cited by applicant

An Office Action dated Dec. 19, 2013, which issued during the prosecution of U.S. Appl. No. 14/027,934. cited by applicant

Notice of Allowance dated Aug. 24, 2022, which issued during the prosecution of U.S. Appl. No. 16/879,952. cited by applicant

Notice of Allowance dated May 12, 2023, which issued during the prosecution of U.S. Appl. No. 16/896,858. cited by applicant

Notice of Allowance dated Nov. 2, 2023, which issued during the prosecution of U.S. Appl. No. 17/466,785. cited by applicant

Notice of Allowance dated Oct. 17, 2023, which issued during the prosecution of U.S. Appl. No. 17/466,785. cited by applicant

An Office Action dated Jun. 10, 2022, which issued during the prosecution of U.S. Appl. No. 16/768,909. cited by applicant

An Office Action dated Jun. 11, 2014, which issued during the prosecution of U.S. Appl. No. 14/027,934. cited by applicant

An Office Action dated Aug. 22, 2014, which issued during the prosecution of U.S. Appl. No. 14/027,934. cited by applicant

An Office Action dated Apr. 2, 2015, which issued during the prosecution of U.S. Appl. No. 14/027,934. cited by applicant

An Office Action dated Jan. 5, 2016, which issued during the prosecution of U.S. Appl. No. 14/027,934. cited by applicant

An Office Action dated Jan. 5, 2016, which issued during the prosecution of U.S. Appl. No. 14/084,426. cited by applicant

An Office Action dated Mar. 16, 2015, which issued during the prosecution of U.S. Appl. No. 14/084,426. cited by applicant

An Office Action dated Jan. 6, 2016, which issued during the prosecution of U.S. Appl. No. 14/128,756. cited by applicant

An Office Action dated May 11, 2016, which issued during the prosecution of U.S. Appl. No. 14/128,756. cited by applicant

Notice of Allowance dated Oct. 20, 2015, which issued during the prosecution of U.S. Appl. No. 12/996,954. cited by applicant

Notice of Allowance dated Feb. 19, 2014, which issued during the prosecution of U.S. Appl. No. 12/795,192. cited by applicant

An Office Action dated Jul. 20, 2012, which issued during the prosecution of U.S. Appl. No. 12/843,412. cited by applicant

An Office Action dated Mar. 27, 2013, which issued during the prosecution of U.S. Appl. No. 12/843,412. cited by applicant

A Restriction Requirement dated May 1, 2012, which issued during the prosecution of U.S. Appl. No. 12/843,412. cited by applicant

A Notice of Allowance dated May 2, 2013, which issued during the prosecution of U.S. Appl. No. 12/843,412. cited by applicant

A Restriction Requirement dated Nov. 19, 2012, which issued during the prosecution of U.S. Appl. No. 12/926,673. cited by applicant

An Office Action dated Feb. 12, 2013, which issued during the prosecution of U.S. Appl. No. 12/926,673. cited by applicant

applicant

An Office Action dated Oct. 22, 2013, which issued during the prosecution of U.S. Appl. No. 12/926,673. cited by applicant

A Notice of Allowance dated Jan. 7, 2014, which issued during the prosecution of U.S. Appl. No. 12/926,673. cited by applicant

An Office Action dated Jul. 6, 2022, which issued during the prosecution of U.S. Appl. No. 16/656,790. cited by applicant

An International Preliminary Report on Patentability dated Jul. 28, 2022, which issued during the prosecution of Applicant's PCT/IL2021/050132. cited by applicant

An Office Action dated Oct. 9, 2013, which issued during the prosecution of U.S. Appl. No. 12/996,954. cited by applicant

An Office Action dated Mar. 24, 2015, which issued during the prosecution of U.S. Appl. No. 12/996,954. cited by applicant

An Office Action dated Oct. 5, 2012, which issued during the prosecution of U.S. Appl. No. 12/996,954. cited by applicant

Notice of Allowance dated Jul. 7, 2015, which issued during the prosecution of U.S. Appl. No. 12/996,954. cited by applicant

An Office Action dated Nov. 16, 2018, which issued during the prosecution of U.S. Appl. No. 16/042,028. cited by applicant

An International Search Report with Written Opinion both dated Feb. 2, 2012, which issued during the prosecution of Applicant's PCT/IL2011/000600. cited by applicant

An International Search Report together with Written Opinion both dated Mar. 30, 2011, which issued during the prosecution of Applicant's PCT/IL2010/001024. cited by applicant

An International Search Report and A Written Opinion both dated Feb. 10, 2011, which issued during the prosecution of Applicant's PCT/IL10/00890. cited by applicant

An Office Action dated May 28, 2015, which issued during the prosecution of U.S. Appl. No. 14/128,756. cited by applicant

An Office Action dated Sep. 6, 2018, which issued during the prosecution of U.S. Appl. No. 15/994,022. cited by applicant

An Office Action dated Sep. 7, 2018, which issued during the prosecution of U.S. Appl. No. 15/995,725. cited by applicant

An Office Action dated Nov. 26, 2018, which issued during the prosecution of U.S. Appl. No. 16/040,831. cited by applicant

An Office Action dated Jul. 11, 2018, which issued during the prosecution of U.S. Appl. No. 15/978,494. cited by applicant

An Office Action dated Nov. 23, 2018, which issued during the prosecution of U.S. Appl. No. 16/041,208. cited by applicant

An Office Action dated Jun. 15, 2018, which issued during the prosecution of U.S. Appl. No. 15/970,314. cited by applicant

An Office Action dated Oct. 12, 2018, which issued during the prosecution of U.S. Appl. No. 15/970,314. cited by applicant

An Office Action dated Jul. 26, 2018, which issued during the prosecution of U.S. Appl. No. 15/979,686. cited by applicant

An Office Action dated Sep. 10, 2018, which issued during the prosecution of U.S. Appl. No. 16/008,618. cited by applicant

An International Preliminary Report on Patentability dated Apr. 28, 2015, which issued during the prosecution of Applicant's PCT/IL2013/050860. cited by applicant

Notice of Allowance dated Jun. 22, 2023, which issued during the prosecution of U.S. Appl. No. 17/141,853. cited by applicant

An Office Action dated Apr. 22, 2019, which issued during the prosecution of U.S. Appl. No. 15/668,559. cited by applicant

Notice of Allowance dated Aug. 30, 2019, which issued during the prosecution of U.S. Appl. No. 15/682,789. cited by applicant

Notice of Allowance dated Mar. 29, 2018, which issued during the prosecution of U.S. Appl. No. 15/541,783. cited by applicant

Dieter RS, "Percutaneous valve repair: Update on mitral regurgitation and endovascular approaches to the mitral valve," Applications in Imaging, Cardiac Interventions, Supported by an educational grant from Amersham Health pp. 11-14 (2003). cited by applicant

An Advisory Action dated Dec. 13, 2013, which issued during the prosecution of U.S. Appl. No. 12/961,721. cited by applicant

An Office Action dated Aug. 7, 2015, which issued during the prosecution of U.S. Appl. No. 14/128,756. cited by applicant

An Office Action dated May 19, 2011, which issued during the prosecution of U.S. Appl. No. 12/706,868. cited by applicant

An Office Action dated Sep. 1, 2011, which issued during the prosecution of U.S. Appl. No. 12/706,868. cited by applicant

An Office Action dated May 30, 2012, which issued during the prosecution of U.S. Appl. No. 12/706,868. cited by applicant

A Notice of Allowance dated Sep. 18, 2012, which issued during the prosecution of U.S. Appl. No. 12/706,868. cited by applicant

Restriction Requirement dated May 5, 2011, which issued during the prosecution of U.S. Appl. No. 12/706,868. cited by applicant

A Restriction Requirement dated Mar. 30, 2012, which issued during the prosecution of U.S. Appl. No. 12/785,717. cited by applicant

An Office Action dated Oct. 5, 2020, which issued during the prosecution of Canadian Patent Application No. 2,973,940. cited by applicant

An Office Action dated Nov. 30, 2020, which issued during the prosecution of U.S. Appl. No. 16/138,129. cited by applicant

An Office Action summarized English translation and Search Report dated Nov. 25, 2020, which issued during the prosecution of Chinese Patent Application No. 201910449820.1. cited by applicant

Notice of Allowance dated Nov. 19, 2020, which issued during the prosecution of U.S. Appl. No. 16/318,025. cited by applicant

An Office Action dated Aug. 2, 2011, which issued during the prosecution of U.S. Appl. No. 12/435,291. cited by applicant

Notice of Allowance dated Dec. 7, 2011, which issued during the prosecution of U.S. Appl. No. 12/435,291. cited by applicant

An Office Action dated Apr. 6, 2010, which issued during the prosecution of Applicant's U.S. Appl. No. 12/484,512. cited by applicant

An Office Action dated Oct. 6, 2010, which issued during the prosecution of Applicant's U.S. Appl. No. 12/484,512. cited by applicant

Notice of Allowance dated Apr. 20, 2011, which issued during the prosecution of U.S. Appl. No. 12/484,512. cited by applicant

Notice of Allowance dated Mar. 23, 2011, which issued during the prosecution of U.S. Appl. No. 12/484,512. cited by applicant

An Office Action dated Jan. 27, 2012, which issued during the prosecution of U.S. Appl. No. 12/548,991. cited by applicant

An Office Action dated Aug. 6, 2012, which issued during the prosecution of U.S. Appl. No. 12/548,991. cited by applicant

An Advisory Action dated Sep. 6, 2012 which issued during the prosecution of U.S. Appl. No. 12/548,991. cited by applicant

Notice of Allowance dated Jun. 23, 2014, which issued during the prosecution of U.S. Appl. No. 12/548,991. cited by applicant

A Restriction Requirement dated Nov. 14, 2011 which issued during the prosecution of U.S. Appl. No. 12/548,991. cited by applicant

Amendment, Terminal Disclaimer and Extension dated Jun. 27, 2012, which issued during the prosecution of U.S. Appl. No. 12/548,991. cited by applicant

A Restriction Requirement dated Jul. 5, 2012, which issued during the prosecution of U.S. Appl. No. 12/563,930. cited by applicant

An Office Action dated Apr. 2, 2013, which issued during the prosecution of U.S. Appl. No. 12/785,717. cited by applicant

An Office Action dated Dec. 27, 2013, which issued during the prosecution of U.S. Appl. No. 12/785,717. cited by applicant

An Office Action dated Nov. 5, 2012, which issued during the prosecution of U.S. Appl. No. 12/795,026. cited by applicant

An Office Action dated May 10, 2012, which issued during the prosecution of U.S. Appl. No. 12/795,026. cited by applicant

Notice of Allowance dated Nov. 13, 2014, which issued during the prosecution of U.S. Appl. No. 12/795,026. cited by applicant

Notice of Allowance dated Dec. 24, 2014, which issued during the prosecution of U.S. Appl. No. 12/795,026. cited by applicant

A Restriction Requirement dated Jan. 6, 2012, which issued during the prosecution of U.S. Appl. No. 12/795,026. cited by applicant

A Restriction Requirement dated Sep. 14, 2012, which issued during the prosecution of U.S. Appl. No. 12/795,192. cited by applicant

An Office Action dated Aug. 15, 2013, which issued during the prosecution of U.S. Appl. No. 12/795,192. cited by applicant

An Office Action dated Jan. 17, 2013, which issued during the prosecution of U.S. Appl. No. 12/795,192. cited by applicant

Notice of Allowance dated Nov. 19, 2013, which issued during the prosecution of U.S. Appl. No. 12/795,192. cited by applicant

A Notice of Allowance dated Jun. 26, 2012, which issued during the prosecution of U.S. Appl. No. 12/608,316. cited by applicant

An Office Action dated Nov. 14, 2011, which issued during the prosecution of U.S. Appl. No. 12/608,316. cited by applicant

A Restriction Requirement dated Apr. 1, 2011, which issued during the prosecution of U.S. Appl. No. 12/608,316. cited by applicant

An Office Action dated Jul. 6, 2012, which issued during the prosecution of U.S. Appl. No. 12/692,061. cited by applicant

An Office Action dated Jan. 23, 2012, which issued during the prosecution of U.S. Appl. No. 12/692,061. cited by applicant

An Office Action dated Mar. 9, 2012, which issued during the prosecution of U.S. Appl. No. 12/689,635. cited by applicant

An Office Action dated Nov. 30, 2012, which issued during the prosecution of U.S. Appl. No. 12/689,635. cited by applicant

A Notice of Allowance dated May 22, 2013, which issued during the prosecution of U.S. Appl. No. 12/689,635. cited by applicant

Restriction Requirement dated Nov. 14, 2011, which issued during the prosecution of U.S. Appl. No. 12/689,635. cited by applicant

An Office Action dated May 6, 2013, which issued during the prosecution of U.S. Appl. No. 12/689,693. cited by applicant

An Office Action dated Feb. 3, 2014, which issued during the prosecution of U.S. Appl. No. 12/689,693. cited by applicant

Notice of Allowance dated Jun. 11, 2014, which issued during the prosecution of U.S. Appl. No. 12/689,693. cited by applicant

A Restriction Requirement dated Sep. 17, 2012, which issued during the prosecution of U.S. Appl. No. 12/689,693. cited by applicant

A Notice of Allowance dated Sep. 3, 2014, which issued during the prosecution of U.S. Appl. No. 12/689,693. cited by applicant

European Search Report dated Jul. 8, 2016, which issued during the prosecution of Applicant's European App No. 13849843.1. cited by applicant

A Supplementary European Search Report dated Dec. 4, 2012, which issued during the prosecution of European Patent Application No. EP 09834225.6. cited by applicant

A Supplementary European Search Report dated Mar. 28, 2013, which issued during the prosecution of European Patent Application No. EP 1077 2091.4. cited by applicant

Search Report in European Patent Application 10772090.6 dated Jan. 17, 2014. cited by applicant

Supplementary European Search Report dated Oct. 23, 2014 which issued during the prosecution of Applicant's European App No. 10826224.7. cited by applicant

Notice of Allowance dated May 6, 2016, which issued during the prosecution of U.S. Appl. No. 14/667,090. cited by applicant

Notice of Allowance dated Apr. 12, 2016, which issued during the prosecution of U.S. Appl. No. 14/667,090. cited by applicant

An Office Action dated Jun. 7, 2013, which issued during the prosecution of U.S. Appl. No. 13/141,606. cited by applicant

An Office Action dated Jun. 13, 2014, which issued during the prosecution of U.S. Appl. No. 13/141,606. cited by applicant

Notice of Allowance dated Sep. 29, 2014, which issued during the prosecution of U.S. Appl. No. 13/141,606. cited by applicant

An Office Action dated Feb. 4, 2013, which issued during the prosecution of U.S. Appl. No. 13/141,606. cited by applicant

An English translation of an Office Action dated Apr. 23, 2014 which issued during the prosecution of Chinese Patent Application No. 201080059948.4. cited by applicant

Communication dated Jul. 25, 2014, issued by the State Intellectual Property Office of the P.R. of China in counterpart Application No. 200980157331.3. cited by applicant

An International Search Report and a Written Opinion both dated Jan. 25, 2016, which issued during the prosecution of Applicant's PCT/IL2015/051027. cited by applicant

An International Search Report dated May 19, 2011, which issued during the prosecution of Applicant's PCT/IL2011/00064. cited by applicant

An International Search Report and a Written Opinion both dated Feb. 22, 2013, which issued during the prosecution of Applicant's PCT/IL201/050451. cited by applicant

An International Search Report & Written Opinion both dated Mar. 21, 2014, which issued during the prosecution of Applicant's PCT/IL13/50992. cited by applicant

An International Search Report and Written Opinion both dated Apr. 9, 2014, which issued during the prosecution of Applicant's PCT/IL13/50860. cited by applicant

An International Search Report and a Written Opinion both dated Apr. 15, 2014, which issued during the prosecution of Applicant's PCT/IL2013/050861. cited by applicant

An International Search Report & Written Opinion both dated May 12, 2015, which issued during the prosecution of Applicant's PCT/IL2014/050914. cited by applicant

An International Search Report and a Written Opinion both dated May 30, 2007, which issued during the prosecution of Applicant's PCT/IL2006/000342. cited by applicant

An International Search Report and a Written Opinion both dated Jun. 10, 2010, which issued during the prosecution of Applicant's PCT/IL09/01209. cited by applicant

An International Search Report and a Written Opinion both dated Aug. 17, 2010, which issued during the prosecution of Applicant's PCT/IL10/00357. cited by applicant

An International Search Report & Written Opinion both dated Sep. 8, 2009, which issued during the prosecution of Applicant's PCT/IL09/00593. cited by applicant

An International Search Report and a Written Opinion both dated Sep. 12, 2008, which issued during the prosecution of Applicant's PCT/IL07/01503. cited by applicant

An International Search Report and Written Opinion dated Nov. 8, 2010, which issued during the prosecution of Applicant's PCT/IL2010/000358. cited by applicant

An International Search Report and a Written Opinion both dated Nov. 23, 2011, which issued during the prosecution of Applicant's PCT/IL2011/000446. cited by applicant

Supplementary European Search Report dated Sep. 25, 2015, which issued during the prosecution of Applicant's European App No. 09794095.1. cited by applicant

A Supplementary European Search Report dated Feb. 1, 2011, which issued during the prosecution of European Patent Application No. EP 07849540. cited by applicant

An English translation of an Office Action dated Dec. 12, 2013 which issued during the prosecution of Chinese Patent Application No. 200980157331.3. cited by applicant

Communication regarding amended claims filed dated Dec. 27, 2012, regarding European App No. 11792047.0. cited by applicant

An Office Action dated Mar. 23, 2015, which issued during the prosecution of European Patent Application No. EP

09834225.6. cited by applicant

An English translation of an Office Action dated Jul. 17, 2015 which issued during the prosecution of Chinese Patent Application No. 201080059948.4. cited by applicant

An English translation of an Office Action dated Dec. 16, 2015 which issued during the prosecution of Chinese Patent Application No. 201080059948.4. cited by applicant

Communication from the European Patent Office dated Jun. 11, 2015, which issued during the prosecution of European Patent Application No. 11811934. cited by applicant

A communication from the European Patent Office dated Sep. 28, 2011 which issued during the prosecution of European Application No. 09834225.6. cited by applicant

A communication from the European Patent Office dated Oct. 19, 2012 which issued during the prosecution of European Application No. 11792047.0. cited by applicant

An Office Action dated Oct. 23, 2012, which issued during the prosecution of Japanese Patent Application No. 2009-539871. cited by applicant

An English Translation of an Office Action dated Nov. 24, 2015, which issued during the prosecution of Israel Patent Application No. 223448. (the relevant part only). cited by applicant

Notice of Allowance dated Nov. 17, 2015, which issued during the prosecution of U.S. Appl. No. 14/486,226. cited by applicant

Notice of Allowance dated Jan. 29, 2016, which issued during the prosecution of U.S. Appl. No. 14/551,951. cited by applicant

An Office Action dated Jun. 18, 2015, which issued during the prosecution of U.S. Appl. No. 14/551,951. cited by applicant

An Office Action dated Jan. 4, 2016, which issued during the prosecution of U.S. Appl. No. 14/589,100. cited by applicant

An Office Action dated May 4, 2016, which issued during the prosecution of U.S. Appl. No. 14/589,100. cited by applicant

An International Search Report and a Written Opinion both dated Nov. 14, 2011, which issued during the prosecution of Applicant's PCT/IL2011/000404. cited by applicant

An International Search Report and a Written Opinion both dated Dec. 6, 2012 which issued during the prosecution of Applicant's PCT/IL2012/000250. cited by applicant

A Notice of Allowance dated Apr. 3, 2013, which issued during the prosecution of U.S. Appl. No. 12/563,930. cited by applicant

An Office Action dated Aug. 24, 2012, which issued during the prosecution of U.S. Appl. No. 12/563,930. cited by applicant

An Office Action dated Dec. 29, 2011, which issued during the prosecution of U.S. Appl. No. 12/563,952. cited by applicant

A Restriction Requirement dated Oct. 27, 2011, which issued during the prosecution of U.S. Appl. No. 12/563,952. cited by applicant

A Notice of Allowance dated May 24, 2012, which issued during the prosecution of U.S. Appl. No. 12/563,952. cited by applicant

An Office Action dated Apr. 1, 2013 which issued during the prosecution of U.S. Appl. No. 13/167,476. cited by applicant

An Office Action dated Nov. 21, 2013, which issued during the prosecution of U.S. Appl. No. 13/167,476. cited by applicant

An Advisory Action dated Feb. 4, 2014, which issued during the prosecution of U.S. Appl. No. 13/167,476. cited by applicant

A Restriction Requirement dated Oct. 25, 2012 which issued during the prosecution of U.S. Appl. No. 13/167,444. cited by applicant

An Office Action dated Jan. 17, 2013, which issued during the prosecution of U.S. Appl. No. 13/167,444. cited by applicant

An Office Action dated Aug. 26, 2014 which issued during the prosecution of U.S. Appl. No. 13/167,444. cited by applicant

An Office Action dated Aug. 23, 2013 which issued during the prosecution of U.S. Appl. No. 13/167,444. cited by applicant

Notice of Allowance dated Nov. 12, 2015, which issued during the prosecution of U.S. Appl. No. 13/319,007. cited by applicant

Notice of Allowance dated Jan. 7, 2016, which issued during the prosecution of U.S. Appl. No. 13/319,007. cited by applicant

An Office Action dated Oct. 2, 2013, which issued during the prosecution of U.S. Appl. No. 13/167,492. cited by applicant

A Restriction Requirement dated Nov. 2, 2012, which issued during the prosecution of U.S. Appl. No. 13/167,492. cited by applicant

An Office Action dated Feb. 14, 2013 which issued during the prosecution of U.S. Appl. No. 13/167,492. cited by applicant

Notice of Allowance dated Nov. 7, 2014, which issued during the prosecution of U.S. Appl. No. 13/167,492. cited by applicant

An Office Action dated Jun. 10, 2014, which issued during the prosecution of U.S. Appl. No. 13/167,492. cited by applicant

Notice of Allowance dated Dec. 9, 2014, which issued during the prosecution of U.S. Appl. No. 13/167,476. cited by applicant

Notice of Allowance dated Jan. 22, 2015, which issued during the prosecution of U.S. Appl. No. 13/167,444. cited by applicant

An International Preliminary Report on Patentability dated May 1, 2012, which issued during the prosecution of Applicant's PCT/IL2010/000890. cited by applicant

An International Preliminary Report on Patentability dated Jun. 9, 2015, which issued during the prosecution of Applicant's PCT/IL2013/050992. cited by applicant

U.S. Appl. No. 60/873,075, filed Dec. 5, 2006. cited by applicant

U.S. Appl. No. 60/902,146, filed Feb. 16, 2007. cited by applicant

An Office Action dated Mar. 29, 2018, which issued during the prosecution of U.S. Appl. No. 15/188,507. cited by applicant

Notice of Allowance dated Sep. 17, 2014, which issued during the prosecution of U.S. Appl. No. 12/961,721. cited by applicant

An Office Action dated Oct. 1, 2015, which issued during the prosecution of U.S. Appl. No. 14/141,228. cited by applicant

A Restriction Requirement dated Jun. 2, 2014, which issued during the prosecution of U.S. Appl. No. 13/319,030. cited by applicant

An Office Action dated Oct. 14, 2014, which issued during the prosecution of U.S. Appl. No. 13/319,030. cited by applicant

An Office Action dated Jun. 18, 2015, which issued during the prosecution of U.S. Appl. No. 13/319,030. cited by applicant

An Office Action dated May 3, 2016, which issued during the prosecution of U.S. Appl. No. 13/319,030. cited by applicant

Notice of Allowance dated Dec. 30, 2016, which issued during the prosecution of U.S. Appl. No. 13/319,030. cited by applicant

An Office Action dated Apr. 7, 2015, which issued during the prosecution of U.S. Appl. No. 13/319,007. cited by applicant

An Office Action dated Apr. 8, 2016, which issued during the prosecution of U.S. Appl. No. 14/141,228. cited by applicant

An Office Action dated Oct. 5, 2015, which issued during the prosecution of U.S. Appl. No. 14/246,417. cited by applicant

An Office Action dated Apr. 7, 2016, which issued during the prosecution of U.S. Appl. No. 14/242,151. cited by applicant

An Office Action dated May 23, 2016, which issued during the prosecution of U.S. Appl. No. 14/209,171. cited by applicant

An Office Action dated Jul. 20, 2016, which issued during the prosecution of U.S. Appl. No. 14/246,417. cited by applicant

An Office Action dated Jun. 14, 2016, which issued during the prosecution of U.S. Appl. No. 14/273,155. cited by applicant

An Office Action dated Jun. 17, 2016, which issued during the prosecution of U.S. Appl. No. 14/357,040. cited by applicant

An Office Action dated Mar. 24, 2015, which issued during the prosecution of U.S. Appl. No. 14/486,226. cited by applicant

applicant

U.S. Appl. No. 61/001,013, filed Oct. 29, 2007. cited by applicant

U.S. Appl. No. 61/132,295, filed Jun. 16, 2008. cited by applicant

U.S. Appl. No. 61/265,936, filed Dec. 2, 2009. cited by applicant

U.S. Appl. No. 61/283,445, filed Dec. 2, 2009. cited by applicant

U.S. Appl. No. 61/207,908, filed Feb. 17, 2009. cited by applicant

U.S. Appl. No. 61/733,979, filed Dec. 6, 2012. cited by applicant

U.S. Appl. No. 61/717,303, filed Oct. 23, 2012. cited by applicant

U.S. Appl. No. 61/820,979, filed May 8, 2013. cited by applicant

U.S. Appl. No. 61/745,848, filed Dec. 6, 2012. cited by applicant

U.S. Appl. No. 61/555,570, filed Nov. 4, 2011. cited by applicant

U.S. Appl. No. 61/557,082, filed Nov. 8, 2011. cited by applicant

U.S. Appl. No. 60/662,616, filed Mar. 17, 2005. cited by applicant

U.S. Appl. No. 60/700,542, filed Jul. 18, 2005. cited by applicant

U.S. Appl. No. 61/782,121, filed Mar. 14, 2013. cited by applicant

European Search Report dated Jul. 15, 2016, which issued during the prosecution of Applicant's European App No. 13849947.0. cited by applicant

European Search Report dated Nov. 4, 2015, which issued during the prosecution of European Patent Application No. EP 1077 2091.4 cited by applicant

Search Report in European Patent Application 10826224.7 dated Nov. 16, 2015. cited by applicant

Supplementary European Search Report dated Dec. 23, 2014 which issued during the prosecution of Applicant's European App No. 10834311. cited by applicant

Supplementary European Search Report dated Jan. 21, 2014 which issued during the prosecution of Applicant's European App No. 11 78 6226. cited by applicant

A Supplementary European Search Report dated Jan. 20, 2015, which issued during the prosecution of European Patent Application No. 12803037.6. cited by applicant

Supplementary European Search Report dated Aug. 4, 2014 which issued during the prosecution of Applicant's European App No. 11 81 1934.6. cited by applicant

European Search Report dated Jun. 24, 2016, which issued during the prosecution of European Patent Application No. EP 12847363. cited by applicant

Supplementary European Search Report dated Apr. 29, 2015, which issued during the prosecution of Applicant's European App No. 14200202. cited by applicant

An Office Action dated Dec. 16, 2013, which issued during the prosecution of U.S. Appl. No. 13/666,262. cited by applicant

An Office Action dated Dec. 18, 2013, which issued during the prosecution of U.S. Appl. No. 13/666,141. cited by applicant

Notice of Allowance dated Jun. 25, 2014, which issued during the prosecution of U.S. Appl. No. 13/666,262. cited by applicant

A Notice of Allowance dated Feb. 2, 2015, which issued during the prosecution of U.S. Appl. No. 13/504,870. cited by applicant

Notice of Allowance dated Aug. 19, 2013, which issued during the prosecution of U.S. Appl. No. 11/908,906. cited by applicant

An Office Action dated Jun. 8, 2012, which issued during the prosecution of U.S. Appl. No. 11/908,906. cited by applicant

An Office Action dated Dec. 21, 2013, which issued during the prosecution of U.S. Appl. No. 11/908,906. cited by applicant

A Restriction Requirement dated Aug. 5, 2011, which issued during the prosecution of U.S. Appl. No. 11/908,906. cited by applicant

An Office Action dated Sep. 16, 2009 which issued during the prosecution of U.S. Appl. No. 11/950,930. cited by applicant

Notice of Allowance dated Sep. 12, 2014, which issued during the prosecution of U.S. Appl. No. 11/950,930. cited by applicant

An Office Action dated Aug. 5, 2010 which issued during the prosecution of U.S. Appl. No. 11/950,930. cited by applicant

An Office Action dated Feb. 17, 2010 which issued during the prosecution of U.S. Appl. No. 11/950,930. cited by applicant

applicant

A Restriction Requirement dated Apr. 19, 2010 which issued during the prosecution of U.S. Appl. No. 12/341,960. cited by applicant

An Office Action dated Sep. 28, 2011, which issued during the prosecution of U.S. Appl. No. 12/437,103. cited by applicant

An Office Action dated Jun. 13, 2012, which issued during the prosecution of U.S. Appl. No. 12/437,103. cited by applicant

A Restriction Requirement dated Jul. 12, 2011, which issued during the prosecution of U.S. Appl. No. 12/437,103. cited by applicant

Notice of Allowance dated Mar. 6, 2014, which issued during the prosecution of U.S. Appl. No. 12/437,103. cited by applicant

Notice of Allowance dated Dec. 20, 2013, which issued during the prosecution of U.S. Appl. No. 12/437,103. cited by applicant

Notice of Allowance dated Apr. 27, 2012, which issued during the prosecution of U.S. Appl. No. 12/341,960. cited by applicant

An Office Action dated Mar. 29, 2011, which issued during the prosecution of U.S. Appl. No. 12/341,960. cited by applicant

An Office Action dated Aug. 4, 2010, which issued during the prosecution of U.S. Appl. No. 12/341,960. cited by applicant

An Interview Summary dated Jul. 27, 2011, which issued during the prosecution of U.S. Appl. No. 12/341,960. cited by applicant

Notice of Allowance dated Aug. 21, 2019, which issued during the prosecution of U.S. Appl. No. 15/703,385. cited by applicant

Notice of Allowance dated Oct. 16, 2019, which issued during the prosecution of U.S. Appl. No. 15/703,385. cited by applicant

Notice of Allowance dated Dec. 24, 2020, which issued during the prosecution of U.S. Appl. No. 15/668,659. cited by applicant

Notice of Allowance dated Oct. 21, 2020, which issued during the prosecution of U.S. Appl. No. 15/668,659. cited by applicant

Declaration of Ivan Vesely, Ph.D., in Support of Petition for Inter Partes Review of U.S. Pat. No. 7,563,267—dated May 29, 2019. cited by applicant

U.S. Appl. No. 60/128,690, filed Apr. 9, 1999. cited by applicant

U.S. Appl. No. 60/613,867, filed Sep. 27, 2004. cited by applicant

An Office Action dated Dec. 24, 2020, which issued during the prosecution of U.S. Appl. No. 16/144,054. cited by applicant

An Office Action dated Feb. 2, 2021, which issued during the prosecution of U.S. Appl. No. 16/811,732. cited by applicant

An Office Action dated Jan. 13, 2021, which issued during the prosecution of European Patent Application No. 15751089.2. cited by applicant

An Office Action together with an English summary dated Mar. 3, 2021, which issued during the prosecution of Chinese Patent Application No. 201780047391.4. cited by applicant

Declaration of Dr. Ivan Vesely, Ph.D. in Support of Petition for Inter Partes Review of U.S. Pat. No. 10,226,341—dated Dec. 17, 2020. cited by applicant

Petition for Inter Partes Review of U.S. Pat. No. 10,226,341 and Exhibits 1001-1013—dated Dec. 29, 2020. cited by applicant

Batista, Randas JV, et al. “Partial left ventriculectomy to treat end-stage heart disease.” *The Annals of thoracic surgery* 64.3 (1997): 634-638. cited by applicant

Beall Jr, Arthur C., et al. “Clinical experience with a dacron velour-covered teflon-disc mitral-valve prosthesis.” *The Annals of thoracic surgery* 5.5 (1968): 402-410. cited by applicant

Kalbacher, D., et al. “1000 MitraClip™ procedures: Lessons learnt from the largest single-centre experience worldwide.” (2019): 3137-3139. cited by applicant

Maisano, F., et al. “The edge-to-edge technique: a simplified method to correct mitral insufficiency.” *European journal of cardio-thoracic surgery* 13.3 (1998): 240-246. cited by applicant

Fucci, C., et al. “Improved results with mitral valve repair using new surgical techniques.” *European journal of cardio-thoracic surgery* 9.11 (1995): 621-627. cited by applicant

Notice of Allowance dated Nov. 19, 2019, which issued during the prosecution of U.S. Appl. No. 15/668,559. cited by applicant

Mitral Valve Academic Research Consortium. "Clinical Trial Design Principles and Endpoint Definitions for Transcatheter Mitral Valve Repair and Replacement: Part 1: Clinical Trial Design Principles A Consensus Document from the Mitral Valve Academic Research Consortium." *Journal of the American College of Cardiology* 66.3 (2015): 278-307. cited by applicant

An Office Action dated Aug. 29, 2018, which issued during the prosecution of U.S. Appl. No. 15/329,920. cited by applicant

An Office Action dated May 8, 2018, which issued during the prosecution of U.S. Appl. No. 15/902,403. cited by applicant

An Office Action dated May 11, 2018, which issued during the prosecution of U.S. Appl. No. 15/899,858. cited by applicant

Notice of Allowance dated Oct. 5, 2018, which issued during the prosecution of U.S. Appl. No. 15/886,517. cited by applicant

Notice of Allowance dated Jul. 19, 2019, which issued during the prosecution of U.S. Appl. No. 15/899,858. cited by applicant

Notice of Allowance dated Nov. 16, 2020, which issued during the prosecution of U.S. Appl. No. 16/324,339. cited by applicant

Notice of Allowance dated Apr. 27, 2020, which issued during the prosecution of U.S. Appl. No. 16/591,330. cited by applicant

An Advisory Action dated Jan. 2, 2020, which issued during the prosecution of U.S. Appl. No. 15/668,659. cited by applicant

Notice of Allowance dated Oct. 17, 2019, which issued during the prosecution of U.S. Appl. No. 15/329,920. cited by applicant

An Office Action dated Dec. 31, 2019, which issued during the prosecution of U.S. Appl. No. 16/591,330. cited by applicant

Notice of Allowance dated Feb. 9, 2021, which issued during the prosecution of U.S. Appl. No. 16/937,216. cited by applicant

An Advisory Action dated Nov. 18, 2020, which issued during the prosecution of U.S. Appl. No. 16/811,732. cited by applicant

An International Search Report and a Written Opinion both dated Mar. 27, 2018, which issued during the prosecution of Applicant's PCT/IL2017/050849. cited by applicant

Notice of Allowance dated Jun. 11, 2021, which issued during the prosecution of U.S. Appl. No. 16/811,732. cited by applicant

Notice of Allowance dated Jul. 16, 2021, which issued during the prosecution of U.S. Appl. No. 16/811,732. cited by applicant

Patent Trial and Appeal Board Decision Granting Institution in U.S. Pat. No. 10,226,341—Dated Jul. 20, 2021. cited by applicant

European Search Report dated Jun. 10, 2021 which issued during the prosecution of Applicant's European App No. 21157988.3. cited by applicant

Notice of Allowance dated Nov. 19, 2018, which issued during the prosecution of U.S. Appl. No. 15/197,069. cited by applicant

Poirier, Nancy C., et al. "A novel repair for patients with atrioventricular septal defect requiring reoperation for left atrioventricular valve regurgitation." *European journal of cardio-thoracic surgery* 18.1 (2000): 54-61. cited by applicant

An Office Action dated Mar. 29, 2021, which issued during the prosecution of U.S. Appl. No. 16/738,516. cited by applicant

Ando, Tomo, et al. "Iatrogenic ventricular septal defect following transcatheter aortic valve replacement: a systematic review." *Heart, Lung and Circulation* 25.10 (2016): 968-974. cited by applicant

Urena, Marina, et al. "Transseptal transcatheter mitral valve replacement using balloon-expandable transcatheter heart valves: a step-by-step approach." *JACC: Cardiovascular Interventions* 10.19 (2017): 1905-1919. cited by applicant

An English summary of an Official Action dated Mar. 29, 2021, which issued during the prosecution of Chinese Patent Application No. 201780061210.3. cited by applicant

An International Search Report and a Written Opinion both dated Jan. 28, 2020, which issued during the

prosecution of Applicant's PCT/IL2019/051031. cited by applicant

An International Preliminary Report on Patentability dated Mar. 9, 2021, which issued during the prosecution of Applicant's PCT/IL2019/051031. cited by applicant

An Office Action dated May 4, 2021, which issued during the prosecution of U.S. Appl. No. 16/636,204. cited by applicant

Notice of Allowance dated May 17, 2021, which issued during the prosecution of U.S. Appl. No. 16/138,129. cited by applicant

Notice of Allowance dated Jun. 4, 2021, which issued during the prosecution of U.S. Appl. No. 16/802,353. cited by applicant

An Office Action dated May 12, 2021, which issued during the prosecution of Canadian Patent Application No. 2,973,940. cited by applicant

Petition for Inter Partes Review of U.S. Pat. No. 10,702,385—dated Jun. 4, 2021. cited by applicant

Declaration of Ivan Vesel Y, Ph.D. In Support of Petition for Inter Partes Review of U.S. Pat. No. 10,702,385—dated Jun. 4, 2021. cited by applicant

Notice of Allowance dated Oct. 30, 2018, which issued during the prosecution of U.S. Appl. No. 15/197,069. cited by applicant

An International Search Report and a Written Opinion both dated Jul. 12, 2021, which issued during the prosecution of Applicant's PCT/IL2021/050132. cited by applicant

Notice of Allowance dated Oct. 3, 2019, which issued during the prosecution of U.S. Appl. No. 15/691,032. cited by applicant

An Office Action dated Sep. 6, 2018, which issued during the prosecution of U.S. Appl. No. 15/213,791. cited by applicant

Condado, José Antonio, et al. "Percutaneous edge-to-edge mitral valve repair: 2-year follow-up in the first human case." Catheterization and cardiovascular interventions 67.2 (2006): 323-325. cited by applicant

Notice of Allowance dated Mar. 18, 2020, which issued during the prosecution of U.S. Appl. No. 16/284,331. cited by applicant

Feldman, Ted, et al. "Percutaneous mitral repair with the MitraClip system: safety and midterm durability in the initial EVEREST (Endovascular Valve Edge-to-Edge REpair Study) cohort." Journal of the American College of Cardiology 54.8 (2009): 686-694. cited by applicant

Notice of Allowance dated Nov. 21, 2018, which issued during the prosecution of U.S. Appl. No. 15/213,791. cited by applicant

Notice of Allowance dated Jul. 3, 2019, which issued during the prosecution of U.S. Appl. No. 15/691,032. cited by applicant

IPR2021-00383 Petitioners' Authorized Reply to Patent Owner's Preliminary Response dated May 27, 2021. cited by applicant

Exhibit 1014—Transcript of proceedings held May 20, 2021 (*Edwards Lifesciences vs. Cardiovalve*). cited by applicant

Exhibit 1015—Facilitate, Meriam-Webster.com, <https://www.merriamwebster.com/dictionary/facilitate> (visited May 26, 2021). cited by applicant

Patent Owner's Authorized Surreply to Petitioner's Reply to Patent Owner's Preliminary Response dated Jun. 4, 2021 (*Edwards Lifesciences vs. Cardiovalve*). cited by applicant

An Invitation to pay additional fees dated May 19, 2021, which issued during the prosecution of Applicant's PCT/IL2021/050132. cited by applicant

An Office Action dated Aug. 18, 2021, which issued during the prosecution of U.S. Appl. No. 17/210,183. cited by applicant

An Office Action dated Sep. 9, 2021, which issued during the prosecution of U.S. Appl. No. 16/768,909. cited by applicant

An Office Action dated Sep. 15, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,599. cited by applicant

Notice of Allowance dated Oct. 14, 2021, which issued during the prosecution of U.S. Appl. No. 16/680,739. cited by applicant

An Office Action dated Oct. 21, 2021, which issued during the prosecution of U.S. Appl. No. 17/335,845. cited by applicant

European Search Report dated Oct. 11, 2021 which issued during the prosecution of Applicant's European App No. 21176010.3. cited by applicant

Fann, James I., et al. "Beating heart catheter-based edge-to-edge mitral valve procedure in a porcine model: efficacy and healing response." *Circulation* 110.8 (2004): 988-993. cited by applicant

IPR2021-00383 Patent Owner's Contingent Motion to Amend Under 37 C.F.R. §42.121 dated Oct. 13, 2021. cited by applicant

IPR2021-00383 Patent Owner's Response Pursuant to 37 C.F.R. §42.120 dated Oct. 13, 2021. cited by applicant

IPR2021-00383 Second Declaration of Dr. Michael Sacks dated Oct. 13, 2021. cited by applicant

An Office Action dated Oct. 21, 2021, which issued during the prosecution of U.S. Appl. No. 17/306,231. cited by applicant

Maisano, Francesco, et al. "The evolution from surgery to percutaneous mitral valve interventions: the role of the edge-to-edge technique." *Journal of the American College of Cardiology* 58.21 (2011): 2174-2182. cited by applicant

An Office Action dated Nov. 6, 2015, which issued during the prosecution of U.S. Appl. No. 14/626,267. cited by applicant

An Office Action dated Jan. 26, 2022, which issued during the prosecution of U.S. Appl. No. 16/888,210. cited by applicant

IPR2021-00383 Deposition of Dr. Ivan Vesely, dated Sep. 22, 2021. cited by applicant

Cardiovalve Exhibit 2009—Percutaneous Mitral Leaflet Repair: MitraClip® Therapy for Mitral Regurgitation (2012). cited by applicant

Feldman, Ted, et al. "Percutaneous mitral valve repair using the edge-to-edge technique: six-month results of the EVEREST Phase I Clinical Trial." *Journal of the American College of Cardiology* 46.11 (2005): 2134-2140. cited by applicant

An Office Action summarized English translation and Search Report dated Oct. 8, 2021, which issued during the prosecution of Chinese Patent Application No. 201780061210.3. cited by applicant

An Office Action dated Nov. 4, 2021, which issued during the prosecution of U.S. Appl. No. 17/366,711. cited by applicant

An Office Action summarized English translation and Search Report dated Aug. 12, 2021, which issued during the prosecution of Chinese Patent Application No. 201880058940.2. cited by applicant

U.S. Appl. No. 63/120,808, filed Dec. 3, 2020. cited by applicant

An Office Action dated Nov. 25, 2021, which issued during the prosecution of European Patent Application No. 18826823.9. cited by applicant

An Office Action dated Oct. 4, 2023, which issued during the prosecution of Canadian Patent Application No. 3,031,187. cited by applicant

IPR2021-01051 Institution decision dated Dec. 10, 2021. cited by applicant

Notice of Allowance dated Dec. 7, 2021, which issued during the prosecution of U.S. Appl. No. 17/394,807. cited by applicant

Notice of Allowance dated Dec. 6, 2021, which issued during the prosecution of U.S. Appl. No. 16/738,516. cited by applicant

Notice of Allowance dated Dec. 29, 2021, which issued during the prosecution of U.S. Appl. No. 17/210,183. cited by applicant

IPR2021-00383 Petitioners' Reply to Patent Owner's Response dated Jan. 5, 2022. cited by applicant

IPR2021-00383 Petitioners' Opposition to Patent Owner's Contingent Motion To Amend dated Jan. 5, 2022. cited by applicant

An Office Action dated Sep. 22, 2021, which issued during the prosecution of European Patent Application No. 20714289.4. cited by applicant

Summary of Examination Notice dated Jan. 6, 2022, which issued during the prosecution of Chinese Patent Application No. 201880064313.X. cited by applicant

An Office Action dated Jan. 12, 2022, which issued during the prosecution of U.S. Appl. No. 17/101,787. cited by applicant

Notice of Allowance dated Jun. 20, 2017, which issued during the prosecution of U.S. Appl. No. 14/626,267. cited by applicant

Notice of Allowance dated Oct. 20, 2021, which issued during the prosecution of U.S. Appl. No. 16/636,204. cited by applicant

Notice of Allowance dated Jan. 31, 2022, which issued during the prosecution of U.S. Appl. No. 17/479,418. cited by applicant

An Office Action dated Jan. 13, 2022, which issued during the prosecution of U.S. Appl. No. 17/473,472. cited by applicant

applicant

An Office Action dated Apr. 11, 2022, which issued during the prosecution of U.S. Appl. No. 17/473,472. cited by applicant

IPR2021-00383 Preliminary Guidance dated Jan. 31, 2022. cited by applicant

An Office Action dated Mar. 18, 2022, which issued during the prosecution of U.S. Appl. No. 16/746,489. cited by applicant

Notice of Allowance dated Mar. 22, 2022, which issued during the prosecution of U.S. Appl. No. 17/366,711. cited by applicant

Notice of Allowance dated Mar. 4, 2022, which issued during the prosecution of U.S. Appl. No. 16/768,909. cited by applicant

An Office Action dated Dec. 9, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,969. cited by applicant

An Office Action dated Jan. 24, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,466. cited by applicant

An Office Action dated Nov. 3, 2023, which issued during the prosecution of Canadian Patent Application No. 3,162,308. cited by applicant

Notice of Allowance dated Jul. 6, 2023, which issued during the prosecution of U.S. Appl. No. 17/004,693. cited by applicant

Notice of Allowance dated Jul. 13, 2023, which issued during the prosecution of U.S. Appl. No. 17/004,693. cited by applicant

Notice of Allowance dated Jul. 10, 2023, which issued during the prosecution of U.S. Appl. No. 17/007,975. cited by applicant

Notice of Allowance dated Mar. 13, 2024, which issued during the prosecution of U.S. Appl. No. 18/216,391. cited by applicant

IPR2021-01051 Preliminary Guidance Patent Owner's Motion to Amend dated Jun. 24, 2022. cited by applicant

Ex Parte Quayle dated May 2, 2022, which issued during the prosecution of U.S. Appl. No. 16/879,952. cited by applicant

An International Search Report and a Written Opinion both dated May 3, 2022, which issued during the prosecution of Applicant's PCT/IL2021/051433. cited by applicant

An Office Action together with an English Summary dated May 7, 2022 which issued during the prosecution of Chinese Patent Application No. 201880058940.2. cited by applicant

Notice of Allowance dated May 4, 2022, which issued during the prosecution of U.S. Appl. No. 16/680,739. cited by applicant

An Office Action dated Jun. 28, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,969. cited by applicant

An Office Action dated Jul. 8, 2022, which issued during the prosecution of U.S. Appl. No. 16/144,054. cited by applicant

Notice of Allowance dated May 1, 2023, which issued during the prosecution of U.S. Appl. No. 17/007,975. cited by applicant

An Office Action dated Jul. 27, 2022, which issued during the prosecution of U.S. Appl. No. 16/881,350. cited by applicant

An Office Action dated Sep. 21, 2022, which issued during the prosecution of U.S. Appl. No. 16/776,581. cited by applicant

An Office Action dated Jul. 20, 2022, which issued during the prosecution of U.S. Appl. No. 17/101,787. cited by applicant

An Office Action dated Sep. 16, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,466. cited by applicant

An Office Action dated Aug. 1, 2022, which issued during the prosecution of European Patent Application No. 18826823.9. cited by applicant

European Search Report dated Sep. 6, 2022 which issued during the prosecution of Applicant's European App No. 22161862.2. cited by applicant

IPR2021-01051 Petitioners' Reply to Preliminary Guidance dated Aug. 2, 2022. cited by applicant

IPR2021-01051 Patent Owner's Sur-Reply To Petitioners' Reply to Preliminary Guidance dated Aug. 23, 2022. cited by applicant

An Office Action dated Aug. 5, 2022, which issued during the prosecution of U.S. Appl. No. 16/760,147. cited by

applicant

An Office Action dated Sep. 8, 2022, which issued during the prosecution of U.S. Appl. No. 16/896,858. cited by applicant

An Office Action dated Nov. 10, 2022, which issued during the prosecution of U.S. Appl. No. 17/007,975. cited by applicant

An Office Action dated Sep. 29, 2022, which issued during the prosecution of U.S. Appl. No. 17/010,886. cited by applicant

An Office Action dated Sep. 29, 2022, which issued during the prosecution of U.S. Appl. No. 16/656,790. cited by applicant

An Office Action dated Nov. 2, 2022, which issued during the prosecution of U.S. Appl. No. 17/004,693. cited by applicant

An Office Action dated Nov. 28, 2022, which issued during the prosecution of U.S. Appl. No. 17/141,853. cited by applicant

An Office Action dated Oct. 19, 2022, which issued during the prosecution of U.S. Appl. No. 17/875,589. cited by applicant

An Office Action dated Oct. 26, 2022, which issued during the prosecution of U.S. Appl. No. 16/746,489. cited by applicant

An International Preliminary Report on Patentability dated May 30, 2023, which issued during the prosecution of Applicant's PCT/IL2021/051433. cited by applicant

Notice of Allowance dated Dec. 21, 2022, which issued during the prosecution of U.S. Appl. No. 17/101,787. cited by applicant

Notice of Allowance dated Oct. 5, 2022, which issued during the prosecution of U.S. Appl. No. 17/101,787. cited by applicant

An Office Action dated Mar. 3, 2023, which issued during the prosecution of European Patent Application No. 17751143.3. cited by applicant

An Office Action dated Mar. 20, 2023, which issued during the prosecution of U.S. Appl. No. 17/181,722. cited by applicant

Notice of Allowance dated Dec. 4, 2023, which issued during the prosecution of U.S. Appl. No. 17/683,875. cited by applicant

European Search Report dated Mar. 20, 2023 which issued during the prosecution of Applicant's European App No. 22204764.9. cited by applicant

Notice of Allowance dated Apr. 6, 2023, which issued during the prosecution of U.S. Appl. No. 16/746,489. cited by applicant

An Office Action dated Apr. 14, 2023, which issued during the prosecution of U.S. Appl. No. 16/144,054. cited by applicant

An Office Action dated May 15, 2023, which issued during the prosecution of U.S. Appl. No. 16/656,790. cited by applicant

An Office Action dated May 16, 2023, which issued during the prosecution of U.S. Appl. No. 17/114,771. cited by applicant

An Office Action dated May 17, 2023, which issued during the prosecution of U.S. Appl. No. 17/466,785. cited by applicant

An Office Action dated May 25, 2023, which issued during the prosecution of U.S. Appl. No. 17/397,235. cited by applicant

Notice of Allowance dated Sep. 20, 2023, which issued during the prosecution of U.S. Appl. No. 17/839,538. cited by applicant

An Office Action dated Oct. 13, 2023, which issued during the prosecution of U.S. Appl. No. 17/181,722. cited by applicant

Grounds of Opposition to European Patent No. EP 2 948 103, filed Sep. 6, 2023. cited by applicant

An Office Action dated Aug. 31, 2023, which issued during the prosecution of U.S. Appl. No. 17/397,235. cited by applicant

An Office Action dated Sep. 8, 2023, which issued during the prosecution of U.S. Appl. No. 18/216,391. cited by applicant

An Office Action dated Sep. 8, 2023, which issued during the prosecution of U.S. Appl. No. 18/218,419. cited by applicant

An Office Action summarized English translation and Search Report dated Sep. 22, 2022, which issued during the

prosecution of Chinese Patent Application No. 201880058940.2. cited by applicant

An Office Action dated Jul. 24, 2024, which issued during the prosecution of U.S. Appl. No. 18/234,745. cited by applicant

Opposition to European Patent No. EP 2 948 103, filed Sep. 6, 2023. cited by applicant

An International Search Report and a Written Opinion both dated Aug. 23, 2023, 2023, which issued during the prosecution of Applicant's PCT/IL2023/050586. cited by applicant

An Office Action dated Aug. 3, 2023, which issued during the prosecution of U.S. Appl. No. 17/683,875. cited by applicant

An International Search Report and a Written Opinion both dated Sep. 13, 2023, which issued during the prosecution of Applicant's PCT/IL2023/050587. cited by applicant

An Invitation to pay additional fees dated Mar. 14, 2019, which issued during the prosecution of Applicant's PCT/IL2018/0051350. cited by applicant

An International Preliminary Report on Patentability dated Jun. 16, 2020, which issued during the prosecution of Applicant's PCT/IL2018/0051350. cited by applicant

Notice of Allowance dated Nov. 12, 2020, which issued during the prosecution of U.S. Appl. No. 16/269,328. cited by applicant

An Advisory Action dated Apr. 9, 2019, which issued during the prosecution of U.S. Appl. No. 15/970,314. cited by applicant

An Advisory Action dated Mar. 13, 2019, which issued during the prosecution of U.S. Appl. No. 15/979,686. cited by applicant

An Advisory Action dated Jul. 8, 2019, which issued during the prosecution of U.S. Appl. No. 15/995,725. cited by applicant

An Office Action dated Nov. 16, 2018, which issued during the prosecution of U.S. Appl. No. 16/042,129. cited by applicant

An Office Action dated Jul. 22, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,447. cited by applicant

An Advisory Action dated Jan. 2, 2020, which issued during the prosecution of U.S. Appl. No. 16/041,208. cited by applicant

IPR2021-00383 Final Written Decision Determining All Challenged Claims Unpatentable Denying Patent Owner's Contingent Motion to Amend Granting-in-Part and Denying-in-Part Petitioner's Motion to Strike Denying Patent Owner's Motion to Exclude dated Jul. 18, 2022. cited by applicant

Notice of Allowance dated May 6, 2020, which issued during the prosecution of U.S. Appl. No. 16/730,090. cited by applicant

Notice of Allowance dated Feb. 2, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,663. cited by applicant

Notice of Allowance dated Feb. 16, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,663. cited by applicant

Notice of Allowance dated Sep. 16, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,770. cited by applicant

Notice of Allowance dated Oct. 22, 2019, which issued during the prosecution of U.S. Appl. No. 15/433,547. cited by applicant

Notice of Allowance dated May 19, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,619. cited by applicant

U.S. Appl. No. 62/295,701, filed Feb. 16, 2016. cited by applicant

An Office Action dated May 15, 2013, which issued during the prosecution of U.S. Appl. No. 12/583,979. cited by applicant

Notice of Allowance dated Oct. 15, 2020, which issued during the prosecution of U.S. Appl. No. 16/740,659. cited by applicant

Notice of Allowance dated Apr. 30, 2020, which issued during the prosecution of U.S. Appl. No. 15/970,314. cited by applicant

Notice of Allowance dated Feb. 28, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,599. cited by applicant

Notice of Allowance dated Jan. 10, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,599. cited by applicant

Notice of Allowance dated May 26, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,599. cited

by applicant

Notice of Allowance dated Feb. 2, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,619. cited by applicant

Notice of Allowance dated Feb. 15, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,619. cited by applicant

Notice of Allowance dated May 12, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,619. cited by applicant

Advisory Action dated Feb. 3, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,505. cited by applicant

An Office Action dated Sep. 8, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,074. cited by applicant

An Office Action dated Jan. 25, 2021, which issued during the prosecution of U.S. Appl. No. 16/136,082. cited by applicant

An Office Action dated Jun. 1, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,082. cited by applicant

An Office Action dated Mar. 16, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,110. cited by applicant

An Office Action dated Mar. 12, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,123. cited by applicant

An Office Action dated Mar. 10, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,150. cited by applicant

An Office Action dated Apr. 7, 2020, which issued during the prosecution of U.S. Appl. No. 16/520,289. cited by applicant

An Office Action dated Jan. 14, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,969. cited by applicant

An Office Action dated May 28, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,969. cited by applicant

An Office Action dated Sep. 9, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,969. cited by applicant

An Office Action dated Mar. 20, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,979. cited by applicant

An Office Action dated Oct. 20, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,979. cited by applicant

An Office Action dated Mar. 6, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,074. cited by applicant

An Office Action dated Mar. 19, 2021, which issued during the prosecution of U.S. Appl. No. 16/136,074. cited by applicant

An Office Action dated Aug. 11, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,619. cited by applicant

An Office Action dated Dec. 21, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,619. cited by applicant

An Office Action dated Apr. 10, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,663. cited by applicant

An Office Action dated Jun. 28, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,663. cited by applicant

An Office Action dated Mar. 8, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,663. cited by applicant

An Office Action dated Sep. 23, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,663. cited by applicant

An Office Action dated Mar. 16, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,770. cited by applicant

An Office Action dated Dec. 10, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,599. cited by applicant

An Office Action dated Jun. 1, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,599. cited by applicant

An Office Action dated Mar. 5, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,843. cited by applicant

An Office Action dated Apr. 9, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,969. cited by applicant

Notice of Allowance dated Feb. 4, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,447. cited by applicant

Notice of Allowance dated Mar. 24, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,447. cited by applicant

Notice of Allowance dated Mar. 14, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,505. cited by applicant

Notice of Allowance dated Mar. 16, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,505. cited by applicant

Notice of Allowance dated Jun. 15, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,843. cited by applicant

Notice of Allowance dated Jun. 23, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,843. cited by applicant

Notice of Allowance dated Aug. 24, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,979. cited by applicant

Notice of Allowance dated Dec. 24, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,979. cited by applicant

Notice of Allowance dated Feb. 22, 2022, which issued during the prosecution of U.S. Appl. No. 16/135,979. cited by applicant

Notice of Allowance dated Mar. 17, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,979. cited by applicant

Notice of Allowance dated Feb. 8, 2022, which issued during the prosecution of U.S. Appl. No. 16/136,074. cited by applicant

An Office Action dated Oct. 25, 2023, which issued during the prosecution of U.S. Appl. No. 16/144,054. cited by applicant

An Office Action dated Sep. 29, 2023, which issued during the prosecution of Chinese Patent Application No. 201880076340.9. cited by applicant

Notice of Allowance dated Jun. 27, 2019, which issued during the prosecution of U.S. Appl. No. 16/042,028. cited by applicant

Notice of Allowance dated Jul. 3, 2019, which issued during the prosecution of U.S. Appl. No. 16/042,129. cited by applicant

Notice of Allowance dated Feb. 16, 2022, which issued during the prosecution of U.S. Appl. No. 16/136,074. cited by applicant

Notice of Allowance dated Jul. 30, 2021, which issued during the prosecution of U.S. Appl. No. 16/136,074. cited by applicant

Notice of Allowance dated Sep. 24, 2021, which issued during the prosecution of U.S. Appl. No. 16/136,074. cited by applicant

Notice of Allowance dated Dec. 23, 2021, which issued during the prosecution of U.S. Appl. No. 16/136,082. cited by applicant

Notice of Allowance dated Feb. 28, 2022, which issued during the prosecution of U.S. Appl. No. 16/136,082. cited by applicant

Advisory Action dated Jan. 14, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,663. cited by applicant

Advisory Action dated Dec. 15, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,969. cited by applicant

Advisory Action dated Jul. 14, 2020, which issued during the prosecution of U.S. Appl. No. 16/520,289. cited by applicant

Notice of Allowance dated Mar. 8, 2019, which issued during the prosecution of U.S. Appl. No. 15/978,494. cited by applicant

Notice of Allowance dated May 22, 2019, which issued during the prosecution of U.S. Appl. No. 15/979,686. cited by applicant

Notice of Allowance dated Apr. 11, 2019, which issued during the prosecution of U.S. Appl. No. 15/994,022. cited by applicant

by applicant

Notice of Allowance dated Feb. 12, 2020, which issued during the prosecution of U.S. Appl. No. 15/995,725. cited by applicant

Notice of Allowance dated Aug. 26, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,150. cited by applicant

Notice of Allowance dated Dec. 10, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,150. cited by applicant

Notice of Allowance dated Aug. 22, 2019, which issued during the prosecution of U.S. Appl. No. 16/507,357. cited by applicant

Notice of Allowance dated Dec. 11, 2019, which issued during the prosecution of U.S. Appl. No. 16/507,357. cited by applicant

Notice of Allowance dated Dec. 16, 2020, which issued during the prosecution of U.S. Appl. No. 16/520,289. cited by applicant

Notice of Allowance dated Sep. 17, 2020, which issued during the prosecution of U.S. Appl. No. 16/520,289. cited by applicant

Notice of Allowance dated Mar. 25, 2020, which issued during the prosecution of U.S. Appl. No. 16/559,365. cited by applicant

An Office Action dated Oct. 7, 2019, which issued during the prosecution of U.S. Appl. No. 16/520,289. cited by applicant

An Office Action dated Oct. 23, 2019, which issued during the prosecution of U.S. Appl. No. 16/559,365. cited by applicant

An Office Action dated Dec. 12, 2019, which issued during the prosecution of U.S. Appl. No. 16/585,349. cited by applicant

An Office Action dated Jul. 16, 2020, which issued during the prosecution of U.S. Appl. No. 16/678,355. cited by applicant

An Office Action dated Mar. 18, 2020, which issued during the prosecution of U.S. Appl. No. 16/740,659. cited by applicant

Notice of Allowance dated Aug. 11, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,110. cited by applicant

Notice of Allowance dated Sep. 30, 2020, which issued during the prosecution of U.S. Appl. No. 16/136,123. cited by applicant

Notice of Allowance dated May 1, 2020, which issued during the prosecution of U.S. Appl. No. 15/995,725. cited by applicant

Notice of Allowance dated Aug. 1, 2019, which issued during the prosecution of U.S. Appl. No. 16/008,618. cited by applicant

Notice of Allowance dated Nov. 13, 2019, which issued during the prosecution of U.S. Appl. No. 16/008,618. cited by applicant

Notice of Allowance dated Aug. 28, 2019, which issued during the prosecution of U.S. Appl. No. 16/040,831. cited by applicant

Notice of Allowance dated Dec. 10, 2019, which issued during the prosecution of U.S. Appl. No. 16/040,831. cited by applicant

Notice of Allowance dated Jan. 23, 2020, which issued during the prosecution of U.S. Appl. No. 16/041,208. cited by applicant

Notice of Allowance dated Mar. 5, 2020, which issued during the prosecution of U.S. Appl. No. 16/041,208. cited by applicant

Notice of Allowance dated May 6, 2020, which issued during the prosecution of U.S. Appl. No. 16/041,208. cited by applicant

An Office Action dated Apr. 19, 2019, which issued during the prosecution of U.S. Appl. No. 16/008,618. cited by applicant

An Office Action dated Feb. 14, 2019, which issued during the prosecution of U.S. Appl. No. 16/008,618. cited by applicant

An Office Action dated Jun. 19, 2019, which issued during the prosecution of U.S. Appl. No. 16/040,831. cited by applicant

An Office Action dated Sep. 30, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,447. cited by applicant

An Office Action dated Mar. 10, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,447. cited by applicant

An Office Action dated Aug. 15, 2019, which issued during the prosecution of U.S. Appl. No. 16/041,208. cited by applicant

An Office Action dated Jun. 21, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,447. cited by applicant

An Office Action dated Apr. 3, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,466. cited by applicant

An Office Action dated Mar. 27, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,505. cited by applicant

An Office Action dated Sep. 8, 2021, which issued during the prosecution of U.S. Appl. No. 16/135,505. cited by applicant

An Office Action dated Sep. 29, 2020, which issued during the prosecution of U.S. Appl. No. 16/135,505. cited by applicant

An Office Action dated Dec. 21, 2018, which issued during the prosecution of U.S. Appl. No. 15/978,494. cited by applicant

An Office Action dated Dec. 28, 2018, which issued during the prosecution of U.S. Appl. No. 15/979,686. cited by applicant

An Office Action dated Feb. 12, 2019, which issued during the prosecution of U.S. Appl. No. 15/994,022. cited by applicant

An Office Action dated Feb. 13, 2019, which issued during the prosecution of U.S. Appl. No. 15/995,725. cited by applicant

An Office Action dated Oct. 22, 2019, which issued during the prosecution of U.S. Appl. No. 15/995,725. cited by applicant

An Office Action dated Jun. 20, 2019, which issued during the prosecution of U.S. Appl. No. 15/970,314. cited by applicant

An Office Action dated Nov. 21, 2019, which issued during the prosecution of U.S. Appl. No. 15/970,314. cited by applicant

An Office Action dated Feb. 22, 2024, which issued during the prosecution of Canadian Patent Application No. 3,096,002. cited by applicant

An Office Action dated Jan. 26, 2024, which issued during the prosecution of Chinese Patent Application No. 201980023291.7. cited by applicant

Notice of Allowance dated Sep. 27, 2023, which issued during the prosecution of U.S. Appl. No. 17/010,886. cited by applicant

Notice of Allowance dated Apr. 10, 2024, which issued during the prosecution of U.S. Appl. No. 17/010,886. cited by applicant

An Office Action dated Apr. 25, 2024, which issued during the prosecution of U.S. Appl. No. 17/397,235. cited by applicant

An Office Action dated Mar. 25, 2024, which issued during the prosecution of U.S. Appl. No. 17/841,912. cited by applicant

An International Search Report and a Written Opinion both dated Oct. 18, 2022, which issued during the prosecution of PCT/US2022/037864. cited by applicant

European Search Report dated Nov. 14, 2023 which issued during the prosecution of Applicant's European App No. 23191562.0. cited by applicant

An Office Action dated Jan. 25, 2024, which issued during the prosecution of U.S. Appl. No. 18/090,058. cited by applicant

Notice of Allowance dated Nov. 8, 2023, which issued during the prosecution of U.S. Appl. No. 16/656,790. cited by applicant

An Office Action dated Oct. 20, 2023, which issued during the prosecution of Canadian Patent Application No. 3,170,042. cited by applicant

An Office Action dated Dec. 19, 2023, which issued during the prosecution of U.S. Appl. No. 17/010,886. cited by applicant

An Office Action dated Dec. 22, 2023, which issued during the prosecution of U.S. Appl. No. 18/216,391. cited by applicant

An International Search Report and a Written Opinion both dated Jan. 18, 2024, which issued during the

prosecution of Applicant's PCT/IL2023/050958. cited by applicant
Notice of Allowance dated May 6, 2020, which issued during the prosecution of U.S. Appl. No. 16/559,365. cited by applicant
Notice of Allowance dated Jul. 23, 2020, which issued during the prosecution of U.S. Appl. No. 16/585,349. cited by applicant
Notice of Allowance dated Nov. 2, 2020, which issued during the prosecution of U.S. Appl. No. 16/585,349. cited by applicant
Notice of Allowance dated Aug. 5, 2020, which issued during the prosecution of U.S. Appl. No. 16/660,231. cited by applicant
Notice of Allowance dated Nov. 18, 2020, which issued during the prosecution of U.S. Appl. No. 16/660,231. cited by applicant
Notice of Allowance dated Dec. 7, 2020, which issued during the prosecution of U.S. Appl. No. 16/678,355. cited by applicant
Notice of Allowance dated Feb. 24, 2020, which issued during the prosecution of U.S. Appl. No. 16/730,090. cited by applicant
Notice of Allowance dated Apr. 10, 2024, which issued during the prosecution of U.S. Appl. No. 17/181,722. cited by applicant
Summons to Attend Oral Proceedings regarding European Patent Application No. 22161862.2, dated Feb. 26, 2024. cited by applicant
A Letter from the opponent regarding European Patent Application No. 14710060.6, dated Apr. 5, 2024. cited by applicant
Summons to Attend Oral Proceedings regarding European Patent Application No. 14710060.6, dated May 10, 2024. cited by applicant
A Brief Communication together with a Letter from the opponent regarding European Patent Application No. 14710060.6, dated Apr. 12, 2024. cited by applicant
An Office Action dated Jun. 18, 2024, which issued during the prosecution of U.S. Appl. No. 17/399,594. cited by applicant
An Office Action dated Aug. 9, 2024, which issued during the prosecution of Chinese Patent Application No. 202210336863.0. cited by applicant
An Office Action dated Aug. 1, 2024, which issued during the prosecution of U.S. Appl. No. 18/368,250. cited by applicant
An Office Action dated Jul. 3, 2024, which issued during the prosecution of U.S. Appl. No. 18/109,937. cited by applicant

Primary Examiner: Nguyen; Tuan V

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation application of U.S. patent application Ser. No. 17/141,853, filed Jan. 5, 2021, entitled, "Techniques for deployment of a prosthetic valve," which issued as U.S. Pat. No. 11,801,135 and which is a Continuation application of U.S. patent application Ser. No. 15/668,659, filed Aug. 3, 2017, entitled, "Techniques for deployment of a prosthetic valve," which issued as U.S. Pat. No. 10,918,481 and which: (1) is a Continuation-In-Part of PCT application IL2016/050125, filed Feb. 3, 2016, entitled "Prosthetic valve with axially-sliding frames," which published as WO 2016/125160, and which claims priority from U.S. Provisional Patent Application 62/112,343 to Hariton et al., filed Feb. 5, 2015, and entitled "Prosthetic valve with axially-sliding frames; and (2) is a Continuation-In-Part of U.S. patent application Ser. No. 15/541,783, entitled "Prosthetic valve with axially-sliding frames," (now U.S. Pat. No. 9,974,651) which is the US National Phase of PCT application IL2016/050125, filed Feb. 3, 2016, entitled "Prosthetic valve with axially-sliding frames," which published as WO 2016/125160, and which claims priority from U.S. Provisional Patent Application

FIELD OF THE INVENTION

(1) Some applications of the present invention relate in general to valve replacement. More specifically, some applications of the present invention relate to prosthetic valves for replacement of a cardiac valve.

BACKGROUND

(2) Ischemic heart disease causes regurgitation of a heart valve by the combination of ischemic dysfunction of the papillary muscles, and the dilatation of the ventricle that is present in ischemic heart disease, with the subsequent displacement of the papillary muscles and the dilatation of the valve annulus.

(3) Dilation of the annulus of the valve prevents the valve leaflets from fully coapting when the valve is closed. Regurgitation of blood from the ventricle into the atrium results in increased total stroke volume and decreased cardiac output, and ultimate weakening of the ventricle secondary to a volume overload and a pressure overload of the atrium.

SUMMARY OF THE INVENTION

(4) For some applications, an implant is provided having a tubular portion, an upstream support portion and one or more flanges. The implant is percutaneously deliverable to a native heart valve in a compressed state, and is expandable at the native valve. The implant and its delivery system facilitate causing the upstream support portion and the flanges to protrude radially outward from the tubular portion without expanding the tubular portion. Expansion of the tubular portion brings the upstream support portion and the flanges closer together, for securing the implant at the native valve by sandwiching tissue of the native valve between the upstream support portion and the flanges.

(5) For some applications, a delivery tool is provided, including a delivery capsule, disposed at a distal portion of the tool. The delivery capsule includes a proximal capsule-portion dimensioned to house a first part of an implant, and a distal capsule-portion dimensioned to house a second part of the implant; and an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion including one or more controllers and one or more locks. The controllers are operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers. The tool has sequential states in which movement of the proximal and distal capsule-portions is variously inhibited or facilitated. For some applications, in response to a movement of a capsule-portion that is facilitated in a given state, the tool unlocks the subsequent state.

(6) There is therefore provided, in accordance with an application of the present invention, apparatus including an implant and a delivery tool, the tool including: a delivery capsule, disposed at a distal portion of the tool, the delivery capsule including a proximal capsule-portion dimensioned to house a first part of the implant, and a distal capsule-portion dimensioned to house a second part of the implant; and an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion including one or more controllers and one or more locks, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers, and the controllers and the locks are mechanically cooperative such that: in a state A of the tool: a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated, a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited, and a proximal movement of the proximal capsule-portion proximally with respect to the implant is facilitated; and in a state B of the tool: the second distal movement of the distal capsule-portion is facilitated.

(7) In an application, in the state A, a lock of the one or more locks is locked, and unlocking of the lock transitions the tool into the state B.

(8) In an application, the lock is not unlockable until the distal capsule-portion has been moved the first distance distally with respect to the implant.

(9) In an application, the lock is not unlockable until the proximal capsule-portion has been moved proximally with respect to the implant.

(10) In an application: the one or more controllers include a first controller and a second controller, and in the state A, actuation of the first controller causes the first distal movement of the distal capsule-portion, and actuation of the second controller causes the proximal movement of the proximal capsule-portion.

(11) In an application, in the state B, further actuation of the first controller causes the second distal movement of the distal capsule-portion.

- (12) In an application, the one or more controllers further include a third controller, and, in the state B, actuation of the third controller causes the second distal movement of the distal capsule-portion.
- (13) In an application, the implant is for use at a native mitral valve of a subject, and: the first part of the implant includes an upstream support portion, configured to be placed against an upstream surface of the native mitral valve, the second part of the implant includes a downstream end of the implant, and a plurality of flanges (i) disposed longitudinally between the upstream support portion and the downstream end, and (ii) configured to engage leaflets of the native mitral valve; and the implant is housed by the delivery capsule such that the first distal movement of the distal capsule-portion unhouses, from the distal capsule-portion, the plurality of flanges, but not the downstream end of the implant.
- (14) In an application, the implant is housed by the delivery capsule such that the second distal movement of the distal capsule-portion unhouses, from the distal capsule-portion, the downstream end of the implant.
- (15) In an application, the implant is housed by the delivery capsule such that the proximal movement of the proximal capsule-portion unhouses, from the proximal capsule-portion, the upstream support portion.
- (16) In an application, the implant includes: a valve frame that includes: a tubular portion shaped to define a lumen therethrough, and the upstream support portion, extending from an upstream end of the tubular portion; an outer frame that circumscribes the tubular portion, and includes the plurality of flanges; and a plurality of prosthetic leaflets, coupled to the tubular portion within the lumen.
- (17) In an application, the second part of the implant includes at least part of the tubular portion.
- (18) There is further provided, in accordance with an application of the present invention, apparatus including an implant and a delivery tool, the tool including: a delivery capsule, disposed at a distal portion of the tool, the delivery capsule including a proximal capsule-portion dimensioned to house a first part of the implant, and a distal capsule-portion dimensioned to house a second part of the implant; and an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion including one or more controllers and one or more locks, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers, and the controllers and the locks are mechanically cooperative such that: in a first state of the tool: a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated, a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited, a proximal movement of the proximal capsule-portion proximally with respect to the implant is inhibited, and responsively to movement of the distal capsule-portion distally, the tool automatically unlocks a second state of the tool; in the second state of the tool: the proximal movement of the proximal capsule-portion is facilitated, the second distal movement of the distal capsule-portion remains inhibited, and responsively to movement of the proximal capsule-portion proximally, the tool automatically unlocks a third state of the tool; and in the third state of the tool: the second distal movement of the distal capsule-portion is facilitated.
- (19) In an application, the tool automatically unlocks the second state responsively to the distal capsule-portion having moved the first distance.
- (20) In an application: the one or more locks include a first lock and a second lock, the one or more controllers include a first controller and a second controller, in the first state of the tool, actuation of the first controller causes the first distal movement of the distal capsule-portion, the tool automatically unlocks the second state by automatically making the first lock unlockable, in the second state of the tool, actuation of the second controller causes the proximal movement of the proximal capsule-portion, and the tool automatically unlocks the third state by automatically making the second lock unlockable.
- (21) In an application, in the third state of the tool, further actuation of the first controller causes the second distal movement of the distal capsule-portion.
- (22) In an application, the one or more controllers further include a third controller, and, in the third state of the tool, actuation of the third controller causes the second distal movement of the distal capsule-portion.
- (23) In an application, the implant is for use at a native mitral valve of a subject, and: the first part of the implant includes an upstream support portion, configured to be placed against an upstream surface of the native mitral valve, the second part of the implant includes a downstream end of the implant, and a plurality of flanges (i) disposed longitudinally between the upstream support portion and the downstream end, and (ii) configured to engage leaflets of the native mitral valve; and the implant is housed by the delivery capsule such that the first distal movement of the distal capsule-portion unhouses, from the distal capsule-portion, the plurality of flanges, but not the downstream end of the implant.
- (24) In an application, the implant is housed by the delivery capsule such that the second distal movement of

the distal capsule-portion unhouses, from the distal capsule-portion, the downstream end of the implant.

(25) In an application, the implant is housed by the delivery capsule such that the proximal movement of the proximal capsule-portion unhouses, from the proximal capsule-portion, the upstream support portion.

(26) In an application, the implant includes: a valve frame that includes: a tubular portion shaped to define a lumen therethrough, and the upstream support portion, extending from an upstream end of the tubular portion; an outer frame that circumscribes the tubular portion, and includes the plurality of flanges; and a plurality of prosthetic leaflets, coupled to the tubular portion within the lumen.

(27) In an application, the second part of the implant includes at least part of the tubular portion.

(28) There is further provided, in accordance with an application of the present invention, a method, including: percutaneously advancing an implant to a native heart valve of a subject, the implant housed in a compressed state within a delivery tool; unhousing a longitudinally-intermediate portion of the implant, the longitudinally-intermediate portion being longitudinally between a longitudinally-proximal portion of the implant and a longitudinally distal portion of the implant; subsequently, unhousing the longitudinally-proximal portion of the implant; and subsequently, unhousing the longitudinally-distal portion of the implant.

(29) In an application: unhousing the longitudinally-intermediate portion of the implant includes unhousing flanges of the implant such that the flanges automatically expand; unhousing the longitudinally-proximal portion of the implant includes unhousing an upstream support portion of the implant such that the upstream support portion automatically expands; and unhousing the longitudinally-distal portion of the implant includes unhousing the longitudinally-distal portion of the implant such that the longitudinally-distal portion of the implant automatically expands.

(30) In an application, the longitudinally-distal portion includes a tubular portion within which a valve member is disposed, and unhousing the longitudinally-distal portion of the implant includes unhousing the longitudinally-distal portion of the implant such that the tubular portion expands to form a lumen and the valve member regulates one-way flow of blood through the lumen.

(31) In an application, unhousing the longitudinally-distal portion of the implant includes facilitating expansion of the implant such that the tubular portion expands and the flanges become longitudinally closer to the upstream support portion.

(32) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a valve frame, including: a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis; and a plurality of arms, extending radially outward from an upstream portion of the tubular portion to define an arm span, and configured to engage tissue in the atrium; a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; and an outer frame: coupled to the valve frame, and including a ring that circumscribes the tubular portion, and a plurality of flanges that extend radially outward from the ring to define a flange span, and are configured to engage tissue of the ventricle,

and: each arm of the plurality of arms defines a rigid portion and a flexible portion that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion, and the majority of the flexible portion is disposed further radially outward than the flange span.

(33) In an application, the implant is configured such that the rigid portion provides resistance against axial movement of the implant, while the flexible portion facilitates conformation of the arm to anatomy of the atrium and/or lateral movement of the implant within the native valve.

(34) In an application, each arm of the plurality of arms defines a serpentine shape in the flexible portion.

(35) In an application, each arm of the plurality of arms defines a plurality of holes in the flexible portion.

(36) In an application, less than 90 percent of the flexible portion is disposed further radially outward than the flange span.

(37) In an application, more than 60 percent of the flexible portion is disposed further radially outward than the flange span.

(38) In an application, for each arm of the plurality of arms, the flexible portion covers an overall surface area including (i) the area occupied by material from which the arm is formed, and (ii) spaces, and the material from which the arm is formed occupies less than 80 percent of the overall surface area.

(39) In an application, the material from which the arm is formed occupies more than 20 percent of the overall surface area.

(40) In an application, the material from which the arm is formed occupies less than 70 percent of the overall surface area.

(41) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis; a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; an outer frame: including a ring defined by a pattern of alternating peaks and troughs, the peaks being longitudinally closer to the upstream end than to the downstream end, and the troughs being longitudinally closer to the downstream end than to the upstream end, and the pattern of the ring having an amplitude longitudinally between the peaks and the troughs, including a plurality of legs, each of the legs coupled to the ring at a respective trough, and shaped to define a plurality of outer-frame coupling elements, each of the outer-frame coupling elements (i) coupled to the ring at a respective peak, and (ii) fixed with respect to a respective valve-frame coupling element, and: the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and the fixation of the outer-frame coupling elements to the valve-frame coupling elements is such that compression of the tubular portion from the expanded state toward the compressed state such that the valve-frame coupling elements pull the outer-frame coupling elements radially inward: (i) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements, and (ii) increases the amplitude of the pattern of the ring.

(42) In an application, the ring circumscribes the tubular portion.

(43) In an application, the valve-frame coupling elements are disposed circumferentially around the longitudinal axis between the upstream end and the downstream end but not at the upstream end nor at the downstream end.

(44) In an application, the upstream support portion includes one or more fabric pockets disposed circumferentially, each pocket of the one or more pockets having an opening that faces a downstream direction.

(45) In an application, the outer frame is coupled to the valve frame only via the fixation of the outer-frame coupling elements to the respective valve-frame coupling elements.

(46) In an application, the apparatus further includes an upstream support portion that includes a plurality of arms that extend radially from the tubular portion, and: the upstream support portion has (i) a constrained-arm state, and (ii) a released-arm state in which the arms extend radially outward from the tubular portion, each leg has a tissue-engaging flange that has (i) a constrained-flange state, and (ii) a released-flange state in which the flange extends radially outward from the tubular portion, and the apparatus has an intermediate state in which (i) the tubular portion is in its compressed state, (ii) the upstream support portion is in its released-arm state, and (iii) the legs are in their released-flange state.

(47) In an application: the apparatus includes an implant that includes the valve frame, the leaflets, and the outer frame, and the apparatus further includes a tool: including a delivery capsule dimensioned (i) to house and retain the implant in a compressed state of the implant in which (a) the tubular portion is in its compressed state, (b) the upstream support portion is in its constrained-arm state, and (c) the legs are in their constrained-flange state, and (ii) to be advanced percutaneously to the heart of the subject while the implant is housed and in its compressed state, and operable from outside the subject to: transition the implant from its compressed state into the intermediate state while retaining the tubular portion in its compressed state, and subsequently, expand the tubular portion toward its expanded state.

(48) In an application, the tool is operable from outside the subject to transition the implant from its compressed state into the intermediate state by (i) releasing the legs into their released-flange state, while retaining the tubular portion in its compressed state, and (ii) subsequently, releasing the upstream support portion into its released-arm state, while retaining the tubular portion in its compressed state.

(49) In an application, the tool is operable from outside the subject to transition the implant from its compressed state into the intermediate state by (i) releasing the upstream support portion into its released-arm state, while retaining the tubular portion in its compressed state, and (ii) subsequently, releasing the legs into their released-flange state, while retaining the tubular portion in its compressed state.

(50) In an application, the fixation of the outer-frame coupling elements to the valve-frame coupling

elements is such that, when the apparatus is in its intermediate state, expansion of the tubular portion from its compressed state toward its expanded state moves the flanges longitudinally away from the valve-frame coupling elements.

(51) In an application, the fixation of the outer-frame coupling elements to the valve-frame coupling elements is such that, when the apparatus is in its intermediate state, expansion of the tubular portion from its compressed state toward its expanded state reduces the amplitude of the pattern of the ring and passes the flanges between the arms.

(52) In an application, the upstream support portion further includes a covering that covers the arms to form an annular shape in the released-arm state, and, when the apparatus is in its intermediate state, expansion of the tubular portion from its compressed state toward its expanded state presses the flanges onto the covering.

(53) In an application, in the compressed state of the tubular portion, a downstream end of each leg is longitudinally closer than the valve-frame coupling elements to the downstream end, and the flange of each leg is disposed longitudinally closer than the valve-frame coupling elements to the upstream end.

(54) In an application, in the expanded state of the tubular portion, the downstream end of each leg is longitudinally closer than the valve-frame coupling elements to the downstream end, and the flange of each leg is disposed longitudinally closer than the valve-frame coupling elements to the upstream end.

(55) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including an implant that includes: a valve frame that includes a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion having an upstream end, a downstream end, a longitudinal length therebetween, and a diameter transverse to the longitudinal axis; a valve member, coupled to the tubular portion, disposed within the lumen, and arranged to provide unidirectional upstream-to-downstream flow of blood through the lumen; an upstream support portion, coupled to the tubular portion; and an outer frame, coupled to the tubular portion, and including a tissue-engaging flange,

and: the implant has a first state and a second state, in both the first state and the second state, (i) the upstream support portion extends radially outward from the tubular portion, and (ii) the tissue-engaging flange extends radially outward from the tubular portion, and the tubular portion, the upstream support portion, and the outer frame are arranged such that transitioning of the implant from the first state toward the second state: increases the diameter of the tubular portion by a diameter-increase amount, decreases the length of the tubular portion by a length-decrease amount, and moves the flange a longitudinal distance toward or toward-and-beyond the upstream support portion, the distance being greater than the length-decrease amount.

(56) In an application, the tubular portion, the upstream support portion, and the outer frame are arranged such that the longitudinal distance is more than 20 percent greater than the length-decrease amount.

(57) In an application, the tubular portion, the upstream support portion, and the outer frame are arranged such that the longitudinal distance is more than 30 percent greater than the length-decrease amount.

(58) In an application, the tubular portion, the upstream support portion, and the outer frame are arranged such that the longitudinal distance is more than 40 percent greater than the length-decrease amount.

(59) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis; a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; an outer frame, including: a ring defined by a pattern of alternating peaks and troughs: the peaks being longitudinally closer than the troughs to the upstream end, the peaks being fixed to respective sites of the tubular portion at respective coupling points disposed circumferentially around the longitudinal axis, and the pattern of the ring having an amplitude longitudinally between the peaks and the troughs; and a plurality of legs, each of the legs coupled to the ring at a respective trough,

and: the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and the fixation of the peaks to the respective sites of the tubular portion is such that compression of the tubular portion from the expanded state toward the compressed state such that the respective sites of the tubular portion pull the peaks radially inward via radially-inward tension on the coupling points: (i) reduces a circumferential distance between each of the coupling points and its adjacent coupling points, and (ii) increases the amplitude of the pattern of the ring.

(60) In an application, the outer frame is coupled to the valve frame only via the fixation of the peaks to the respective sites of the tubular portion at the respective coupling points.

(61) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the valve frame defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis; a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; an outer frame: including a ring defined by a pattern of alternating peaks and troughs, the peaks being longitudinally closer to the upstream end than to the downstream end, and the troughs being longitudinally closer to the downstream end than to the upstream end, and the pattern of the ring having an amplitude longitudinally between the peaks and the troughs, including a plurality of legs, each of the legs coupled to the ring at a respective trough, and shaped to define a plurality of outer-frame coupling elements, each of the outer-frame coupling elements (i) coupled to the ring at a respective peak, and (ii) fixed with respect to a respective valve-frame coupling element, and: the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and the fixation of the outer-frame coupling elements with respect to the valve-frame coupling elements is such that compression of the tubular portion from the expanded state toward the compressed state (i) pulls the outer-frame coupling elements radially inward via radially-inward pulling of the valve-frame coupling elements on the outer-frame coupling elements, (ii) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements, and (iii) increases the amplitude of the pattern of the ring, without increasing a radial gap between the valve frame and the ring by more than 1.5 mm.

(62) In an application, the outer frame is coupled to the valve frame only via the fixation of the outer-frame coupling elements to the respective valve-frame coupling elements.

(63) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis; a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; an outer frame, including: a ring defined by a pattern of alternating peaks and troughs: the peaks being longitudinally closer than the troughs to the upstream end, the peaks being fixed to respective sites of the tubular portion at respective coupling points disposed circumferentially around the longitudinal axis, and the pattern of the ring having an amplitude longitudinally between the peaks and the troughs; and a plurality of legs, each of the legs coupled to the ring at a respective trough, and: the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and the fixation of the peaks to the respective sites of the tubular portion is such that compression of the tubular portion from the expanded state toward the compressed state (i) pulls the peaks radially inward via radially-inward pulling of the respective sites of the tubular portion on the peaks, (ii) reduces a circumferential distance between each of the coupling points and its adjacent coupling points, and (iii) increases the amplitude of the pattern of the ring, without increasing a radial gap between the valve frame and the ring by more than 1.5 mm.

(64) In an application, the outer frame is coupled to the valve frame only via the fixation of the peaks to the respective sites of the tubular portion at the respective coupling points.

(65) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion having an upstream end, a downstream end, and defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis between the upstream end and the downstream end but not at the upstream end nor at the downstream end; a plurality of prosthetic leaflets, disposed within the lumen, and arranged to provide unidirectional flow of blood through the lumen; an outer frame: including a ring defined by a pattern of alternating peaks and troughs, the peaks being longitudinally closer to the upstream end than to the downstream end, and the

troughs being longitudinally closer to the downstream end than to the upstream end, including a plurality of legs, each of the legs coupled to the ring at a respective trough, and shaped to define a plurality of outer-frame coupling elements, each of the outer-frame coupling elements (i) coupled to the ring at a respective peak, and (ii) fixed with respect to a respective valve-frame coupling element at a respective coupling point, and: the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and expansion of the tubular portion from the compressed state toward the expanded state (i) increases a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements, and (ii) moves the plurality of legs in a longitudinally upstream direction with respect to the tubular portion.

(66) In an application, the outer frame is coupled to the valve frame only via the fixation of the outer-frame coupling elements to the respective valve-frame coupling elements.

(67) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion having an upstream end and a downstream end; a plurality of prosthetic leaflets, disposed within the lumen, and arranged to provide unidirectional flow of blood through the lumen; an outer frame, including: a ring defined by a pattern of alternating peaks and troughs: the peaks being longitudinally closer than the troughs to the upstream end, the peaks being fixed to respective sites of the tubular portion at respective coupling points disposed circumferentially around the longitudinal axis between the upstream end and the downstream end but not at the upstream end nor the downstream end; and a plurality of legs, each of the legs coupled to the ring at a respective trough, and: the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and expansion of the tubular portion from the compressed state toward the expanded state (i) increases a circumferential distance between each of the coupling points and its adjacent coupling points, and (ii) moves the plurality of legs in a longitudinally upstream direction with respect to the tubular portion.

(68) In an application, the outer frame is coupled to the valve frame only via the fixation of the peaks to the respective sites of the tubular portion at the respective coupling points.

(69) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including: a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, and including: a valve frame, including: a tubular portion having an upstream end and a downstream end, and shaped to define a lumen therebetween, and an upstream support portion, extending from the upstream end of the tubular portion; and at least one leg, coupled to the valve frame at a coupling point, and having a tissue-engaging flange; and a valve member disposed within the lumen, and configured to facilitate one-way liquid flow through the lumen from the upstream end of the tubular portion to the downstream end of the tubular portion, and the frame assembly: has a compressed state, for percutaneous delivery to the heart, in which the tubular portion has a compressed diameter, is biased to assume an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and is configured such that increasing the diameter of the tubular portion toward the expanded diameter causes longitudinal movement: of the upstream support portion toward the coupling point, and of the tissue-engaging flange away from the coupling point.

(70) In an application: the apparatus includes an implant that includes the frame assembly and the valve member, and the apparatus further includes a tool: including a delivery capsule dimensioned (i) to house and retain the implant in the compressed state, and (ii) to be advanced percutaneously to the heart of the subject while the implant is housed and in the compressed state, and operable from outside the subject to facilitate an increase of the diameter of the tubular portion from the compressed diameter toward the expanded diameter such that the increase of the diameter actuates longitudinal movement: of the upstream support portion toward the coupling point, and of the tissue-engaging flange away from the coupling point.

(71) In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes longitudinal movement of the upstream end of the tubular portion toward the coupling point.

(72) In an application, the coupling point is disposed closer to the downstream end of the frame assembly than are either the tissue-engaging flange or the upstream support portion.

(73) In an application, in the expanded state of the frame assembly, the leg extends away from the central longitudinal axis.

(74) In an application: the expanded state of the frame assembly is a fully-expanded state of the frame assembly, the leg is expandable into an expanded state of the leg, independently of increasing the diameter of the tubular portion, and in the expanded state of the leg, the leg extends away from the central longitudinal axis.

(75) In an application: in the expanded state of the frame assembly, the leg extends away from the central longitudinal axis, and in the compressed state of the frame assembly, the leg is generally parallel with the central longitudinal axis.

(76) In an application, the frame assembly is configured such that the longitudinal movement of the tissue-engaging flange away from the coupling point is a translational movement of the tissue-engaging flange that does not include rotation of the tissue-engaging flange.

(77) In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes 1-20 mm of longitudinal movement of the tissue-engaging flange away from the coupling point.

(78) In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes 1-20 mm of longitudinal movement of the upstream support portion toward the coupling point.

(79) In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state reduces a distance between the upstream support portion and the tissue-engaging flange by 5-30 mm.

(80) In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state moves the tissue-engaging flange longitudinally past the upstream support portion.

(81) In an application: the tubular portion is defined by a plurality of cells of the valve frame, and increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state: includes (i) increasing a width, orthogonal to the longitudinal axis of the frame assembly, of each cell, and (ii) reducing a height, parallel with the longitudinal axis of the frame assembly, of each cell, and causes longitudinal movement of the upstream support portion toward the coupling point by reducing a height, parallel with the longitudinal axis of the frame assembly, of the tubular portion, by reducing the height of each cell.

(82) In an application, the leg is disposed on an outside of the tubular portion.

(83) In an application: the at least one leg includes a plurality of legs, the coupling point includes a plurality of coupling points, and the frame assembly includes a leg frame that circumscribes the tubular portion, includes the plurality of legs, and is coupled to the valve frame at the plurality of coupling points, such that the plurality of legs is distributed circumferentially around the tubular portion.

(84) In an application, the plurality of coupling points is disposed circumferentially around the frame assembly on a transverse plane that is orthogonal to the longitudinal axis of the frame assembly.

(85) In an application, the plurality of legs is coupled to the valve frame via a plurality of struts, each strut: having a first end that is coupled to a leg of the plurality of legs, and a second end that is coupled to a coupling point of the plurality of coupling points, in the compressed state of the frame assembly, being disposed at a first angle in which the first end is disposed closer to the downstream end of the frame assembly than is the second end, and being deflectable with respect to the coupling point of the plurality of coupling points, such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes the strut to deflect to a second angle in which the first end is disposed further from the downstream end of the frame assembly than is the first end in the compressed state of the frame assembly.

(86) In an application, the leg frame is structured such that each leg of the plurality of legs is coupled to two struts of the plurality of struts, and two struts of the plurality of struts are coupled to each coupling point of the plurality of coupling points.

(87) In an application, the leg is coupled to the valve frame via a strut, the strut: having a first end that is coupled to the leg, and a second end that is coupled to the coupling point, in the compressed state of the frame assembly, being disposed at a first angle in which the first end is disposed closer to the downstream end of the frame assembly than is the second end, and being deflectable with respect to the coupling point, such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes the strut to deflect to a second angle in which the first end is disposed further from the

downstream end of the frame assembly than is the first end in the compressed state of the frame assembly.

(88) In an application, the at least one leg includes at least a first leg and a second leg.

(89) In an application, the first leg and the second leg are both coupled to the valve frame at the coupling point.

(90) In an application, the first leg is coupled to the coupling point via a respective first strut, and the second leg is coupled to the coupling point via a respective second strut.

(91) In an application, the first and second legs, the first and second struts, and the coupling point are arranged such that, in the expanded state of the frame assembly: the coupling point is disposed, circumferentially with respect to the tubular portion, between the first strut and the second strut, the first strut is disposed, circumferentially with respect to the tubular portion, between the coupling point and the first leg, and the second strut is disposed, circumferentially with respect to the tubular portion, between the coupling point and the second leg.

(92) In an application, the coupling point includes at least a first coupling point and a second coupling point.

(93) In an application, the leg is coupled to the valve frame at the first coupling point and at the second coupling point.

(94) In an application, the leg is coupled to the first coupling point via a respective first strut, and to the second coupling point via a respective second strut.

(95) In an application, the first and second legs, the first and second struts, and the coupling point are arranged such that, in the expanded state of the frame assembly: the leg is disposed, circumferentially with respect to the tubular portion, between the first strut and the second strut, the first strut is disposed, circumferentially with respect to the tubular portion, between the leg and the first coupling point, and the second strut is disposed, circumferentially with respect to the tubular portion, between the leg and the second coupling point.

(96) In an application, in the expanded state of the frame assembly, the upstream support portion extends radially outward from the tubular portion.

(97) In an application: the expanded state of the frame assembly is a fully-expanded state of the frame assembly, the upstream support portion is expandable into an expanded state of the upstream support portion, independently of increasing the diameter of the tubular portion, and in the expanded state of the upstream support portion, the upstream support portion extends radially outward from the tubular portion.

(98) In an application, in the compressed state of the frame assembly, the upstream support portion is generally tubular, collinear with the tubular portion, and disposed around the central longitudinal axis.

(99) In an application, in the expanded state of the frame assembly, an inner region of the upstream support portion extends radially outward from the tubular portion at a first angle with respect to the tubular portion, and an outer region of the upstream support portion extends, from the inner region of the upstream support portion, further radially outward from the tubular portion at a second angle with respect to the tubular portion, the second angle being smaller than the first angle.

(100) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including: a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, and including: a valve frame, including: a tubular portion having an upstream end and a downstream end, and shaped to define a lumen therebetween, and an upstream support portion, extending from the upstream end of the tubular portion; and at least one leg, coupled to the valve frame at a coupling point, and having a tissue-engaging flange; and a valve member disposed within the lumen, and configured to facilitate one-way liquid flow through the lumen from the upstream end of the tubular portion to the downstream end of the tubular portion, and the frame assembly: has a compressed state, for percutaneous delivery to the heart, in which the tubular portion has a compressed diameter, is biased to assume an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and is configured such that reducing the diameter of the tubular portion toward the compressed diameter causes longitudinal movement: of the upstream support portion away from the coupling point, and of the tissue-engaging flange toward the coupling point.

(101) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including: a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, including: a valve frame, including: a tubular portion having an upstream end and a downstream end, and shaped to define a lumen therebetween, and an upstream support portion, extending from the upstream end of the tubular portion; and at least one

leg, coupled to the valve frame at a coupling point, and having a tissue-engaging flange; and a valve member disposed within the lumen, and configured to facilitate one-way liquid flow through the lumen from the upstream end of the tubular portion to the downstream end of the tubular portion, and the frame assembly: has a compressed state, for percutaneous delivery to the heart, is intracorporeally expandable into an expanded state in which a diameter of the tubular portion is greater than in the compressed state, and is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes longitudinal movement of the tissue-engaging flange away from the coupling point.

(102) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including: a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, and including: an inner frame including an inner-frame tubular portion that circumscribes the central longitudinal axis, has an upstream end and a downstream end, and defines a channel therebetween, the inner frame defining a plurality of inner-frame couplings disposed circumferentially at a longitudinal location of the inner frame, an outer frame including an outer-frame tubular portion that coaxially circumscribes at least a portion of the inner-frame tubular portion, the outer frame defining a plurality of outer-frame couplings disposed circumferentially at a longitudinal location of the outer frame, and a plurality of connectors, each connector connecting a respective inner-frame coupling to a respective outer-frame coupling; a liner, disposed over at least part of the inner-frame tubular portion; and a plurality of prosthetic leaflets, coupled to the inner-frame tubular portion and disposed within the channel,

and: the frame assembly: (i) is compressible by a radially-compressive force into a compressed state in which the inner frame is in a compressed state thereof and the outer frame is in a compressed state thereof, (ii) is configured, upon removal of the radially-compressive force, to automatically expand into an expanded state thereof in which the inner frame is in an expanded state thereof and the outer frame is in an expanded state thereof, in the expanded state of the frame assembly, the prosthetic leaflets are configured to facilitate one-way fluid flow, in a downstream direction, through the channel, and the connection of the inner-frame couplings to the respective outer-frame couplings is such that expansion of the frame assembly from the compressed state to the expanded state causes the inner-frame tubular portion to slide longitudinally in a downstream direction with respect to the outer-frame tubular portion.

(103) There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve disposed between an atrium and a ventricle of a heart of a subject, the apparatus including: a tubular portion, having an upstream portion that includes an upstream end, and a downstream portion that includes a downstream end, and shaped to define a lumen between the upstream portion and the downstream portion; a plurality of prosthetic leaflets, disposed within the lumen, and arranged to provide unidirectional flow of blood from the upstream portion to the downstream portion; an annular upstream support portion: having an inner portion that extends radially outward from the upstream portion, and including one or more fabric pockets disposed circumferentially around the inner portion, each pocket of the one or more pockets having an opening that faces a downstream direction.

(104) In an application: the upstream support portion includes (i) a plurality of arms that extend radially outward from the tubular portion, and (ii) a covering, disposed over the plurality of arms, each arm has (i) a radially-inner part at the inner portion of the upstream support portion, and (ii) a radially-outer part at the outer portion of the upstream support portion, at the inner portion of the upstream support portion, the covering is closely-fitted between the radially-inner parts of the arms, and at the outer portion of the upstream support portion, the pockets are formed by the covering being loosely-fitted between the radially-outer parts of the arms.

(105) In an application: the upstream support portion includes (i) a plurality of arms that extend radially outward from the tubular portion, and (ii) a covering, disposed over the plurality of arms, each arm has (i) a radially-inner part at the inner portion of the upstream support portion, and (ii) a radially-outer part at the outer portion of the upstream support portion, the radially-outer part being more flexible than the radially-inner part.

(106) In an application: the upstream support portion includes (i) a plurality of arms that extend radially outward from the tubular portion, and (ii) a covering, disposed over the plurality of arms, each arm has (i) a radially-inner part at the inner portion of the upstream support portion, and (ii) a radially-outer part at the outer portion of the upstream support portion, at the outer portion of the upstream support portion, the pockets are formed by each arm curving to form a hook shape.

(107) In an application, each pocket is shaped and arranged to billow in response to perivalvular flow of blood in an upstream direction.

(108) In an application, the apparatus is configured to be transluminally delivered to the heart, and implanted at the native valve by expansion of the apparatus, such that the upstream support portion is disposed in the atrium and the tubular portion extends from the upstream support portion into the ventricle, and each pocket is shaped and arranged such that perivalvular flow of blood in an upstream direction presses the pocket against tissue of the atrium.

(109) There is further provided, in accordance with an application of the present invention, apparatus including: a plurality of prosthetic valve leaflets; and a frame assembly, including: a tubular portion defined by a repeating pattern of cells, the tubular portion extending circumferentially around a longitudinal axis so as to define a longitudinal lumen, the prosthetic valve leaflets coupled to the inner frame and disposed within the lumen; an outer frame, including a plurality of legs, distributed circumferentially around the tubular portion, each leg having a tissue-engaging flange; an upstream support portion that includes a plurality of arms that extend radially outward from the tubular portion; and a plurality of appendages, each having a first end that defines a coupling element via which the tubular portion is coupled to the outer frame, and a second end; and the frame assembly defines a plurality of hubs, distributed circumferentially around the longitudinal axis on a plane that is transverse to the longitudinal axis, each hub defined by convergence and connection of, (i) two adjacent cells of the tubular portion, (ii) an arm of the plurality of arms, and (iii) an appendage of the plurality of appendages.

(110) In an application, each hub has six radiating spokes, two of the six spokes being part of a first cell of the two adjacent cells, two of the six spokes being part of a second cell of the two adjacent cells, one of the six spokes being the arm, and one of the six spokes being the second end of the appendage.

(111) In an application, the appendages are in-plane with the tubular portion.

(112) In an application, the appendages are in-plane with the outer frame.

(113) There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including: percutaneously advancing to heart, an implant: including a valve frame, a valve member disposed within a lumen defined by the valve frame, and at least one leg, coupled to the valve frame at a coupling point, and having an upstream end, a downstream end, and a central longitudinal axis therebetween; positioning the implant within the heart such that a tissue-engaging flange of the leg is disposed downstream of the valve, and thereafter causing the flange to protrude radially outward from the axis; subsequently, while an upstream support portion of the valve frame is disposed upstream of the valve, causing the upstream support portion to protrude radially outward from the axis, such that tissue of the valve is disposed between the upstream support portion and the flange; and subsequently, sandwiching the tissue between the upstream support portion and the flange by reducing a distance between the upstream support portion and the flange by causing longitudinal movement (i) of the upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point.

(114) In an application, causing the longitudinal movement (i) of the upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point, includes causing the longitudinal movement by increasing a diameter of the lumen.

(115) There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including: percutaneously advancing to heart, an implant: including a valve frame, a valve member disposed within a lumen defined by the valve frame, and at least one leg, coupled to the valve frame at a coupling point, and having an upstream end, a downstream end, and a central longitudinal axis therebetween; positioning the implant within the heart such that an upstream support portion of the valve frame is disposed upstream of the valve, and thereafter causing the upstream support portion to protrude radially outward from the axis; subsequently, while a tissue-engaging flange of the leg is disposed downstream of the valve, causing the tissue-engaging flange to protrude radially outward from the axis, such that tissue of the valve is disposed between the upstream support portion and the flange; and subsequently, sandwiching the tissue between the upstream support portion and the flange by reducing a distance between the upstream support portion and the flange by causing longitudinal movement (i) of the upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point.

(116) In an application, causing the longitudinal movement (i) of the upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point, includes causing the

longitudinal movement by increasing a diameter of the lumen.

(117) There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including: percutaneously advancing an implant to the heart, the implant: having an upstream end, a downstream end, and a central longitudinal axis therebetween, and including a tubular portion, an upstream support portion, and a plurality of tissue-engaging flanges; positioning the implant within the heart such that the upstream support portion is disposed upstream of the valve, positioning the implant within the heart such that the tissue-engaging flanges are disposed downstream of the valve, without increasing a diameter of the tubular portion: causing the upstream support portion to extend radially outward from the axis so as to have a first support-portion span, and causing the flanges to extend radially outward from the axis so as to have a first flange span; and subsequently, causing the upstream support portion and the flanges move toward each other by at least 5 mm by increasing a diameter of the tubular portion such that: the upstream support portion extends radially outward so as to have a second support-portion span, the first support-portion span being at least 40 percent as great as the second support-portion span, and the flanges extend radially outward so as to have a second flange span, the first flange span being at least 30 percent as great as the second flange span.

(118) There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including: percutaneously advancing an implant to the heart, the implant: having an upstream end, a downstream end, and a central longitudinal axis therebetween, and including a tubular portion, an upstream support portion, and a plurality of tissue-engaging flanges; positioning the implant within the heart such that the upstream support portion is disposed upstream of the valve, positioning the implant within the heart such that the tissue-engaging flanges are disposed downstream of the valve, without increasing a diameter of the tubular portion: causing the upstream support portion to extend radially outward from the axis, and causing the flanges to extend radially outward from the axis so as to have a first flange span; and subsequently, by increasing a diameter of the tubular portion: causing the upstream support portion and the flanges move toward each other by at least 5 mm, causing the upstream support portion to move further radially outward from the axis, and causing each flange of the plurality of flanges to translate radially outward so as to have a second flange span that is greater than the first flange span.

(119) The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-B and 2A-H are schematic illustrations of an implant for use with a native valve of a heart of a subject, in accordance with some applications of the invention;

(2) FIGS. 3A-C are schematic illustrations that show structural changes in a frame assembly during transitioning of the assembly between its compressed and expanded states, in accordance with some applications of the invention;

(3) FIGS. 4A-F are schematic illustrations of implantation of the implant at the native valve, in accordance with some applications of the invention;

(4) FIG. 5 is a schematic illustration of a step in the implantation of the implant, in accordance with some applications of the invention;

(5) FIG. 6 is a schematic illustration of the implant, in accordance with some applications of the invention;

(6) FIGS. 7A-B and 8A-B are schematic illustrations of frame assemblies of respective implants, in accordance with some applications of the invention; and

(7) FIGS. 9A-C are schematic illustrations of an implant comprising a frame assembly, in accordance with some applications of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) Reference is made to FIGS. 1A-B and 2A-H, which are schematic illustrations of an implant **20** for use with a native valve of a heart of a subject, in accordance with some applications of the invention. Implant **20** comprises a frame assembly **22** that has an upstream end **24**, a downstream end **26**, and a central longitudinal axis ax1 therebetween. Frame assembly **22** comprises a valve frame **30** that comprises a tubular portion **32** that has an upstream end **34** and a downstream end **36**, and is shaped to define a lumen **38** through the

tubular portion from the upstream end to the downstream end. Tubular portion 32 circumscribes axis ax1, and thereby defines lumen 38 along the axis. Valve frame 30 further comprises an upstream support portion 40, extending from upstream end 34 of tubular portion 32. Frame assembly 22 further comprises at least one leg 50, coupled to valve frame 30 at (e.g., via) a coupling point 52, and having a tissue-engaging flange 54. (9) Typically, and as described hereinbelow, leg 50 is part of an outer frame (or “leg frame”) 60, and frames 30 and 60 define respective coupling elements 31 and 61, which are fixed with respect to each other at coupling points 52. Typically, frames 30 and 60 are coupled to each other only at coupling points 52 (e.g., only via the fixation of coupling elements 31 and 61 with respect to each other).

(10) Implant 20 further comprises a valve member 58 (e.g., one or more prosthetic leaflets) disposed within lumen 38, and configured to facilitate one-way liquid flow through the lumen from upstream end 34 to downstream end 36 (e.g., thereby defining the orientation of the upstream and downstream ends of tubular portion 32). FIG. 1A shows implant 20 in a fully-expanded state, in which frame assembly 22 is in a fully-expanded state. FIG. 1B shows an exploded view of frame assembly 22 in its fully-expanded state. FIGS. 2A-E show respective states of implant 20, which will be discussed in more detail hereinbelow with respect to the implantation of the implant and the anatomy in which the implant is implanted. FIG. 2A shows implant 20 in a compressed state (in which frame assembly 22 is in a compressed state), for percutaneous delivery of the implant to the heart of the subject. Typically, in the compressed state, leg 50 (including flange 54 thereof) is in a constrained-flange state in which the flange is generally parallel with axis ax1. Further typically, in the compressed state, upstream support portion 40 is generally tubular, collinear with tubular portion 32 (e.g., extending collinearly from the tubular portion), and disposed around axis ax1.

(11) FIG. 2B shows a state of implant 20 in which tissue-engaging flange 54 of each leg 50 extends radially away from axis ax1 (e.g., radially away from tubular portion 32). FIG. 2C shows a state of implant 20 in which upstream-support portion 40 extends radially away from axis ax1 (and thereby radially away from tubular portion 32). FIG. 2D shows a state of implant 20 in which both flange 54 and portion 40 extend away from axis ax1. In the fully-expanded state (FIGS. 1A-B) both upstream support portion 40 and flange 54 extend radially away from axis ax1. Typically, frame assembly 22 is biased (e.g., shape-set) to assume its fully-expanded state, which is shown in FIG. 2E. Transitioning of implant 20 between the respective states is typically controlled by delivery apparatus, such as by constraining the implant in a compressed state within a delivery tube and/or against a control rod, and selectively releasing portions of the implant to allow them to expand.

(12) In the compressed state of frame assembly 22, tubular portion 32 has a diameter d1, and in the expanded state, the tubular portion has a diameter d2 that is greater than diameter d1. For some applications, diameter d1 is 4-15 mm, (e.g., 5-11 mm) and diameter d2 is 20-50 mm, (e.g., 23-33 mm). Frame assembly 22 is configured such that increasing the diameter of tubular portion 32 (e.g., from d1 to d2) causes longitudinal movement of flange 54 away from coupling point 52. In the same way, reducing the diameter of tubular portion 32 (e.g., from d2 to d1) causes longitudinal movement of flange 54 toward coupling point 52. It is to be noted that the term “longitudinal movement” (including the specification and the claims) means movement parallel with central longitudinal axis ax1. Therefore, longitudinal movement of flange 54 away from coupling point 52 means increasing a distance, measured parallel with longitudinal axis ax1, between flange 54 and coupling point 52. An example of such a configuration is described in more detail with respect to FIG. 3A.

(13) Thus, expansion of tubular portion 32 from its compressed state toward its expanded state (i) increases a circumferential distance between each of coupling points 52 and its adjacent coupling points (e.g., between each of outer-frame coupling elements 61 and its adjacent outer-frame coupling elements) (e.g., from d8 to d9), and (ii) moves legs 50 in a longitudinally upstream direction with respect to the tubular portion.

(14) Typically, frame assembly 22 is configured such that increasing the diameter of tubular portion 32 also causes longitudinal movement of upstream support portion 40 toward coupling point 52, e.g., as described in more detail with respect to FIGS. 3B-C. Typically, frame assembly 22 is configured such that increasing the diameter of tubular portion 32 also causes longitudinal movement of upstream end 34 of tubular portion 32 toward coupling point 52. In the same way, reducing the diameter of tubular portion 32 causes longitudinal movement of upstream end 34 away from coupling point 52.

(15) For some applications, upstream support portion 40 comprises a plurality of arms 46 that each extends radially outward from tubular portion 32 (e.g., from upstream end 34 of the tubular portion). Arms 46 are typically flexible. For some such applications, arms 46 are coupled to tubular portion 32 such that each arm may deflect independently of adjacent arms during implantation (e.g., due to anatomical topography).

(16) For some applications, upstream support portion **40** comprises a plurality of barbs **48** that extend out of a downstream surface of the upstream support portion. For example, each arm **46** may comprise one or more of barbs **48**. Barbs **48** press into tissue upstream of the native valve (e.g., into the valve annulus), thereby inhibiting downstream movement of implant **20** (in addition to inhibition of downstream movement provided by the geometry of upstream support portion **40**).

(17) One or more surfaces of frame assembly **22** are covered with a covering **23**, which typically comprises a flexible sheet, such as a fabric, e.g., comprising polyester. Typically, covering **23** covers at least part of tubular portion **32**, typically lining an inner surface of the tubular portion, and thereby defining lumen **38**.

(18) Further typically, upstream support portion **40** is covered with covering **23**, e.g., extending between arms **46** to form an annular shape. It is hypothesized that this reduces a likelihood of paravalvular leakage. For such applications, excess covering **23** may be provided between arms **46** of upstream support portion **40**, so as to facilitate their independent movement. Although FIG. 1A shows covering **23** covering an upstream side of upstream support portion **40**, the covering typically additionally (or alternatively) covers the downstream side of the upstream support portion. For example, covering **23** may extend over the tips of arms **46** and down the outside of the arms, or a separate piece of covering may be provided on the downstream side of the upstream support portion.

(19) Alternatively, each arm **46** may be individually covered in a sleeve of covering **23**, thereby facilitating independent movement of the arms.

(20) For some applications, at least part of legs **50** (e.g., flanges thereof) is covered with covering **23**.

(21) Typically, frame assembly **22** comprises a plurality of legs **50** (e.g., two or more legs, e.g., 2-16 legs, such as 4-12 legs, such as 6-12 legs), arranged circumferentially around valve frame **30** (e.g., around the outside of tubular portion **32**). Typically, frame assembly **22** comprises a plurality of coupling points **52** at which the legs are coupled to valve frame **30**.

(22) As described in more detail hereinbelow (e.g., with reference to FIG. 3A), each leg **50** is typically coupled to a coupling point **52** via a strut **70**. For some applications, each leg **50** is coupled to a plurality of (e.g., two) coupling points **52** via a respective plurality of (e.g., two) struts **70**. For some such applications, frame assembly **22** is arranged such that, in the expanded state of the frame assembly, leg **50** is disposed, circumferentially with respect to tubular portion **32**, between two struts, and each of the two struts are disposed, circumferentially with respect to the tubular portion, between the leg and a respective coupling point **52**.

(23) For some applications, a plurality of (e.g., two) legs are coupled to each coupling point **52** via a respective plurality of (e.g., two) struts **70**. For some such applications, frame assembly **22** is arranged such that, in the expanded state of the frame assembly, coupling point **52** is disposed, circumferentially with respect to tubular portion **32**, between two struts **70**, and each of the two struts are disposed, circumferentially with respect to the tubular portion, between the coupling point and a respective leg **50**.

(24) For some applications, frame assembly **22** comprises an outer frame (e.g., a leg frame) **60** that circumscribes tubular portion **32**, comprises (or defines) the plurality of legs **50** and the plurality of struts **70**, and is coupled to valve frame **30** at the plurality of coupling points **52**, such that the plurality of legs is distributed circumferentially around the tubular portion. For such applications, outer frame **60** comprises a ring **66** that is defined by a pattern of alternating peaks **64** and troughs **62**, and that typically circumscribes tubular portion **32**. For example, the ring may comprise struts **70**, extending between the peaks and troughs. Peaks **64** are longitudinally closer to upstream end **34** of tubular portion **32** than to downstream end **36**, and troughs **62** are longitudinally closer to the downstream end than to the upstream end. (It is to be noted that throughout this patent application, including the specification and the claims, the term “longitudinally” means with respect to longitudinal axis ax1. For example, “longitudinally closer” means closer along axis ax1 (whether positioned on axis ax1 or lateral to axis ax1), and “longitudinal movement” means a change in position along axis ax1 (which may be in addition to movement toward or away from axis ax1).)

Therefore, peaks **64** are closer than troughs **62** to upstream end **34**, and troughs **62** are closer than peaks **64** to downstream end **36**. For applications in which frame **60** comprises ring **66**, each leg **50** is coupled to the ring (or defined by frame **60**) at a respective trough **62**.

(25) In the embodiment shown, the peaks and troughs are defined by ring **66** having a generally zig-zag shape. However, the scope of the invention includes ring **66** having another shape that defines peaks and troughs, such as a serpentine or sinusoid shape.

(26) For applications in which frame assembly **22** has a plurality of coupling points **52**, the coupling points (and therefore coupling elements **31** and **61**) are disposed circumferentially around the frame assembly (e.g.,

around axis ax1), typically on a transverse plane that is orthogonal to axis ax1. This transverse plane is illustrated by the position of section A-A in FIG. 2B. Alternatively, coupling points 52 may be disposed at different longitudinal heights of frame assembly 22, e.g., such that different flanges 54 are positioned and/or moved differently to others. Typically, coupling points 52 (and therefore coupling elements 31 and 61) are disposed longitudinally between upstream end 24 and downstream end 26 of frame assembly 22, but not at either of these ends. Further typically, coupling points 52 are disposed longitudinally between upstream end 34 and downstream end 36 of tubular portion 32, but not at either of these ends. For example, the coupling points may be more than 3 mm (e.g., 4-10 mm) both from end 34 and from end 36. It is hypothesized that this advantageously positions the coupling points at a part of tubular portion 32 that is more rigid than end 34 or end 36.

(27) It is to be noted that leg 50 is typically expandable into its expanded state (e.g., a released-flange state) such that flange 54 extends away from axis ax1, independently of increasing the diameter of tubular portion 32 (e.g., as shown in FIGS. 2B & 2D). Similarly, upstream support portion 40 is typically expandable into its expanded state (e.g., a released-arm state) such that it (e.g., arms 46 thereof) extends away from axis ax1, independently of increasing the diameter of tubular portion 32 (e.g., as shown in FIGS. 2C & 2D). The state shown in FIG. 2D may be considered to be an intermediate state. Therefore, implant 20 is typically configured such that legs 50 (e.g., flanges 54 thereof) and upstream support portion 40 are expandable such that they both extend away from axis ax1, while retaining a distance d3 therebetween. This distance is subsequently reducible to a distance d4 by expanding tubular portion 32 (e.g., shown in FIG. 2E).

(28) For some applications, while tubular portion 32 remains in its compressed state, flange 54 can extend away from axis ax1 over 40 percent (e.g., 40-80 percent, such as 40-70 percent) of the distance that it extends from the axis subsequent to the expansion of the tubular portion. For example, for applications in which implant 20 comprises a flange on opposing sides of the implant, a span d15 of the flanges while tubular portion 32 is in its compressed state may be at least 40 percent (e.g., 40-80 percent, such as 40-70 percent) as great as a span d16 of the flanges subsequent to the expansion of the tubular portion. For some applications, span d15 is greater than 15 mm and/or less than 50 mm (e.g., 20-30 mm). For some applications, span d16 is greater than 30 mm and/or less than 60 mm (e.g., 40-50 mm). It is to be noted that flange 54 is effectively fully expanded, with respect to other portions of leg 50 and/or with respect to tubular portion 32, before and after the expansion of the tubular portion.

(29) Similarly, for some applications, while tubular portion 32 remains in its compressed state, upstream support portion 40 (e.g., arms 46) can extend away from axis ax1 over 30 percent (e.g., 30-70 percent) of the distance that it extends from the axis subsequent to the expansion of the tubular portion. That is, for some applications, a span d17 of the upstream support portion while tubular portion 32 is in its compressed state may be at least 30 percent (e.g., 30-70 percent) as great as a span d18 of the upstream support portion subsequent to the expansion of the tubular portion. For some applications, span d17 is greater than 16 mm (e.g., greater than 20 mm) and/or less than 50 mm (e.g., 30-40 mm). For some applications, span d18 is greater than 40 mm and/or less than 65 mm (e.g., 45-56 mm, such as 45-50 mm). It is to be noted that upstream support portion 40 is effectively fully expanded, with respect to tubular portion 32, before and after the expansion of the tubular portion.

(30) It is to be noted that when tubular portion 32 is expanded, flanges 54 typically translate radially outward from span d15 to span d16 (e.g., without deflecting). Typically, upstream support portion 40 behaves similarly (e.g., arms 46 translated radially outward from span d17 to span d18, e.g., without deflecting). That is, an orientation of each flange 54 and/or each arm 46 with respect to tubular portion 32 and/or axis ax1 is typically the same in the state shown in FIG. 2D as it is in the state shown in FIG. 2E. Similarly, for some applications an orientation of each flange 54 with respect to upstream support portion 40 (e.g., with respect to one or more arms 46 thereof) is the same before and after expansion of tubular portion 32.

(31) For some applications, increasing the diameter of tubular portion 32 from d1 to d2 causes greater than 1 mm and/or less than 20 mm (e.g., 1-20 mm, such as 1-10 mm or 5-20 mm) of longitudinal movement of flange 54 away from coupling point 52. For some applications, increasing the diameter of tubular portion 32 from d1 to d2 causes greater than 1 mm and/or less than 20 mm (e.g., 1-20 mm, such as 1-10 mm or 5-20 mm) of longitudinal movement of upstream support portion 40 toward coupling point 52. For some applications, distance d3 is 7-30 mm. For some applications, distance d4 is 0-15 mm (e.g., 2-15 mm). For some applications, increasing the diameter of tubular portion 32 from d1 to d2 reduces the distance between the upstream support portion and flanges 54 by more than 5 mm and/or less than 30 mm, such as 5-30 mm (e.g., 10-30 mm, such as 10-20 mm or 20-30 mm). For some applications, the difference between d3 and d4

is generally equal to the difference between d1 and d2. For some applications, the difference between d3 and d4 is more than 1.2 and/or less than 3 times (e.g., 1.5-2.5 times, such as about 2 times) greater than the difference between d1 and d2.

(32) For some applications, flanges **54** curve such that a tip of each flange is disposed at a shallower angle with respect to inner region **42** of upstream support portion **40**, than are portions of leg **50** that are closer to downstream end **26** of frame assembly **22**. For some such applications, a tip of each flange may be generally parallel with inner region **42**. For some such applications, while tubular portion **32** is in its expanded state, a tip portion **55** of each flange **54** that extends from the tip of the flange at least 2 mm along the flange, is disposed within 2 mm of upstream support portion **40**. Thus, for some applications, while tubular portion **32** is in its expanded state, for at least 5 percent (e.g., 5-8 percent, or at least 8 percent) of span **18** of upstream support portion **40**, the upstream support portion is disposed within 2 mm of a flange **54**.

(33) For some applications, in the absence of any obstruction (such as tissue of the valve or covering **23**) between flange **54** and upstream support portion **40**, increasing the diameter of tubular portion **32** from d1 to d2 causes the flange and the upstream support portion to move past each other (e.g., the flange may move between arms **46** of the upstream support portion), such that the flange is closer to the upstream end of implant **20** than is the upstream support portion, e.g., as shown hereinbelow for frame assemblies **122** and **222**, mutatis mutandis. (For applications in which upstream support portion **40** is covered by covering **23**, flanges **54** typically don't pass the covering. For example, in the absence of any obstruction, flanges **54** may pass between arms **46**, and press directly against covering **23**.) It is hypothesized that for some applications this configuration applies greater force to the valve tissue being sandwiched, and thereby further facilitates anchoring of the implant. That is, for some applications, distance d3 is smaller than the sum of distance d5 and a distance d14 (described with reference to FIG. 3C). For some applications, increasing the diameter of tubular portion **32** from d1 to d2 advantageously causes flanges **54** and upstream support portion **40** to move greater than 3 mm and/or less than 25 mm (e.g., greater than 5 mm and/or less than 15 mm, e.g., 5-10 mm, such as about 7 mm) with respect to each other (e.g., toward each other and then past each other).

(34) For some applications, in the expanded state of frame assembly **22**, upstream support portion **40** has an inner region (e.g., an inner ring) **42** that extends radially outward at a first angle with respect to axis ax1 (and typically with respect to tubular portion **32**), and an outer region (e.g., an outer ring) **44** that extends, from the inner region, further radially outward from the tubular portion at a second angle with respect to the tubular portion, the second angle being smaller than the first angle. For example, for some applications inner region **42** extends radially outward at an angle α_1 of 60-120 degrees (e.g., 70-110 degrees) with respect to axis ax1, and outer region **44** extends radially outward at an angle α_2 of 5-70 degrees (e.g., 10-60 degrees) with respect to axis ax1.

(35) It is to be noted that angles α_1 and α_2 are measured between the respective region support portion **40**, and the portion of axis ax1 that extends in an upstream direction from the level of frame assembly **22** at which the respective region begins to extend radially outward.

(36) For some applications in which implant **20** is configured to be placed at an atrioventricular valve (e.g., a mitral valve or a tricuspid valve) of the subject, region **42** is configured to be placed against the upstream surface of the annulus of the atrioventricular valve, and region **44** is configured to be placed against the walls of the atrium upstream of the valve.

(37) For some applications, each arm **46** defines a rigid portion **43** and a flexible portion **45** that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion. Typically, at least part of flexible portion **45** is disposed in outer region **44**, and at least part of rigid portion **43** is disposed in inner region **42**. Therefore, typically overall, outer region **44** is more flexible than inner region **42**. To achieve this differential flexibility, each arm **46** may have a different structure in portion **45** and/or region **44**, than it has in portion **43** and/or region **42**.

(38) FIG. 2G shows an arm **46a**, which is a variant of arm **46**. Arm **46a** defines flexible portion **45a**, which is a variant of flexible portion **45**. In flexible portion **45a**, the material from which arm **46a** is formed is thinner than in rigid portion **43**, but defines a serpentine or zig-zag pattern that increases the overall thickness of portion **45a**, and as shown, may make the overall thickness of portion **45a** equal to or greater than the overall thickness of portion **43**.

(39) FIG. 2H shows an arm **46b**, which is a variant of arm **46**. Arm **46b** defines flexible portion **45b**, which is a variant of flexible portion **45**. In flexible portion **45b**, arm **46b** defines holes **47** in the material from which arm **46b** is formed. Nonetheless, as shown, the overall thickness of portion **45b** may be equal to or greater than the overall thickness of portion **43**.

(40) Therefore, as shown for arms **46a** and **46b**, flexible portion **45** covers an overall surface area including (i) the area in which the material from which the arm is formed occupies, and (ii) spaces, such as holes, or gaps between turns. For some applications, the material from which the arm is formed occupies less than 80 percent and/or more than 20 percent of the overall surface area (e.g., 20-80 percent, e.g., 40-80 percent, e.g., 40-70 percent of the overall surface area).

(41) It is hypothesized that the relative rigidity of portion **43** and/or region **42** provides resistance against axial movement (e.g., ventricular migration) of implant **20**, while the relative flexibility of portion **45** and/or region **44** facilitates conformation of upstream support portion **40** to the atrial anatomy and/or lateral movement of the implant within the native valve.

(42) For some applications, and as shown (e.g., in FIGS. **1A** and **2E-F**), in the expanded state of frame assembly **22**, the majority of flexible portion **45** (e.g., more than 60 percent and/or less than 90 percent) is disposed further radially outward than span **d16** of flanges **54**.

(43) For some applications, two or more of arms **46** are connected by a connector (not shown), reducing the flexibility, and/or the independence of movement of the connected arms relative to each other. For some applications, arms **46** are connected in particular sectors of upstream support portion **40**, thereby making these sectors more rigid than sectors in which the arms are not connected. For example, a relatively rigid sector may be provided to be placed against the posterior portion of the mitral annulus, and a relatively flexible sector may be provided to be placed against the anterior side of the mitral annulus, so as to reduce forces applied by upstream support portion **40** on the aortic sinus.

(44) For some applications, and as shown, coupling points **52** are disposed closer to downstream end **26** of frame assembly **22** than are flanges **54**, or is upstream support portion **40**.

(45) As described in more detail with respect to FIGS. **4A-F**, the movement of flange **54** away from coupling point **52** (and the typical movement of upstream support portion **40** toward the coupling point) facilitates the sandwiching of tissue of the native valve (e.g., leaflet and/or annulus tissue) between the flange and the upstream support portion, thereby securing implant **20** at the native valve.

(46) Typically, in the compressed state of tubular portion **32**, a downstream end of each leg **50** is longitudinally closer than valve-frame coupling elements **31** to downstream end **36**, and flange **54** of each leg is disposed longitudinally closer than the valve-frame coupling elements to upstream end **34**. Typically, this is also the case in the expanded state of tubular portion **32**.

(47) FIGS. **3A-C** show structural changes in frame assembly **22** during transitioning of the assembly between its compressed and expanded states, in accordance with some applications of the invention. FIGS. **3A-C** each show a portion of the frame assembly, the structural changes thereof being representative of the structural changes that occur in other portions of the frame assembly. FIG. **3A** shows a leg **50** and struts **70** (e.g., a portion of outer frame **60**), and illustrates the structural changes that occur around outer frame **60**. FIG. **3B** shows a portion of valve frame **30**, and illustrates the structural changes that occur around the valve frame. FIG. **3C** shows valve frame **30** as a whole. In each of FIGS. **3A-C**, state (A) illustrates the structure while frame assembly **22** (and in particular tubular portion **32**) is in its compressed state, and state (B) illustrates the structure while the frame assembly (and in particular tubular portion **32**) is in its expanded state.

(48) FIG. **3A** shows structural changes in the coupling of legs **50** to coupling point **52** (e.g., structural changes of outer frame **60**) during the transitioning of frame assembly **22** (and in particular tubular portion **32**) between its compressed and expanded states. Each leg **50** is coupled to valve frame **30** via at least one strut **70**, which connects the leg to coupling point **52**. Typically, each leg **50** is coupled to valve frame **30** via a plurality of struts **70**. A first end **72** of each strut **70** is coupled to leg **50**, and a second end **74** of each strut is coupled to a coupling point **52**. As described hereinabove, for applications in which frame **60** comprises ring **66**, each leg **50** is coupled to the ring at a respective trough **62**. Ring **66** may comprise struts **70**, extending between the peaks and troughs, with each first end **72** at (or close to) a trough **62**, and each second end **74** at (or close to) a peak **64**.

(49) In the compressed state of frame assembly **22** (and in particular of tubular portion **32**), each strut **70** is disposed at a first angle in which first end **72** is disposed closer than second end **74** to the downstream end of the frame assembly. Expansion of frame assembly **22** (and in particular of tubular portion **32**) toward its expanded state causes strut **70** to deflect to a second angle. This deflection moves first end **72** away from the downstream end of frame assembly **22**. That is, in the expanded state of frame assembly **22**, first end **72** is further from the downstream end of the frame assembly than it is when the frame assembly is in its compressed state. This movement is shown as a distance **d5** between the position of end **72** in state (A) and

its position in state (B). This movement causes the above-described movement of flanges **54** away from coupling points **52**. As shown, flanges **54** typically move the same distance $d5$ in response to expansion of frame assembly **22**.

(50) For applications in which outer frame **60** comprises ring **66**, the pattern of alternating peaks and troughs may be described as having an amplitude longitudinally between the peaks and troughs, i.e., measured parallel with central longitudinal axis $ax1$ of frame assembly **22**, and the transition between the compressed and expanded states may be described as follows: In the compressed state of frame assembly **22** (and in particular of tubular portion **32**), the pattern of ring **66** has an amplitude $d20$. In the expanded state frame assembly **22** (and in particular of tubular portion **32**), the pattern of ring **66** has an amplitude $d21$ that is lower than amplitude $d20$. Because (i) it is at peaks **64** that ring **66** is coupled to valve frame **30** at coupling points **52**, and (ii) it is at troughs **62** that ring **66** is coupled to legs **50**, this reduction in the amplitude of the pattern of ring **66** moves legs **50** (e.g., flanges **54** thereof) longitudinally further from the downstream end of the frame assembly. The magnitude of this longitudinal movement (e.g., the difference between magnitudes $d20$ and $d21$) is equal to $d5$.

(51) Typically, distance $d5$ is the same distance as the distance that flange **54** moves away from coupling point **52** during expansion of the frame assembly. That is, a distance between flange **54** and the portion of leg **50** that is coupled to strut **70**, typically remains constant during expansion of the frame assembly. For some applications, the longitudinal movement of flange **54** away from coupling point **52** is a translational movement (e.g., a movement that does not include rotation or deflection of the flange).

(52) For some applications, a distance $d6$, measured parallel to axis $ax1$ of frame assembly **22**, between coupling point **52** and first end **72** of strut **70** while assembly **22** is in its compressed state, is 3-15 mm. For some applications, a distance $d7$, measured parallel to axis $ax1$, between coupling point **52** and first end **72** of strut **70** while assembly **22** is in its expanded state, is 1-5 mm (e.g., 1-4 mm).

(53) For some applications, amplitude $d20$ is 2-10 mm (e.g., 4-7 mm). For some applications, amplitude $d21$ is 4-9 mm (e.g., 5-7 mm).

(54) For some applications, and as shown, in the expanded state, first end **72** of strut **70** is disposed closer to the downstream end of frame assembly **22** than is coupling point **52**. For some applications, in the expanded state, first end **72** of strut **70** is disposed further from the downstream end of frame assembly **22** than is coupling point **52**.

(55) For applications in which frame assembly **22** comprises a plurality of legs **50** and a plurality of coupling points **52** (e.g., for applications in which the frame assembly comprises outer frame **60**) expansion of the frame assembly increases a circumferential distance between adjacent coupling points **52**, and an increase in a circumferential distance between adjacent legs **50**. FIG. 3A shows such an increase in the circumferential distance between adjacent coupling points **52**, from a circumferential distance $d8$ in the compressed state to a circumferential distance $d9$ in the expanded state. For some applications, distance $d8$ is 1-6 mm. For some applications, distance $d9$ is 3-15 mm.

(56) For some applications, in addition to being coupled via ring **66** (e.g., struts **70** thereof) legs **50** are also connected to each other via connectors **78**. Connectors **78** allow the described movement of legs **50** during expansion of frame assembly **22**, but typically stabilize legs **50** relative to each other while the frame assembly is in its expanded state. For example, connectors **78** may bend and/or deflect during expansion of the frame assembly.

(57) FIGS. 3B-C show structural changes in valve frame **30** during the transitioning of frame assembly **22** between its compressed and expanded states. Tubular portion **32** of valve frame **30** is defined by a plurality of cells **80**, which are defined by the repeating pattern of the valve frame. When frame assembly **22** is expanded from its compressed state toward its expanded state, cells **80** (i) widen from a width $d10$ to a width $d11$ (measured orthogonal to axis $ax1$ of the frame assembly), and (ii) shorten from a height $d12$ to a height $d13$ (measured parallel to axis $ax1$ of the frame assembly). This shortening reduces the overall height (i.e., a longitudinal length between upstream end **34** and downstream end **36**) of tubular portion **32** from a height $d22$ to a height $d23$, and thereby causes the above-described longitudinal movement of upstream support portion **40** toward coupling points **52** by a distance $d14$ (shown in FIG. 3C). For some applications, and as shown, coupling points **52** are disposed at the widest part of each cell.

(58) Due to the configurations described herein, the distance by which flanges **54** move with respect to (e.g., toward, or toward-and-beyond) upstream support portion **40** (e.g., arms **46** thereof), is typically greater than the reduction in the overall height of tubular portion **32** (e.g., more than 20 percent greater, such as more than 30 percent greater, such as more than 40 percent greater). That is, implant **20** comprises: a valve frame (**30**)

that comprises a tubular portion (32) that circumscribes a longitudinal axis (ax1) of the valve frame so as to define a lumen (38) along the axis, the tubular portion having an upstream end (34), a downstream end (36), a longitudinal length therebetween, and a diameter (e.g., d1 or d2) transverse to the longitudinal axis; a valve member (58), coupled to the tubular portion, disposed within the lumen, and arranged to provide unidirectional upstream-to-downstream flow of blood through the lumen; an upstream support portion (40), coupled to the tubular portion; and an outer frame (60), coupled to the tubular portion, and comprising a tissue-engaging flange (54), wherein: the implant has a first state (e.g., as shown in FIG. 2D and FIG. 4D) and a second state (e.g., as shown in FIG. 2E and FIG. 4E), in both the first state and the second state, (i) the upstream support portion extends radially outward from the tubular portion, and (ii) the tissue-engaging flange extends radially outward from the tubular portion, and the tubular portion, the upstream support portion, and the outer frame are arranged such that transitioning of the implant from the first state toward the second state: increases the diameter of the tubular portion by a diameter-increase amount (e.g., the difference between d1 and d2), decreases the length of the tubular portion by a length-decrease amount (e.g., the difference between d22 and d23), and moves the flange a longitudinal distance with respect to (e.g., toward or toward-and-beyond) the upstream support portion (e.g., the difference between d3 and d4), this distance being greater than the length-decrease amount.

(59) As shown in the figures, valve frame 30 is typically coupled to outer frame 60 by coupling between (i) a valve-frame coupling element 31 defined by valve frame 30, and (ii) an outer-frame coupling element 61 defined by outer frame 60 (e.g., an outer-frame coupling element is coupled to end 74 of each strut).

Typically, elements 31 and 61 are fixed with respect to each other. Each coupling point 52 is thereby typically defined as the point at which a valve-frame coupling element and a corresponding outer-frame coupling element 61 are coupled (e.g., are fixed with respect to each other). For some applications, and as shown, elements 31 and 61 are eyelets configured to be coupled together by a connector, such as a pin or suture. Alternatively or additionally, elements 31 and 61 are soldered or welded together. For example, elements 31 and 61 may be coupled together by a pin that passes through the eyelets, and is welded in place.

(60) Typically, and as shown, valve-frame coupling elements 31 are defined by tubular portion 32, and are disposed circumferentially around central longitudinal axis ax1. Outer-frame coupling elements 61 are coupled to ring 66 (or defined by frame 60, such as by ring 66) at respective peaks 64.

(61) As shown (e.g., in FIGS. 2A-E), valve frame 30 (e.g., tubular portion 32 thereof) and outer frame 60 (e.g., ring 66 thereof) are arranged in a close-fitting coaxial arrangement, in both the expanded and compressed states of frame assembly 22. Ignoring spaces due to the cellular structure of the frames, a radial gap d19 between valve frame 30 (e.g., tubular portion 32 thereof) and outer frame 60 (e.g., ring 66 thereof) is typically less than 2 mm (e.g., less than 1 mm), in both the compressed and expanded states, and during the transition therebetween. This is facilitated by the coupling between frames 30 and 60, and the behavior, described hereinabove, of frame 60 in response to changes in the diameter of tubular portion 32 (e.g., rather than solely due to delivery techniques and/or tools). For some applications, more than 50 percent (e.g., more than 60 percent) of ring 66 is disposed within 2 mm of tubular portion 32 in both the compressed and expanded states, and during the transition therebetween. For some applications, more than 50 percent (e.g., more than 60 percent) of outer frame 60, except for flanges 54, is disposed within 2 mm of tubular portion 32 in both the compressed and expanded states, and during the transition therebetween.

(62) The structural changes to frame assembly 22 (e.g., to outer frame 60 thereof) are described hereinabove as they occur during (e.g., as a result of) expansion of the frame assembly (in particular tubular portion 32 thereof). This is the natural way to describe these changes because, as described hereinbelow with respect to FIGS. 4A-6, assembly 22 is in its compressed state during percutaneous delivery to the implant site, and is subsequently expanded. However, the nature of implant 20 may be further understood by describing structural changes that occur during compression of the frame assembly (e.g., a transition from the expanded state in FIG. 2E to the intermediate state in FIG. 2D), in particular tubular portion 32 thereof (including if tubular portion 32 were compressed by application of compressive force to the tubular portion, and not to frame 60 except via the tubular portion pulling frame 60 radially inward). Such descriptions may also be relevant because implant 20 is typically compressed (i.e., “crimped”) soon before its percutaneous delivery, and therefore these changes may occur while implant 20 is in the care of the operating physician.

(63) For some applications, the fixation of peaks 64 to respective sites of tubular portion 32 is such that compression of the tubular portion from its expanded state toward its compressed state such that the respective sites of the tubular portion pull the peaks radially inward via radially-inward tension on coupling points 52: (i) reduces a circumferential distance between each of the coupling points and its adjacent

coupling points (e.g., from d9 to d8), and (ii) increases the amplitude of the pattern of ring **66** (e.g., from d21 to d20).

(64) For some applications, the fixation of outer-frame coupling elements **61** to valve-frame coupling elements **31** is such that compression of tubular portion **32** from its expanded state toward its compressed state such that the valve-frame coupling elements pull the outer-frame coupling elements radially inward: (i) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements (e.g., from d9 to d8), and (ii) increases the amplitude of the pattern of ring **66** (e.g., from d21 to d20).

(65) For some applications, the fixation of peaks **64** to the respective sites of tubular portion **32** is such that compression of the tubular portion from its expanded state toward its compressed state (i) pulls the peaks radially inward via radially-inward pulling of the respective sites of the tubular portion on the peaks, (ii) reduces a circumferential distance between each of coupling points **52** and its adjacent coupling points (e.g., from d9 to d8), and (iii) increases the amplitude of the pattern of ring **66** (e.g., from d21 to d20), without increasing radial gap d19 between valve frame **30** (e.g., tubular portion **32** thereof) and the ring by more than 1.5 mm.

(66) For some applications, the fixation of outer-frame coupling elements **61** with respect to valve-frame coupling elements **31** is such that compression of tubular portion **32** from its expanded state toward its compressed state (i) pulls outer-frame coupling elements **61** radially inward via radially-inward pulling of valve-frame coupling elements **31** on outer-frame coupling elements **61**, (ii) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements (e.g., from d9 to d8), and (iii) increases the amplitude of the pattern of ring **66** (e.g., from d21 to d20), without increasing radial gap d19 between valve frame **30** (e.g., tubular portion **32** thereof) and the ring by more than 1.5 mm.

(67) Reference is made to FIGS. **4A-F**, which are schematic illustrations of implantation of implant **20** at a native valve **10** of a heart **4** of a subject, in accordance with some applications of the invention. Valve **10** is shown as a mitral valve of the subject, disposed between a left atrium **6** and a left ventricle **8** of the subject. However, implant **20** may be implanted at another heart valve of the subject, mutatis mutandis. Similarly, although FIGS. **4A-F** show implant **20** being delivered transeptally via a sheath **88**, the implant may alternatively be delivered by any other suitable route, such as transatrially, or transapically.

(68) Implant **20** is delivered, in its compressed state, to native valve **10** using a delivery tool **89** that is operable from outside the subject (FIG. **4A**). Typically, implant **20** is delivered within a delivery capsule **90** of tool **89**, which retains the implant in its compressed state. Delivery capsule **90** is disposed at a distal end of tool **89**. A transeptal approach, such as a transfemoral approach, is shown. Typically, implant **20** is positioned such that at least flanges **54** are disposed downstream of the native valve (i.e., within ventricle **8**). At this stage, frame assembly **22** of implant **20** is as shown in FIG. **2A**.

(69) Subsequently, flanges **54** are allowed to protrude radially outward, as described hereinabove, e.g., by releasing them from capsule **90** (FIG. **4B**). For example, and as shown, capsule **90** may comprise a distal capsule-portion **92** and a proximal capsule-portion **94**, and the distal capsule-portion may be moved distally with respect to implant **20**, so as to expose flanges **54**. At this stage, frame assembly **22** of implant **20** is as shown in FIG. **2B**.

(70) Subsequently, implant **20** is moved upstream, such that upstream support portion **40**, in its compressed state, is disposed upstream of leaflets **12** (i.e., within atrium **6**). For some applications, the upstream movement of implant **20** causes flanges **54** to engage leaflets **12**. However, because of the relatively large distance d3 provided by implant **20** (described hereinabove), for some applications it is not necessary to move the implant so far upstream that flanges **54** tightly engage leaflets **12** and/or pull the leaflets upstream of the valve annulus. Upstream support portion **40** is then allowed to expand such that it protrudes radially outward, as described hereinabove, e.g., by releasing it from capsule **90** (FIG. **4D**). For example, and as shown, proximal capsule-portion **94** may be moved proximally with respect to implant **20**, so as to expose upstream support portion **40**. At this stage, frame assembly **22** of implant **20** is as shown in FIG. **2D**, in which: (i) distance d3 exists between upstream support portion **40** and flanges **54**, (ii) the flanges have span d15, (iii) the upstream support portion has span d17, and (iv) tubular portion **32** has diameter d1.

(71) Typically, expansion of frame assembly **22** is inhibited by distal capsule-portion **92**, e.g., by inhibiting expansion of tubular portion **32**. For example, a downstream region **33** of frame assembly **22**, downstream of flanges **54**, may remain compressed within distal capsule-portion **92**. Alternatively or additionally, the expansion may be inhibited by another portion of delivery tool **89** (e.g., a portion of the delivery tool that is

disposed within lumen 38).

(72) Subsequently, implant **20** is allowed to expand toward its expanded state, such that tubular portion **32** widens to diameter d_2 , and the distance between upstream support portion **40** and flanges **54** reduces to distance d_4 (FIG. 4E). This is typically achieved by deploying downstream region **33** from capsule-portion **92**. This expansion sandwiches tissue of valve **10** (typically including annular tissue and/or leaflets **12**) between upstream support portion **40** and flanges **54**, thereby securing implant **20** at the valve. FIG. 4F shows delivery capsule **90** having been removed from the body of the subject, leaving implant **20** in place at valve **10**.

(73) As described hereinabove, implant **20** is configured such that when tubular portion **32** is expanded, flanges **54** and upstream support portion **40** move a relatively large distance toward each other. This enables distance d_3 to be relatively large, while distance d_4 is sufficiently small to provide effective anchoring. As also described hereinabove, implant **20** is configured such that flanges **54** and upstream support portion **40** can extend radially outward a relatively large distance while tubular portion **32** remains compressed. It is hypothesized that for some applications, these configurations (independently and/or together) facilitate effective anchoring of implant **20**, by facilitating placement of a relatively large proportion of valve tissue (e.g., leaflets **12**) between the flanges and the upstream support portion prior to expanding tubular portion **32** and sandwiching the valve tissue.

(74) It is further hypothesized that the relatively great radially-outward extension of flanges **54** and upstream support portion **40** prior to expansion of tubular portion **32**, further facilitates the anchoring/sandwiching step by reducing radially-outward pushing of the valve tissue (e.g., leaflets **12**) during the expansion of the tubular portion, and thereby increasing the amount of valve tissue that is sandwiched.

(75) It is yet further hypothesized that this configuration of implant **20** facilitates identifying correct positioning of the implant (i.e., with upstream support portion **40** upstream of leaflets **12** and flanges **54** downstream of the leaflets) prior to expanding tubular portion **32** and sandwiching the valve tissue.

(76) It is to be noted that the deployment of implant **20** is therefore achieved by: unhousing a longitudinally-intermediate portion of the implant (to allow flanges **54** to expand), the longitudinally-intermediate portion being longitudinally between a longitudinally-proximal portion of the implant and a longitudinally distal portion of the implant; subsequently, unhousing the longitudinally-proximal portion of the implant (to allow upstream support portion **40** to expand); and subsequently, unhousing the longitudinally-distal portion of the implant (to allow the longitudinally-distal portion of the implant, typically including the tubular portion within which valve member **58** is disposed, to expand).

(77) It is to be further noted that the deployment of implant **20** may be described with respect to delivery tool **89**. For example: distal capsule-portion **92** is moved distally with respect to the implant (to allow flanges **54** to expand); subsequently, proximal capsule-portion **94** is moved proximally with respect to the implant (to allow upstream support portion **40** to expand); and subsequently, distal capsule-portion **92** is again moved distally with respect to the implant (to allow the longitudinally distal portion of the implant, typically including the tubular portion within which valve member **58** is disposed, to expand).

(78) For some applications, delivery tool **89** (e.g., an extracorporeal control portion **110** thereof, disposed at a proximal portion of the delivery tool) is configured to ensure that the operator follows the particular sequence of movement of capsule-portions **92** and **94** described. For example, control portion **110** may comprise a series of locks **114**, each lock becoming unlockable only after the previous step in the sequence has been completed, the unlocking of the lock allowing the subsequent step in the sequence to be performed.

For some such applications, control portion **110** may further comprise a series of controllers **112**, via which the operator causes the movement of a respective capsule-portion for each respective step. An example of how control portion **110** may be used, in accordance with some applications of the invention, is as follows:

(1) Operator moves distal capsule-portion **92** distally with respect to implant **20** by actuating a first controller **112a**. Once capsule-portion **92** has been moved a given distance, a first lock **114a** automatically becomes unlockable (and typically capsule-portion **92** is prevented from moving further than the given distance). (2) Operator unlocks lock **114a**, and then moves proximal capsule-portion **94** proximally with respect to implant **20** by actuating a second controller **112b**. Once capsule-portion **94** has been moved a given distance, a second lock **114b** automatically becomes unlockable. (3) Operator unlocks lock **114b**, and then moves distal capsule-portion **92** further distally with respect to implant **20** by again actuating first controller **112a**. Alternatively, control portion **110** may comprise a third controller (not shown) for this second movement of distal capsule-portion **92**.

(79) It is to be noted that throughout this patent application (including in the specification and in the claims)

the term “unlockable” means able to be unlocked (rather than meaning not lockable).

(80) Therefore, apparatus is provided, in accordance with some applications of the invention, the apparatus comprising an implant (e.g., implant **20**) and a delivery tool (e.g., tool **89**), the tool comprising: (1) a delivery capsule, disposed at a distal portion of the tool, the delivery capsule comprising a proximal capsule-portion **94** dimensioned to house a first part of the implant, and a distal capsule-portion **92** dimensioned to house a second part of the implant; and (2) an extracorporeal control portion **110**, disposed at a proximal portion of the tool, the control portion comprising one or more controllers **112** and one or more locks **114**, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the housed implant via actuation of the controllers, wherein the controllers and the locks are mechanically cooperative such that: in a first state of the tool: a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated, a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited, a proximal movement of the proximal capsule-portion proximally with respect to the implant is inhibited, and responsively to movement of the distal capsule-portion distally, the tool automatically unlocks a second state of the tool; in the second state of the tool: the proximal movement of the proximal capsule-portion is facilitated, the second distal movement of the distal capsule-portion remains inhibited, and responsively to movement of the proximal capsule-portion proximally, the tool automatically unlocks a third state of the tool; and in the third state of the tool: the second distal movement of the distal capsule-portion is facilitated.

(81) For some applications, rather than having three states, tool **89** has two states. For clarity, these will be referred to as state A and state B. However, state A may be similar to the second state described hereinabove, *mutatis mutandis*. (State A may alternatively be called a primary state, and state B may alternatively be called a secondary state.) For applications in which tool **89** has two states, the tool typically has only one lock **114**, the unlocking of which transitions the tool from state A to state B.

(82) In state, A, it is possible to move distal capsule-portion **92** up to the first distance distally with respect to implant **20** (e.g., by actuating controller **112a**), and to move proximal capsule-portion **94** proximally with respect to the implant (e.g., by actuating controller **112b**), but it is not possible to move the distal capsule-portion further distally, which would deploy downstream region **33**. Typically, in state A, it is also possible to return proximal capsule-portion **94** distally and distal capsule-portion **92** proximally. In state B, it is possible to move distal capsule-portion **92** further distally, thereby deploying downstream region **33**. For some applications, lock **114** is not unlockable until distal capsule-portion **92** has been moved the first distance distally, and/or until proximal capsule-portion **94** has been moved proximally.

(83) Therefore, apparatus is provided, in accordance with some applications of the invention, the apparatus comprising an implant and a delivery tool, the tool comprising: a delivery capsule, disposed at a distal portion of the tool, the delivery capsule comprising a proximal capsule-portion dimensioned to house a first part of the implant, and a distal capsule-portion dimensioned to house a second part of the implant; and an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion comprising one or more controllers and one or more locks, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers, wherein the controllers and the locks are mechanically cooperative such that: in a state A of the tool: (i) a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated, (ii) a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited, and (iii) a proximal movement of the proximal capsule-portion proximally with respect to the implant is facilitated; and in a state B of the tool, the second distal movement of the distal capsule-portion is facilitated.

(84) As shown in FIG. 1A, for some applications, in the expanded state of frame assembly **22**, implant **20** defines a toroidal space **49** between flanges **54** and upstream support portion **40** (e.g., a space that is wider than distance d_4). For example, space **49** may have a generally triangular cross-section. It is hypothesized that for some such applications, in addition to sandwiching tissue of the native valve between upstream support portion **40** and flanges **54** (e.g., the tips of the flanges), space **49** advantageously promotes tissue growth therewithin (e.g., between leaflet tissue and covering **23**), which over time further secures implant **20** within the native valve.

(85) Reference is now made to FIG. 5, which is a schematic illustration of a step in the implantation of implant **20**, in accordance with some applications of the invention. Whereas FIGS. 4A-F show an implantation technique in which flanges **54** are expanded prior to upstream support portion **40**, for some

applications the upstream support portion is expanded prior to the flanges. FIG. 5 shows a step in such an application.

(86) Reference is again made to FIGS. 2A-5. As noted hereinabove, implant **20** may be implanted by causing flanges **54** to radially protrude before causing upstream support portion **40** to radially protrude, or may be implanted by causing the upstream support portion to protrude before causing the flanges to protrude. For some applications, implant **20** is thereby configured to be deliverable in a downstream direction (e.g., transseptally, as shown, or transapically) or in an upstream direction (e.g., transapically or via the aortic valve). Thus, for some applications, an operating physician may decide which delivery route is preferable for a given application (e.g., for a given subject, and/or based on available equipment and/or expertise), and implant **20** is responsively prepared for the chosen delivery route (e.g., by loading the implant into an appropriate delivery tool).

(87) It is to be noted that for some applications, downstream delivery of implant **20** may be performed by expanding flanges **54** first (e.g., as shown in FIGS. 4A-F) or by expanding upstream support portion **40** first (e.g., as shown in FIG. 5). Similarly, for some applications upstream delivery of implant **20** may be performed by upstream support portion **40** first, or by expanding flanges **54** first.

(88) Reference is now made to FIG. 6, which is a schematic illustration of implant **20**, in the state and position shown in FIG. 4D, in accordance with some applications of the invention. For some applications, while implant **20** is in the state and position shown in FIG. 4D, leaflets **12** of valve **10** are able to move, at least in part in response to beating of the heart. Frame (A) shows leaflets **12** during ventricular systole, and frame (B) shows the leaflets during ventricular diastole. For some such applications, blood is thereby able to flow from atrium **6** to ventricle **8**, between leaflets **12** and implant **20**. It is hypothesized that this advantageously facilitates a more relaxed implantation procedure, e.g., facilitating retaining of implant **20** in this state and position for a duration of greater than 8 minutes. During this time, imaging techniques may be used to verify the position of implant **20**, and/or positioning of leaflets **12** between upstream support portion **40** and flanges **54**.

(89) Reference is made to FIGS. 7A-B and 8A-B, which are schematic illustrations of frame assemblies **122** and **222** of respective implants, in accordance with some applications of the invention. Except where noted otherwise, frame assemblies **122** and **222** are typically identical to frame assembly **22**, mutatis mutandis. Elements of frame assemblies **122** and **222** share the name of corresponding elements of frame assembly **22**. Additionally, except where noted otherwise, the implants to which frame assemblies **122** and **222** belong are similar to implant **20**, mutatis mutandis.

(90) Frame assembly **122** comprises (i) a valve frame **130** that comprises a tubular portion **132** and an upstream support portion **140** that typically comprises a plurality of arms **146**, and (ii) an outer frame (e.g., a leg frame) **160** that circumscribes the valve frame, and comprises a plurality of legs **150** that each comprise a tissue-engaging flange **154**. Typically, outer frame **160** comprises a ring **166** to which legs **150** are coupled. Ring **166** is defined by a pattern of alternating peaks and troughs, the peaks being fixed to frame **130** at respective coupling points **152**, e.g., as described hereinabove for frame assembly **22**, mutatis mutandis.

(91) Frame assembly **222** comprises (i) a valve frame **230** that comprises a tubular portion **232** and an upstream support portion **240** that typically comprises a plurality of arms **246**, and (ii) an outer frame (e.g., a leg frame) **260** that circumscribes the valve frame, and comprises a plurality of legs **250** that each comprise a tissue-engaging flange **254**. Typically, outer frame **260** comprises a ring **266** to which legs **250** are coupled. Ring **266** is defined by a pattern of alternating peaks and troughs, the peaks being fixed to frame **230** at respective coupling points **252**, e.g., as described hereinabove for frame assembly **22**, mutatis mutandis.

(92) Whereas arms **46** of frame assembly **22** are shown as extending from upstream end **34** of tubular portion **32**, arms **146** and **246** of frame assemblies **122** and **222**, respectively, extend from sites further downstream. (This difference may also be made to frame assembly **22**, mutatis mutandis.) Tubular portions **32**, **132** and **232** are each defined by a repeating pattern of cells that extends around the central longitudinal axis.

Typically, and as shown, tubular portions **32**, **132** and **232** are each defined by two stacked, tessellating rows of cells. In the expanded state of each tubular portion, these cells are typically narrower at their upstream and downstream extremities than midway between these extremities. For example, and as shown, the cells may be roughly diamond or astroid in shape. In frame assembly **22**, each arm **46** is attached to and extends from a site **35** that is at the upstream extremity of cells of the upstream row. In contrast, in frame assemblies **122** and **222**, each arm **146** or **246** is attached to and extends from a site **135** (assembly **122**) or **235** (assembly **222**) that is at the connection between two adjacent cells of the upstream row (alternatively described as being at the upstream extremity of cells of the downstream row).

(93) It is hypothesized by the inventors that this lower position of the arms, while maintaining the length of the lumen of the tubular portion, advantageously reduces the distance that the tubular portion (i.e., the downstream end thereof) extends into the ventricle of the subject, and thereby reduces a likelihood of inhibiting blood flow out of the ventricle through the left ventricular outflow tract. It is further hypothesized that this position of the arms reduces radial compression of the tubular portion by movement of the heart, due to greater rigidity of the tubular portion at sites **135** and **235** (which is supported by two adjacent cells) than at site **35** (which is supported by only one cell).

(94) As shown, in the expanded state of frame assemblies **22**, **122** and **222**, the legs (**50**, **150** and **250**, respectively) are circumferentially staggered with the arms of the upstream support portion (**46**, **146** and **246**, respectively). This allows the legs to move in an upstream direction between the arms during expansion of the tubular portion (**32**, **132** and **232**, respectively), facilitating application of greater sandwiching force on tissue of the native valve. The lower position of the arms of assemblies **122** and **222** includes circumferentially shifting the position of the arms by the width of half a cell. In order to maintain the circumferential staggering of the arms and legs, rings **166** and **266** (and thereby legs **150** and **250**) are circumferentially shifted correspondingly. As a result, whereas the peaks of ring **66** generally align with connections between adjacent cells of the downstream row of cells of tubular portion **32** (and are fixed to these sites), the peaks of rings **166** and **266** are generally aligned midway between these sites (i.e., at spaces of the cellular structure of the tubular portion). Appendages **168** (for assembly **122**) or **268** (for assembly **222**) facilitate fixing of the peak with respect to the tubular structure.

(95) For assembly **122**, appendages **168** are defined by valve frame **130** (e.g., by tubular portion **132** thereof) and extend (in a downstream direction) to the peaks of ring **166**, to which they are fixed. For example, each appendage **168** may define a valve-frame coupling element **131** that is fixed to a respective outer-frame coupling element **161** defined by outer frame **260**. Typically, appendages **168** extend from sites **135**. Typically, appendages **168** are integral with tubular portion **132** and/or in-plane with the tubular portion (e.g., are part of its tubular shape).

(96) For assembly **222**, appendages **268** are defined by outer frame **260**, and extend (e.g., in an upstream direction) from the peaks of ring **266**. Typically, appendages **268** extend to sites **235**, to which they are fixed. For example, each appendage **268** may define an outer-frame coupling element **261** that is fixed to a respective valve-frame coupling element **231** defined by valve frame **230** (e.g., by tubular portion **232** thereof). Typically, appendages **268** are integral with outer frame **260** and/or in-plane with adjacent portions of outer frame **260**, such as ring **266**.

(97) Therefore, frame assembly **122** defines a hub at site **135**, and frame assembly **222** defines a hub at site **235**. For some applications, apparatus therefore comprises: a plurality of prosthetic valve leaflets; and a frame assembly, comprising: a tubular portion (**132** or **232**) defined by a repeating pattern of cells, the tubular portion extending circumferentially around longitudinal axis ax1 so as to define a longitudinal lumen, the prosthetic valve leaflets coupled to the inner frame and disposed within the lumen; an outer frame (**160** or **260**), comprising a plurality of legs (**150** or **250**), distributed circumferentially around the tubular portion, each leg having a tissue-engaging flange (**154** or **254**); an upstream support portion (**140** or **240**) that comprises a plurality of arms (**146** or **246**) that extend radially outward from the tubular portion; and a plurality of appendages (**168** or **268**), each having a first end that defines a coupling element (**161** or **261**) via which the tubular portion is coupled to the outer frame, and a second end; wherein the frame assembly defines a plurality of hubs (**135** or **235**), distributed circumferentially around the longitudinal axis on a plane that is transverse to longitudinal axis ax1, each hub defined by convergence and connection of, (i) two adjacent cells of the tubular portion, (ii) an arm of the plurality of arms, and (iii) an appendage of the plurality of appendages.

(98) Reference is made to FIGS. **9A-C**, which are schematic illustrations of an implant **320** comprising a frame assembly **322**, in accordance with some applications of the invention. Except where noted otherwise, frame assembly **322** is identical to frame assembly **122**, and implant **300** is identical to the implant to which frame assembly **122** belongs, mutatis mutandis. FIG. **9A** is a side-view of implant **320**, and FIG. **9B** is an isometric bottom-view of the implant.

(99) Frame assembly **122** comprises (i) a valve frame **330** that comprises a tubular portion **332** and an upstream support portion **340** that typically comprises a plurality of arms **346**, and (ii) an outer frame (e.g., a leg frame) **360** that circumscribes the valve frame, and comprises a plurality of legs **350** that each comprise a tissue-engaging flange **354**. Typically, outer frame **360** comprises a ring **366** to which legs **350** are coupled. Ring **366** is defined by a pattern of alternating peaks and troughs, the peaks being fixed to frame **330** at

respective coupling points **352**, e.g., as described hereinabove for frame assembly **22** and/or frame assembly **122**, *mutatis mutandis*.

(100) Frame assembly **322** comprises an annular upstream support portion **340** that has an inner portion **342** that extends radially outward from the upstream portion (e.g., the upstream end) of tubular portion **332**. Upstream support portion **340** further comprises one or more fabric pockets **344** disposed circumferentially around inner portion **342**, each pocket of the one or more pockets having an opening that faces a downstream direction (i.e., generally toward the downstream end of implant **320**). In the figures, upstream support portion **340** has a single toroidal pocket **344** that extends circumferentially around inner portion **342**.

(101) Typically, a covering **323** (e.g., similar to covering **23**, described hereinabove, *mutatis mutandis*) is disposed over arms **346**, thereby forming pocket **344**. Further typically, arms **346** are shaped to form pocket **344** from covering **323**. For example, and as shown, arms **346** may curve to form a hook-shape.

(102) For some applications, portion **340** has a plurality of separate pockets **344**, e.g., separated at arms **346**. For some such applications, covering **323** is loosely-fitted (e.g., baggy) between radially-outward parts of arms **346**, e.g., compared to inner portion **342**, in which the covering is more closely-fitted between radially-inward parts of the arms.

(103) FIG. **9C** shows implant **320** implanted at native valve **10**. Pocket **344** is typically shaped and arranged to billow in response to perivalvular flow **302** of blood in an upstream direction. If ventricular systole forces blood in ventricle **8** between implant **320** and native valve **10**, that blood inflates pocket **344** and presses it (e.g., covering **323** and/or the radially-outward part of arm **346**) against tissue of atrium **6** (e.g., against the atrial wall), thereby increasing sealing responsively. It is hypothesized by the inventors that the shape and orientation of pocket **344** (e.g., the hook-shape of arms **346**) facilitates this pressing radially-outward in response to the pocket's receipt of upstream-flowing blood.

(104) Pocket(s) **344** may be used in combination with any of the implants described herein, *mutatis mutandis*.

(105) Reference is again made to FIGS. **1A-9C**. It is to be noted that unless specifically stated otherwise, the term “radially outward” (e.g., used to describe upstream support portion **40** and flanges **54**) means portions of the element are disposed progressively further outward from a central point (such as longitudinal axis **ax1** or tubular portion **32**), but does not necessarily mean disposed at 90 degrees with respect to longitudinal axis **ax1**. For example, flanges **54** may extend radially outward at 90 degrees with respect to longitudinal axis **ax1**, but may alternatively extend radially outward at a shallower angle with respect to the longitudinal axis.

(106) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

Claims

1. Apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus comprising: a valve frame, comprising: a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis; and a plurality of arms, extending radially outward from an upstream portion of the tubular portion to define an arm span, and configured to engage tissue in the atrium; a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; and an outer frame: coupled to the valve frame, and comprising a ring that circumscribes the tubular portion, and a plurality of flanges that extend radially outward from the ring to define a flange span, and are configured to engage tissue of the ventricle, wherein: each arm of the plurality of arms defines a rigid portion and a flexible portion that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion, and the majority of the flexible portion is disposed further radially outward than the flange span.
2. The apparatus according to claim 1, wherein the apparatus is configured such that the rigid portion provides resistance against axial movement of the apparatus while the flexible portion facilitates conformation of each arm of the plurality of arms to anatomy of the atrium and/or lateral movement of the

apparatus within the native valve.

3. The apparatus according to claim 1, wherein each arm of the plurality of arms defines a serpentine shape in the flexible portion.

4. The apparatus according to claim 1, wherein each arm of the plurality of arms defines a plurality of holes in the flexible portion.

5. The apparatus according to claim 1, wherein less than 90 percent of the flexible portion is disposed further radially outward than the flange span.

6. The apparatus according to claim 1, wherein more than 60 percent of the flexible portion is disposed further radially outward than the flange span.

7. The apparatus according to claim 1, wherein, for each arm of the plurality of arms, the flexible portion covers an overall surface area including (i) the area occupied by material from which the arm is formed, and (ii) spaces, and the material from which the arm is formed occupies less than 80 percent of the overall surface area.

8. The apparatus according to claim 7, wherein the material from which the arm is formed occupies more than 20 percent of the overall surface area.

9. The apparatus according to claim 7, wherein the material from which the arm is formed occupies less than 70 percent of the overall surface area.
